X
1
I
′ R′2 X′
2 2 I2
I0
R
′ L
V
1
R
0
X
0
V
′ 2
X
′
L

Figure 4.16 *Equivalent circuit, with secondary quantities transferred to primary.*

I
1
I
′ R1 X1 R′ X′ R01 X012 2 2 I1 I′
2
I0 I0
R′L R′L R0 X0 V1 R0 X0V1
X
′ X′
L L

Figure 4.17 *Approximate equivalent circuit.*

4.7.1. Efficiency of transformer

The efficiency η = $\frac{\text{Output}}{\text{Input}}$. Here, output refers to the output active power supplied by the transformer to the load (in Watts) and input refers to the active power drawn

from the supply (in Watts). In percentage we can say,

$\%\eta = \frac{\text{Output}}{\text{Input}} \times 100$ (4.28)

Input = Output + Losses. (4.29) The losses in the transformer are,

• *Cu loss* in primary and secondary winding. The total Cu loss = $I^2R_1 + I^2R_2 = I$

2

1 2

$R01 = I_2R02._1$

The copper loss is proportional to (current)^2 or (kVA)^2. This means that at half full load, the copper loss would be one fourth that at full load.

• The *iron loss* in the core of the transformer. It is also called core loss. The core loss is made of the *hysteresis loss* and *eddy current loss*. Hysteresis loss,

$W_h = PB^{1.6} f$; P is a constant (4.30)$_{max}$

(4.30) is the Steinmetz's empirical relation to compute hysteresis loss.

$^2 f^2$; Q is a constant (4.31)$W_e = QB_{max}$

Because the core flux is practically a constant, the core losses remain constant for all loads. These losses are minimized by using steel of high silicon content for the core and using very thin laminations.

As discussed earlier, the no-load power input is equal to the core losses.

4.7.2. Regulation of a transformer

When the transformer is loaded, keeping the primary voltage a constant, the secondary voltage decreases because of the voltage drop in the winding. Let $_0V_2$, be the no-load secondary terminal voltage. The *regulation* is defined as

$\%R = \frac{0V2 - V2}{0V2} \times 100$ (4.32)

where V_2 is the full-load, secondary terminal voltage. The change in the voltage from no-load to full-load can be calculated as follows.

When there is no-load, the voltage drop in the primary winding is negligible, since I_0 is very small. Therefore,

V1 E1 and E2 = KE1 = KV1 Further, $E_2 = {}_0V_2$, the secondary terminal no-load voltage. We now draw the phasor diagram as shown in Fig. 4.18.

E

$E_2 = 0V^2 \varphi 2$

A C Dφ_2 V_2 φ_2
I2 I2R02$_B$
I2X02
Figure 4.18 *Approximate voltage drop (lagging p.f.).*
0V2 I2X02
I2 φ2 V_2 I_2R_{02}
Figure 4.19 *Approximate voltage drop (leading p.f.).*
The no-load terminal voltage = $_0V_2$.

The approximate voltage drop can be taken to be AD = AC + CD = I_2R_{02} cosφ_2 + I_2X_{02} sinφ_2. In Fig. 4.18, the load current is assumed to be lagging the terminal voltage. For a leading p.f., the figure is redrawn as shown in Fig. 4.19.

The approximate voltage drop is given by
I2R02 cosφ2 − I2X02 sinφ2. In general the approximate voltage drop for a transformer is given by
I_2R_{02} cosφ_2 ± I_2X_{02} sinφ_2 (4.33)
The % voltage drop, which is nothing but % regulation is given by
%R = I2R02 cosφ2 ± I2X02 sinφ2× 100 (4.34)
0V2
100× I2R02$_{\cos\varphi2 \pm}$ 100I2X02$_{\sin\varphi2=}$ 0V2 0V2
= v_r cosφ_2 ± v_x sinφ_2 (4.35)
Here v_r is the *percentage resistive drop*,
100× I2R02$_{(4.36)vr =}$
0V2
and v_x is the *percentage reactive drop*,
v
x
=
100× I2X02$_{(4.37)}$
0V2

The regulation thus defined is called % *regulation down* (this is normally used). If the voltage drop is taken as a percentage of full load voltage instead of no-load voltage, it is called *regulation up*.

%Regulation (up) = $\frac{^{0}V2 - V2}{V2} \times 100$ (4.38)

The value of the regulation should be low for a good transformer. A low regulation, implies that the secondary voltage is a constant over varying load conditions.

4.8. Transformer tests

The performance of the transformer, in terms of efficiency and regulation can be pre determined by performing suitable tests. The two tests are (i) no-load test (ii) short-circuit test. These tests, do not require direct loading of the transformer and are economical and convenient.

4.8.1. Open-circuit test or no-load test

The purpose of this test is to determine the no-load or core losses and the no-load current I_0, which are used to find R_0 and X_0 of the equivalent circuit.

The circuit connection for this test is shown in Fig. 4.20.

In this test,

• One of the windings, usually the high voltage winding is left open.

• The other winding, usually the low voltage winding is connected to its normal rated voltage.

• The wattmeter, ammeter and voltmeter are connected as shown in the figure.

W_0 ^{I0} A

$^{V}1$ V V V_2

LV HV **Figure 4.20** *No-load test.*

• With rated voltage applied to the primary, normal flux will be set up in the core, and we will have normal iron losses in the core, as indicated by the wattmeter.

• The ammeter reads the no-load current I_0.

• The voltmeters read the respective terminal voltages and the ratio of V_2/V_1 gives the turns ratio K.

The calculations are as follows:

core loss/iron loss) = W_0 (4.39a)W_i(

cos W0 = V1I0 cosφ0 (4.39b) ^{W0} (4.39c)$\varphi_0 = _{V1I0}$

$I\mu$ = I0 sinφ0; X0 = $\frac{^{V1}}{I\mu}$ (4.39d)

^{V1} (4.39e)$I_w = I_0 \cos\varphi_0$; $R_0 = _{Iw}$

Example 4.14 The no-load test on a single phase transformer gave the following

results:
V_1 = 220 V; V_2 = 110 V; I_0 = 0.5 A; W_0 = 40 W. Find the turns ratio, I_w,I_μ,R_o and X_0 and the iron loss.

Solution:

$^{N2}_{N1} = \frac{110}{220} = \frac{1}{2}$ (i) K =

(ii) cos

φ

0

=

$W0 = 40$

$220 \times 0.5 = 0.3636_{V1I0}$

$\sin \varphi_0 = 0.9315$

$I_\mu = I_0 \sin\varphi_0 = 0.5 \times 0.9315 = 0.466$ A $I_w = I_0 \sin\varphi_0 = 0.5 \times 0.3636 = 0.181866$ A

(iii) $^{V1}_{Iw} = \frac{220}{0.1818} = 1210.12$ $R_0 =$

$^{V1}_{I\mu} = \frac{220}{0.466} = 472.1$ $X0 =$

(iv) Iron loss W_0 = 40 W.

4.8.2. Short-circuit text

This test is used to compute the equivalent resistance and reactance as referred to one of the windings and the full load copper loss. The test set up is shown in Fig. 4.21 In this test,

• One of the windings, usually the low voltage winding is solidly short-circuited.

W_{sc}

^{Isc} A

V_1 V V_{sc}

HV LV **Figure 4.21** *Short circuit test.*

I1

R_1 X_1 R_2'' $_{2\ V' = 02}$ Short $_{V1 = Vsc}$ circuit

Figure 4.22 *Equivalent circuit under short circuited secondary.*

• A low voltage (around 5–10% of normal primary voltage) at rated frequency is applied such that rated current flows through the primary winding

• Since the voltage is only around 5 to 10% of the rated value, the mutual fluxφ produced in the core is very small. Hence the core losses can be neglected. The equivalent circuit under this condition is shown in Fig. 4.22.

• The output terminals being short circuited, there is no power supplied from the secondary to any external load. Also, the core losses are negligible. Hence, the input power drawn is equal to the full load copper loss of the transformer.

Let W_{sc} be the wattmeter reading, I_{sc} be the ammeter reading and V_{sc} be the voltmeter reading , of Fig. 4.21. Then the calculations are as follows:

Full load copper loss = W_{sc} (4.40a) Wsc = I^2R01 (4.40b)$_{sc}$

^{Wsc} (4.40c)R_{01} =$_{I2}$

sc

^{Vsc} (4.40d)Z_{01} =$_{Isc}$

$X01 = Z^2 - R^2$ (4.40e)$_{01\ 01}$

It is to be noted that the applied voltage is consumed in the impedance drop of the windings.

4.8.3. Predetermination of efficiency and regulation from OC and SC test data

The test data can be used to predetermine the efficiency and regulation of the transformer, without actually loading it. One way to calculate the efficiency and regulation, under given load conditions, is to actually load the transformer, measure the input power, output power and the voltages, to calculate the efficiency and regulation by using,

% ^{output} 100 (4.41)η =$_{Input}$ ×

%R = $^{0V2\ -\ V2}$ × 100 (4.42)

0V2

In predetermination, we do not load the transformer, but make a prediction of the efficiency and regulation for different loads.

Let W_0 and W_{sc} be the wattmeter readings for the OC and SC tests respectively. Similarly, let I_0 and I_{sc} be the ammeter readings of the OC and SC tests respectively. Let x be the fraction of the full load output for which %η and %R have to be calculated and cosφ be the power factor of the load Now,

output =Rated kVA× x × cosφW

↓↓ ↓ (4.43) Full load percentage power factor

Iron losses (core losses) = W_0 (4.44)

Copper loss is = x^2W_{sc} (4.45) (Copper loss is proportional to square of the current. Hence, when load current changes by a fraction x, the copper loss changes by a factor x^2)

From (4.43), (4.44), (4.45) we calculate %η as

$\%\ \text{output} \times 100\ (4.46)\eta = \text{output} + \text{Iron loss} + \text{Cu loss}$

To calculate regulation we find R_{01} and X_{01} using (4.40a and 4.40c).

$Z_{01} = \frac{V_{sc}}{I_{sc}}$

$W_{sc}R_{01} = I_{2sc}$

$X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$

These values are transferred to the secondary side as $R02 = K^2R01$; $X02 = K^2X01$

The voltage drop for a secondary currentI_2 is given by $R02I2\cos\varphi \pm X02I2\sin\varphi$ where I_2 is x times the rated full load current. Then

$\%\ R02I2\cos\varphi \pm X02I2\sin\varphi_{(4.47)R = 0V2}$

In (4.47) $_0V_2$ is the rated secondary voltage and I_2 = x × full load current.

4.8.4. Condition for maximum efficiency of a transformer

Full load Cu loss = $I^2R_{01} = W_{Cu1}$

Iron loss = $W0 = W_i$

Primary input = $V_1I_1\cos\varphi_1$

Output Input − losses$_=$ $V1I1\cos\varphi1 - I^2R01 - W_{i\eta = \text{Input} = 1}$ Input $V_1I_1\cos\varphi_1$

$= 1 - WiI1R01 - V_1I_1\cos\varphi_1V_1\cos\varphi_1$

The current I_1 changes with load, while V_1 is normally maintained constant at the rated value. Differentiating η, with respect to I_1,weget

R01 W_i

dI1

$^{d\eta}=-_{V1\ \cos\varphi 1}\ +_{V1I2\ \cos\varphi 11}$

The efficiency is maximum when $^{d\eta}$ = 0. Hence,$_{dI1}$

W_i

$V_1\ \cos\varphi_1$

$^{R01\ =}\ V1I^2\ \cos\varphi 1_1$

or Wi = I^2R01 = Wcu.$_1$

Therefore, the efficiency is maximum at the load where
Cu loss = Iron loss (4.48)
The output current corresponding to this is
$I_2\ =^{Wi}$ (4.49)$_{R02}$
The load at which maximum efficiency occurs is
full load
×
W_i Wcu

The transformer being a static device, the efficiency is very high. The copper loss is dependent on the current and the iron loss on the voltage. Hence, the total loss depends on (VA) and not on the power factor. Therefore the rating of the transformer is in kVA.

4.8.5. Separation of core losses

The total core loss is given by (4.30) and (4.31).

$PB^{1.6}\ f^{\,2}$ + $QB^2\ f^{\,2}$ (4.50)$W_i\ =_{max\ max}$

If we find the iron loss (from no-load test) at two different frequencies but with same flux density, then we can evaluate P and Q in (4.50) and individually find eddy current and hysteresis losses.

Example 4.15 A 5 kVA, 220/1100 V, 50 Hz single phase transformer gave the following test results:
OC test (LV side): 220 V; 0.9 A; 75 W SC test (HV side): 50 V; 4.545 A; 120 W
Calculate the parameters of the equivalent circuit referred to LV side.
Solution:
cos

φ

0
=
W0 = 75
220× 0.9 = 0.379 A_{I1I0}
$I_w = I_0 \cos\varphi_0 = 0.9\times 0.379 = 0.341$ A $I_\mu = I_0 \sin\varphi_0 = 0.9\times 0.925 = 0.833$ A
V1 =220 = 645.2R0 =Iw 0.341
V1 =220 = 264.1X0 =Iμ 0.833
Z_{02} (since in SC test measurements are on HV side) Vsc =50 = $11_{Isc\ 4.545}$
R
02
=
W_{sc} = 100
(
4.545
)
2
= 5.81_{I2}
sc
$X02 = Z^2 - R^2 = 11^2 - 5.81^2 = 9.34_{02\ 02}$
R02 =5.81 = 0.2324R01 =K2 1100 2
220
X02 9.34 0.3736 .X01 =K2 =25 =
The equivalent circuit is shown in Fig. 4.15.
R_{01} X_{01}
0.2324Ω 0.3736Ω
220V 645.2Ω 264.1Ω Load (referred to primary)
Figure 4.23 *Example 4.15.*

Note that when we draw the equivalent circuit, all parameters must be referred to the same side. Since, generally in the OC test and SC test we make measurements on LV side and HV side respectively, ensure that the equivalent values are transferred to the same side.

Example 4.16 A 25 kVA, 2200/220 V, 50 Hz transformer, has the following test data: OC test (LV side): 220 V, 4 A, 150 W
SC test (HV side): 90 V, 10 A, 350 W

Determine

(i) The efficiency and regulation at full load, unify power factor.
(ii) The efficiency and regulation at half full-load, 0.8 p.f. lag
(iii) The efficiency and regulation at half full load, 0.8 p.f. lead.

Solution:

Full load current on HV side = $\frac{25,000}{2,200}$ = 11.36 A.

The SC test is conducted at 10 A. Therefore the full load Cu loss is given by

$W_{Cu} = 350 \times 11.36^2$

$_{10}$ = 451.7 W W_i = 150 W

$W_{sc} = \frac{350}{I2\ 102sc} = 3.5R02 =$

$V_{sc} = \frac{90}{Isc\ 10} = 9Z02 =$

$X02 = Z^2 - R^2 = 8.29_{02\ 02}$

(i) Efficiency and regulation at full load u.p.f. Output = x × kVA× p.f. = 1× 25× 1 = 25 kW = 25,000 W W_i = 150 W;

W_{Cu} = 451.7 W

$\% \eta = \frac{25,000}{25,000+ 150+ 451.7} = 97.65\%$

$\%R = \frac{R02I2 \cos\varphi + X02I2 \sin\varphi \times 100}{0V2}$

$\frac{11.36(3.5\times 1+ 8.29\times 0) \times 100}{2,200} = 1.8\%$

(ii) The efficiency and regulation at half full load 0.8 p.f. lag.

Current at half full load = $\frac{11.36}{2}$ = 5.68 A

Output = $\frac{1}{4} \times 25\times 0.8 = 10$ kW = 10,000 W

W_i = 150 W;

$W_{Cu} = \frac{451.7}{4} = 112.925$ W

$\eta = \frac{10,000}{\%}$

$10{,}000 + 150 + 112.925 \times 100 = 97.43\%$ % $5.68(3.5 \times 0.8 + 8.29 \times 0.6) \times 100 = 2.01\%R = 2{,}200$

(iii) The efficiency and regulation at half full load 0.8 p.f. load.

The efficiency does not change if the p.f. is changed from 0.8 lag to 0.8 lead, as the losses remain the same.

$\therefore \%\eta = 97.43\%$

$\%R = R_{02}I_2 \cos\varphi - X_{02}I_2 \sin\varphi \times 100$ (for leading p.f.)$_{0V2}$

$5.68(3.5 \times 0.8 - 8.29 \times 0.6) \times 100 = -0.561\% = 2{,}200$

Example 4.17 The test results on a 250/500 V transformer is given below: OC test (LV side): 250 V, 1 A, 80 W
SC test (HV side): 20 V, 12 A, 100 W.

Determine the equivalent circuit referred to primary LV side. Also determine the input voltage applied and the efficiency when the output is 10 A at 500 V, 0.8 p.f. lag.

Solution:

$K = 500 = 2_{250}$

OC test data

$\cos^{80} = 0.32\varphi 0 = W0 =_{250V1I0}$

$I_w = I_0 \cos\varphi_0 = 1 \times 0.32 = 0.32$ A $I_\mu = I_0 \sin\varphi_0 = 1 \times 0.95 = 0.95$ A $V1 = 250 = 781.25R0 =_{Iw\ 0.32}$

$V1 = 250 = 263.16X0 =_{I\mu\ 0.95}$

SC test data $20 = 1.57Z02 = Vsc = 12Isc$

$Wsc = 100 = 0.694R02 =_{I2\ 122sc}$

$X\ 02 = Z^2 - R^2 = 1.519_{02\ 02}$

These are values referred to HV side we now refer them to LV side.

$R02 = 0.694 = 0.174R01 =_{K2\ 22}$

$X02 = 1.519 = 0.38X01 =_{K2\ 22}$

The equivalent circuit is shown in Fig. 4.24

(ii) To calculate the applied input voltage we draw the phasor diagram by using the equivalent circuit.

The output current is 10 A at 500 V, 0.8 p.f. log.

I_2 = 10 A
I_2 = KI2 = 2× 10 = 20 A

The phasor diagram is shown in Fig. 4.25.
I
1
I
′ 2
R_{01} X01 0.174Ω 0.38Ω
250V
R
0
X
0
V
′
2
781.25Ω 263.16Ω
Figure 4.24 *Equivalent circuit, example 4.17.*
B
V
′ I_1X_{01} 1
V_1 I_1R_{01} $V_1\sin\varphi_1$ $\varphi_1 = \cos^{-1}0.8$
IAOC $V_1\cos\varphi_1$ I_1R_{01} **Figure 4.25** *Phasor diagram, Example 4.17.*
In the figureV_1 is the no-load primary voltage, of 250 V. As the secondary is loaded,
• If primary voltage is kept constant, the secondary terminal voltage drops.

• If secondary terminal voltage drops, the primary voltage has to be increased to keep it a constant. This increase must be equal to the drop in the equivalent impedance referred to the primary. V_1 is the increased voltage.

Referring to Fig. 4.25,

$OB^2 = OC^2 + BC^2$ OB = V_1 = $(V1\cos\varphi1 + I1R01)^2 + (V1\sin\varphi1 + I1X01)^2$ = $(250\times 0.8 + 20\times 0.174)^2 + (250\times 0.6 + 20\times 0.38)^2$ = 257.4 V

Output = $V_2 I_2 \cos\varphi = 500 \times 10 \times 0.8 = 4000$ W

$W_i = 80$ W; $W_{Cu} = I^2 R_{02} = 10^2 \times 0.694 = 69.4$ W_2

%

η

=

4000

$4000 + 80 + 69.4 \times 100 = 96.4\%$

Example 4.18 The test results on a 20 kVA, 2200/220 V 50 Hz transformer is as follows:

OC test: 220 V, 4.2 A, 150 W (LV side)

SC test: 85 V, 11 A, 350 W (HV side)

Find the regulation at full load (i) 0.8 p.f. lag (ii) 0.8 p.f. lead (iii) u.p.f. What is the p.f. on short-circuit?

Solution: The OC test data is not necessary to compute the regulation.

$\frac{350}{112} = 2.892$ (on HV side) $R_{01} =$

$\frac{85}{11} = 7.727$ $Z01 =$

$X01 = Z^2 - R^2 = 7.165_{01\ 01}$

Full load current $= \frac{20{,}000}{2{,}200} = 9.09$ A on HV side

$\%R(0.8\text{ p.f. lag}) = \frac{9.09(2.892 \times 0.8 + 7.165 \times 0.6)}{2{,}200} \times 100 = 2.73\%$

$\%\ \frac{9.09(2.892 \times 0.8 - 7.165 \times 0.6)}{2{,}200} \times 100 = -0.82\% R(0.8\text{ p.f. lead})$

$\%\ \frac{9.09(2.892 \times 1 + 7.165 \times 0)}{2{,}200} \times 100 = 1.195\% R(\text{u.p.f.})$

p.f. on short circuit $\frac{R01}{Z01} = \frac{2.892}{7.727} = 0.374$ lag. $=$

Example 4.19 A 230/230 V, 3 kVA transformer gave the following results. OC test: 230 V, 2 A, 90 W

SC test: 15 V, 13 A, 100 W

Determine the regulation and efficiency at full load 0.8 p.f. lag.

Solution:

Rated current $= \frac{3000}{230} = 13$ A.

W_{Cu}(full load) = 100 W $W_i = 90$ W $\%\eta = \frac{3000 \times 0.8}{3000 \times 0.8 + 100 + 90} \times 100 = 92.66\%$

$\frac{15}{13} = 1.154$ Z_{02}(from SC test) $=$

$\frac{100}{132} = 0.592$ $R02 =$

$X_{02} = 1.154^2 - 0.592^2 = 0.99$

$\%\ \frac{13[0.592 \times 0.8 + 0.99 \times 0.6]}{230} \times 100 = 6.03\%.R$

Example 4.20 In a 25 kVA, 2200/220 V, single-phase transformer, the iron loss

is 350 W and full load copper loss is 400 W. Plot the curve of efficiency vs. output for (i) upf (ii) 0.8 p.f. lag. Find the load kVA corresponding to maximum efficiency.

Solution:

(i) *u.p.f.*

Let's assume different fractions of full load and calculate efficiency.

1th 1^2

$_4$ *FL* : W_i = 350 W; W_{Cu} = 400_4 = 25 W; output = 25× 1000× ^{1} = 6, 250 W.$_4$

%

η

=

6,250

6,250+ 350+ 25 $^{\times 100 = 94.34\%}$

1 1^2

2 FL : W_i = 350 W; W_{Cu} = 400×$_2$ = 100 W;

output = 25× 1000× ^{1} = 12, 500 W_2

$^{\%\eta = 12,\ 500 \times 100 = 96.52\%}$12, 500+ 350+ 100

3

th 3 2

FL : W_i = 350 W; W_{Cu} = 400×$_4$ = 225 W;$_4$

output = 25× 1000× ^{3} = 18, 750 W.$_4$

$^{\%\eta = 18,750 = 97.02\%}$18,750+ 350+ 225

FL : W_i = 350 W; W_{Cu} = 400 W; Output = 25, 000 W. %η = 97.08%

We repeat the calculations for a p.f. of 0.8 lag.

(ii) *0.8 p.f. lag*

^{1} *FL* : W_i = 350 W; W_{Cu} = 25 W; output = 25, 000× ^{1} × 0.8 = 5000 W.$_{4\ 4}$

%

η

=

5000

5000+ 350+ 25 $^{= 93.02\%}$

^{1} *FL* : W_i = 350 W; W_{Cu} = 100 W; Output = 18, 500× 0.8 = 10,000 W.$_2$

10,000$^{\%\eta\ =}$ 10,000+ 350+ 100 $^{= 95.69\%}$ ^{3}*thFL* : W_i = 350 W; W_{Cu} = 225 W;

Output = 18, 750× 0.8 = 15,000 W.4
15,000%η = 15,000+ 350+ 225 = 96.31%
Table 4.1 Output kW %η Output kW %η

6.25 94.34 5 97.02
12.5 96.52 10 95.69
18.75 97.02 15 96.37
23.38 97.09 18.7 96.39
25.0 97.08 20 96.38

FL : W_i = 350 W; W_{Cu} = 400 W; Output = 25, 000× 0.8 = 20, 000 W.
% 20,000 = 96.38%η = 20,000+ 350+ 400
The load at which maximum efficiency occurs is
FL kVAIron loss = 25350 = 23.38 kVA$_{\text{FL Cu loss 400}}$
At upf, 23.38 kVA = 23.38 kW.
At 0.8 p.f., 23.38 kVA = 18.704 kW.
The maximum efficiency is computed for both cases. At maximum efficiency, W_i = WCu.
Maximum efficiency at u.p.f. =23, 380 × 100 = 97.09%$_{\text{23, 380+ 350+ 350}}$
Maximum efficiency at 0.8 p.f.18704 = 96.39%. =$_{\text{18704+ 350+ 350}}$
The results are tabulated in Table 4.1.
The efficiency vs output is plotted in Fig. 4.26.
Example 4.21 IfW_1 andW_2 are the iron loss and full load copper loss of a transformer, find their ratio if maximum efficiency occurs at 80% FL.
100%
p.f. = 1.0
%η p.f. = 0.8 95%
10 Output (Kw)20 **Figure 4.26** *Example 4.21*.
Solution: At 0.8 FL, Copper loss = $W_2(0.8)^2$ since maximum efficiency occurs here, iron loss is equal to copper loss.
W1 = W2(0.8)2 W1 = 0.64.W2
Example 4.22 A 230 V, 2.5 kVA transformer has an iron loss of 100 W at 40-Hz and 70 W at 30 Hz. Find the loss at 50 Hz.
Solution:
$W_i = W_e + W_h$ (eddy currenty loss) $W_e \propto f^2$ (hysteresis loss) $W_h \propto f$
$\therefore Wi = Af + Bf^2$

At 40 Hz

$100 = A \times 40 + B \times 40^2$

$40A + 1600B = 100.$

At 30 Hz

$70 = A \times 30 + B \times 30^2$

$30A + 900B = 70.$

Solving we get A

=

$11_{\text{and B} =} 1$

6 60

Loss at 50 Hz = ^{11} 50+ $^{1} \times (50)^2 = 133.33$ W.$_{6 \times 60}$

Example 4.23 A 10 kVA, 2500/250 V, single phase transformer gave the following test results.

OC test: 250 V, 0.8 A, 50 W SC test: 60 V, 3 A, 45 W.

Calculate (i) Efficiency at half FL, 0.8 p.f. lag

(ii) Load kVA at which maximum efficiency occurs and maximum efficiency at 0.8 p.f. lag.

(iii) Regulation at 0.8 p.f. lag and 0.8 p.f. lead, FL.

Solution:

(i) Full load current = ^{10,000} 4A.$_{2,500 =}$

Full load copper loss

W

Cu

=

45

×

4^2

$_3$ = 80 W

(This is because SC test has been conducted at 3 A). At half FL, 0.8 p.f. lag, the efficiency is given by

%

η

=

10

×
1000
×
0.8
×
1
^{2} × 100 10
×
1000
×
0.8
×
^{1} + 50+ 80× ^{12}
2 2
^{4000} × 100 = 98.28%.=$_{4000+\ 50+\ 20}$

(ii) Load kVA at which maximum η occurs

FL kVA^{Wi} = 10^{50} = 7.9 kVA.$_{WCu\ 80}$

$^{7.9\times\ 1000\times\ 0.8\ \times\ 100\ =\ 98.44\%\eta max(0.8\ p.f.lag)\ =}$ (7.9× 1000× 0.8) + 50+ 50

(iii)

R02 = ^{45} 5 ; Z02 = ^{60} 20 ;$_{9}$ =$_{3}$ = X_{02} = $20^2 - 5^2$ = 19.36 .

% $^{4(5\times\ 0.8+\ 19.36\times\ 0.6)}$ × 100 = 2.5%R(0.8 p.f. lag) =$_{2500}$

%R(0.8 p.f. lead) = $^{4(5\times\ 0.8-\ 19.36\times\ 0.6)}$ × 100=−1.21%$_{2500}$

Example 4.24 A50 kVA,1− φ transformer has an iron loss of 500 W and full load copper loss of 900 W(i).

(i) Find the load at which maximum efficiency is achieved at u.p.f. (ii) At what load will efficiency be 96.8% at u.p.f.

(iii) What is maximum efficiency at 0.9 p.f.?

Solution:

Load at which maximum efficiency occurs

=

FL kVA Wi WCu

= 50×^{500} = 37.26 kVA$_{900}$

(i) At u.p.f., the efficiency at 37.26 kVA, is

$^{\%\eta\ =\ 37.26\times\ 1000\times\ 1\ \times\ 100\ =\ 97.38\%.}$37.26× 1000+ 500+ 500

(ii) Let x be the fraction of full load at which η = 96.8% at u.p.f.

50× 1000× x$^{0.968\ =}$ 50× 1000× x + 500+ x^{2}900. Solving, x = 0.3822 and x = 1.45.

Thus, we get an η of 96.8% at 38.22% and 145% of full load. (iii) Maximum efficiency at 0.9 p.f. is obtained similar to (i).

$$\%\eta = \frac{37.26\times 1000\times 0.9}{37.26\times 1000\times 0.9+ 500+ 500} \times 100 = 97.1\%.$$

Example 4.25 A transformer has a percentage resistive drop of 1.8% and a reactance drop of 5.4%. What is the power at which regulation is (i) zero (ii) positive maximum.

Solution:

(i) $\%R = v_r \cos\varphi \pm V_x \sin\varphi$ Regulation can be zero only at leading pf.

$$0 = V_r \cos\varphi - v_x \sin\varphi$$

$$0 = 1.8 \cos\varphi - 5.4 \sin\varphi$$

$$\tan\varphi = \frac{1.8}{5.4} = 0.333 \text{ leading; } \varphi = 18.44^\circ$$

p.f. = $\cos\varphi = 0.949$ lead

(ii) The maximum positive regulation occurs at lagging pf.

$$\%R = V_r \cos\varphi + v_x \sin\varphi$$

$$\frac{dR}{d\varphi} = -v_x \sin\varphi + v_x \cos\varphi = 0$$

$$\tan\varphi = \frac{5.4}{1.8} = 3; \varphi = 71.56^\circ$$

p.f. = $\cos\varphi = 0.316$ lag

Example 4.26 A 600 kVA transformer, has an efficiency 92% at both full load and half load u.p.f. Determine its efficiency at 75% full load at 0.8 pf lag.

Solution: $$\eta = \frac{x\text{kVA}\times \cos\varphi}{x\text{kVA}\times \cos\varphi + W_i + x^2 W_{Cu}}$$ *At FL u.p.f.*, x = 1,

$$0.92 = \frac{1\times 600\times 1}{1\times 600\times 1+ W_i + W_{Cu}} \quad W_i + W_{Cu} = 52.174 \text{ kW} \quad (4.51)$$

At $\frac{1}{2}$ *FL u.p.f.*, x = $\frac{1}{2}$

$$\frac{\frac{1}{2}\times 600\times 1}{\frac{1}{2}\times 600\times 1+ W_i + (\frac{1}{2})^2 W_{Cu}} = 0.92$$

$$W_i + \frac{1}{4} W_{Cu} = 26.087 \text{ kW} \quad (4.52)$$

Solving (4.51) and (4.52) we get, W_i = 17.39 kW; W_{Cu} = 34.78 kW.

η *at 75% FL, 0.8 pf lag*

$$\% \eta = \frac{3}{4} \times 600 \times 0.8 \times 100 \Big/ \left(\frac{3}{4} \times 600 \times 0.8 + 17.39 + \left(\frac{3}{4}\right)^2 \times 34.78\right)$$

$$= 90.69\%$$

Example 4.27 A 200 kVA transformer has an efficiency of 98% at full load. If the maximum efficiency occurs at $\frac{3}{4}$ full load, find the iron and copper losses. The p.f. of the load is 0.8.

Solution: Full load output at 0.8 pf = 200 × 0.8 = 160 kW. Since η = 98%, full load input = $\frac{160}{0.98}$ = 163.265 kW.

Losses = 163.26− 160 = 3.265 kW

Full load losses = $W_i + W_{Cu}$ = 3.265 kW (4.53)

Maximum η occurs at $\frac{3}{4}$ FL = $160 \times \frac{3}{4}$ = 120 kW.

At this load the copper loss is equal to iron loss.

$\therefore$

$$\left(\frac{3}{4}\right)^2 W_{Cu} = W_i$$

$$\frac{9}{16} W_{Cu} = W_i \quad (4.54)$$

Substituting (4.54) in (4.53), we get

$$\frac{9}{16} W_{Cu} + W_{Cu} = 3.265 \text{ kW}$$

or W_{Cu} = 2.896 kW W_i = 1.1754 kW

Example 4.28 A 25 kVA, single phase transformer, 2200/220 V, has a primary resistance of 1.0 and secondary resistance of 0.01 . Find the equivalent resistance referred to secondary. If the iron loss is 80% of the full load copper loss, find FL efficiency at 0.8 pf.

Solution:

$$K = \frac{220}{2200} = \frac{1}{10}$$

$$R_2$$

1
=
0.01
+
1
×
1
2
R02 = R2 + K_{10} = 0.02

Full load current = $\frac{25\times 1000}{220}$ = 113.63 A

Full load Cu loss = $(113.63)^2 \times 0.02$ = 258.23 W

Iron loss = 80% of 258.23 W = 206.59 W

$\eta = \frac{25\times 0.8}{25\times 0.8+ 0.258+ 0.206} \times 100 = 97.7\%$

Example 4.29 A 10 kVA; 500/250 V, single-phase transformer gave following test result:

SC test (HV side): 60 V, 20 A, 150 W.

The maximum efficiency occurs at 1.2 times full load, u.p.f. Find the maximum efficiency.

Solution:

FL current

=

$\frac{10\times 10^3}{500}$ = 20 A

Since SC test has been conducted at rated current, the full load Cu loss = 150 W. The maximum efficiency occurs at 1.2 times full load. The Cu loss at that load= $(1.2)^2\times 150$ = 216 W. since, at maximum efficiency iron loss= copper loss, iron loss = 216 W.

3

Maximum efficiency

=

$\frac{1.2\times 10\times 10}{1.2\times 10\times 10^3 + 216+ 216} \times 100 = 96.53\%.$

Example 4.30 A 20 kVA, 440/220 V, 1− φ, 50 Hz, transformer has an iron loss of

324 W. The copper loss is 100 W at half full load. Determine efficiency at full load

0.8 pf and load at which maximum efficiency occurs.

Solution:
Cu loss at 12 FL = 100 W
Cu loss at FL = 100× 4 = 400 W.∵.
Iron loss = 324. 20× 0.8 %η at 0.8 p.f., FL $= \frac{20\times 0.8}{20\times 0.8+ 0.4+ 0.324} \times 100 = 95.67\%$
Load at which maximum
η
occurs
=
20

×
324
400
= 18 kVA.

4.9. All-Day Efficiency

The commercial efficiency of a transformer is defined as
$\eta = \frac{\text{Output in Watts}}{\text{Input in Watts}}$.

In distribution transformers, typically, the load keeps changing during the day and the transformer also runs at no-load for a couple of hours in a day. This means that the core loss occurs throughout the day, as the transformer is energised and copper loss keeps varying. In such cases, the efficiency is defined in terms of energy consumed.

$\eta_{all-day} = \frac{\text{Output in kWh}_{(\text{for 24 hours})}}{\text{Input in kWh}}$
This is always less than the commercial efficiency. The load pattern during the day has to be known to evaluate the all-day efficiency.

Example 4.31 Find the all-day efficiency of 500 kVA distribution transformer whose copper loss and iron loss at full load are 4.5 kW and 3.5 kW respectively. During 24 hours, the load pattern is as follows:

No. of hours Load (kW) pf

6 400 0.8
10 300 0.75
4 100 0.8
40

• 400 kW at 0.8 pf = ^{400} = 500 $kVA_{0.8}$
Cu loss at 500 kVA = 4.5 kW.
^{300} = 400 kVA• 300 kW at 0.75 pf =$_{0.75}$
Cu loss at 400 KVA
=
4.5
×
400^2 $_{500}$ = 2.88 kW
• 100 kW at 0.8 pf = 125 kVA
Cu loss at 125 kVA
=
4.5

×
125^2
$_{500}$ = 0.28 kW

Total Cu loss in 24 hrs = (4.5× 6) + (2.88× 10) + (0.281× 4) = 56.924 kWh
Total iron loss in 24 hrs = 3.5× 24 = 84 kWh.
Total output = (400× 6) + (300× 10) + (100× 4) = 5800 kWh

$5800^{\%\eta all-day =}$ 5800+ 56.924+ 84 $^{\times 100 = 97.6\%.}$ **Example 4.32** A 100 kVA transformer has a full load Cu loss of 1.6 kW and iron loss of 1.5 kW. It operates on FL for 3 hours, half load for 4 hours and negligible load rest of the time, during the day. Calculate all-day efficiency.

Solution:
Cu loss during day
=
1.6
×
3
+
1.6×1^2 $_2 \times 4$ = 6.4 kWh
Compare this with FL Iron loss = 1.5× 24 = 36 kWh
Output = 100× 3+ 50× 4 = 500 kWh
$^{\% 500 = 92.2\%\eta all-day =}$ 500+ 36+ 6.4

$100 = 97\%\ \eta = 100 + 1.6 + 1.5$

4.10. Auto transformer

It is a single winding transformer, with a part of the winding common to both primary and secondary. The two windings are not electrically isolated from each other. The principle of operation is same as a two winding transformer. It is generally used, when transformation ratio is close to unity. The auto transformer is shown in Fig. 4.27.

As compared to a two winding transformer, an auto transformer has higher efficiency and smaller size for the same output.

The volume and hence weight of copper required depends on length and area of cross section. The length of the winding depends on the number of turns and the area of cross section on the current. In Fig. 4.27(a) Wt of Cu in section AC is$\alpha(N_1 - N_2)I_1$,WtofCu in section BC is $\propto N_2(I_2 - I_1)$

$\therefore$ Total weight of Cu $\propto$ (N1 − N2)I1 + N2(I2 − I1)
If it were a two winding transformer, Wt of Cu required$\propto N_1I_1 + N_2I_2$
$\therefore$ Wt of Cu in auto-transformer = (N1 − N2)I1 + N2(I2 − I1) = $1 - K$Wt of Cu in 2 winding transformer $N_1I_1 + N_2I_2$
I_1 AI_2
V_1 C I2 I1 Load V_2
Load V2 V1
$(I_2 - I_1)$ $(I_1 - I_2)$
I_1 B I_2 I_1 I_2 (a) (b)
Figure 4.27 *Auto transformer*.
(using $\frac{I2}{I1} = \frac{1}{K}$ and $\frac{N2}{N1} = K$)
$\therefore$ Wt of Cu in auto transformer = (1− K) Wt in 2 winding transformer Saving = kWt(2 winding Transformer) The power transformed inductively is (1− K) input and power transferred conductively is K × input. Auto transformers are used when

- K is close to unity.
- As starters for induction motors to give a voltage of 50–60% rated voltage.
- As furnace transformer.
- Control transformer.

Questions

1 Briefly explain the working principle of a transformer.
2 Give the constructional details of the two main types of transformers.
3 Derive the e.m.f. equation of a transformer.
4 Prove that the flux through the core is almost a constant.
5 Draw and explain the phasor diagram of a transformer, on no-load.

6 Derive an expression for the equivalent resistance and reactance of the transformer referred to a single winding.

7 What are core losses? How can they be minimized?
8 Explain the OC and SC tests conducted on the transformer.

9 Clearly explain how the equivalent circuit of the transformer can be obtained from the test results.
10 Why does the transformer draw current even under no-load?

11 Justify the fact that the open circuit test wattmeter reading gives the iron loss in the transformer and the SC test the full load copper loss.
12 What is regulation? Derive an expression for regulation of the transformer.

13 Derive the condition for maximum efficiency of the transformer.
14 At what load does the maximum efficiency occur.
15 How would you separate the core loss components?
16 What is all-day efficiency? How is it different from commercial efficiency?
17 Explain the phasor diagram of a transformer, when supplying a lagging load.
18 Show that the regulation can be zero under leading pf loads.
20 What is a step-up transformer?

21 The primary of a transformer takes1Aatapfof 0.4, when connected to 200 V, 50 Hz, with secondary open.$N_1 = 2N_2$. If a load of 50 A at 0.8 pf lag is connected, find the primary current drawn.
Ans 25.9 A.

22 A 400/200 V, 1 − φ transformer is supplying a load of 25 A at a pf of 0.866 lag on no load the current and power factor are 2 A and 0.208 respectively. Find primary current on load.
Ans 13.9 a, lagging V_1 by 36.1°.

23 A 30 kVA, 2400/120 V, 50 Hz transformer has a high voltage winding resistance of 0.1 and a leakage reactance of 0.22 . The LV winding resistance and reactance are 0.035 and 0.012 respectively. Find the equivalent resistance and reactance referred to (i) HV side (ii) LV side.
Ans (i) 14.1 ; 5.02 ; (ii) 0.03525 ; 0.01255 .

24 A transformer is connected to 2200 V, 40 Hz supply. The core loss is 800 W, with 600 W due to hysteresis and 200 W due to eddy current losses. What is the core loss if supply voltage is 3300 V, 60 Hz.
Ans 1350 W.
25 The following is the test results on a 25 kVA, 2200/220 V, 50 Hz transformer:

OC test (LV side): 220 V, 4 A, 150 W
SC test (HV side): 90 V, 10 A, 350 W.

Determine the regulation at 0.8 p.f. lagging current and % η at full load and half load at 0.8 p.f. lag.
Ans R = 4.045%; η_{FL} = 97.07%; η_{1FL} = 97.44%.$_2$

26 A 10 kVA, 400/200 V single phase 50 Hz transformer, has a maximum efficiency of 92% at 80% FL, u.p.f. Determine the efficiency at full load 0.8 p.f. lag. **Ans** 90%.

27 In a 50 kVA, 11 KV/400 V transformer, the iron and copper losses are 500 W and 600 W respectively under full load. Calculate the %η at FL u.p.f., and the iron and copper losses at the load where maximum efficiency occurs.
Ans %η = 97.85%; $W_i = W_{Cu}$ = 500 W.

28 A 50 kVA, 2200/110 V transformer when tested gave following results: OC test (LV side): 400 W, 10 A, 110 V
SC test (HV side): 808 W, 20.5 A, 90 V.
Compute all parameters of equivalent circuit referred to HV side.
Ans R_0 = 12.1 K; X_0 = 4.72 K; R_{01} = 1.923 ; X_{01} = 4.39 .

CHAPTER - 5

Three phase induction motors



Learning objectives

AC motors
General principle of induction motor Construction: Squirrel cage and slip ring
Rotating magnetic field
Torque expression
Torque - slip characteristics
Losses in induction motor
Calculation of efficiency
Methods of starting

5.1. Introduction

AC motors are becoming widespread in industrial drives in the recent past. Broadly a.c. motors can be classified into synchronous motors and asynchronous motors. Synchronous motors run at synchronous speed given by $N_S = \frac{120f}{P}$, where f is the frequency of the supply voltage and P is the number of poles. Asynchronous motors run at speeds lower than the synchronous speed. They are basically induction motors and commutator motors. Various constructions are possible in these motors.

Of all the motors, the three phase induction motor is extensively used. It offers many advantages such as,

- Simple and rugged construction
- Low cost
- Reliable
- Reasonably good power factor and efficiency
- Does not need additional starting motor
- Minimum maintenance.

The main disadvantages are

- Speed cannot be varied widely
- Speed decreases with increase in load
- Starting torque is lower than d.c. shunt motor.

In the following sections we study the three phase induction motor in detail.

5.2. Construction

The three phase induction motor essentially consists of (a) rotor (b) stator.

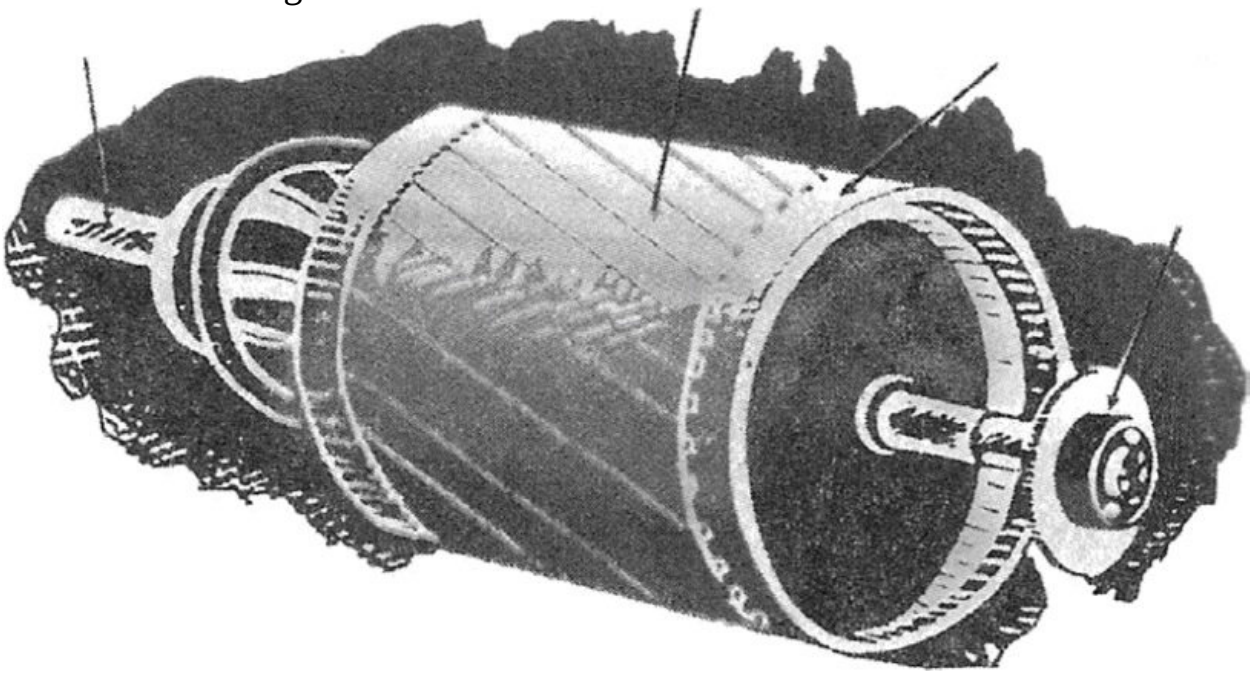
The stator is made up of a number of stampings, slotted to house the windings. The stator has a three phase winding fed from a three phase supply. It is wound for a definite number of poles. The rotors are of two types: squirrel cage and wound rotor.

5.2.1. Squirrel - Cage rotor

This rotor has the simplest construction and is the most rugged. The construction is shown in Fig. 5.1.

Rotor Skewed shaft

rotor slots End-ring



Ball bearings

Figure 5.1 *Squirrel cage rotor.*

The rotor consists of a cylindrical laminated core with parallel slots to carry rotor conductors. The rotor slots are given a slight skew. The skew has the following effects.

• It helps the motor run quietly

• Helps in reducing the locking tendency of the rotor *i.e.* the tendency of the rotor teeth to remain under the stator teeth due to direct attraction between the two.

• Increases rotor resistance

• Increases slip for a given torque

• Increases impedance at a given slip.

The rotor conductors are heavy bars of copper, aluminium or alloys. One bar is placed in each slot. The bars are brazed/welded/bolted to two short-circuiting end-rings. The rotor bars are permanently short circuited on themselves. Hence, no external resistance can be added to them.

5.2.2. Wound rotor or phase-wound rotor

The rotor is also called a slip-ring rotor. The rotor consists of 3-phase double layer distributed winding. It is wound for same number of poles as the stator. The three phase windings are connected internally in a star. The other three

The three phase windings are connected internally in a star. The other three terminals are brought out and connected to three insulated slip-rings mounted on the shaft with brushes resting on them. These brushes are externally connected to a 3-phase star-connected rheostat. Thus additional resistance can be connected in the rotor as shown in Fig. 5.2.

Stator
Slip rings
Rotor
Starting resistance
Figure 5.2 *Slip ring induction motor.*
The three phase stator winding, supplied with three phase voltage, produces a rotating magnetic field.

5.3. Production of rotating magnetic field

Consider three-phase windings displaced in space by 120° and supplied by three-phase voltages displaced in time by 120°. These voltages produce currents in the stator windings which produce time varying flux. The three fluxes are described as follows

$1 = {}_m \sin wt$

$2 = {}_m \sin(wt - 120^\circ)$

$3 = {}_m \sin(wt + 120^\circ)$ (5.1)

[Remember $_1 - {}_2 - {}_3$ are produced in three windings displaced in space by 120°].
We now consider different points to evaluate the total flux.
2
120°
1
120° φ_2 φ_3 60° 3
(a) (b)$_{\varphi r}$
Figure 5.3 *(a) Axis of three phase windings. (b) Fluxes when* ωt = 0°.
(i) wt = 0°. Substituting in (5.1)
√
3
$_m$; $3 = {}_{2\,m}.1 = 0$; 2=−
√$_3$
2

The three vectors are plotted as in Fig. 5.3.

In Fig. 5.3(a) the positive axis of the three fluxes is shown. In Fig. 5.3(b) we plot the fluxes. If a flux is negative, it is plotted in a direction opposite to the positive axis. The total flux$_r$ is given by

$\sqrt{3}$ cos 60° 3 .r = 2× 2 m 2 = 2 m

(ii) ωt = 60° substituting in (5.1)

√

3

m; 2=−2 m; 3 = 0.1 =

√3

2

These fluxes are plotted in Fig. 5.4.

$\sqrt{3}$ cos 60° 3 .r = 2× 2 m 2 = 2 m

Comparing with the case whenωt = 0°, the resultant flux is the same. But has rotated by 60°.

(iii) ωt = 120°

√

3

m; 2 = 0; 3=−2 m.1 =

√3

2

φ_1

φ_r φ_2

Figure 5.4 *Fluxes when*wt = 60°.

φ_r φ_2

φ_1

Figure 5.5 *Fluxes when*wt = 120◦.

The fluxes are plotted in Fig. 5.5.

$\sqrt{3}$ cos 60◦ 3.r = 2× 2 m 2 = 2 m

(iv)

wt

=

180

◦ √ 3

$1 = 0$; $2 =$
$\sqrt{3}_{m}$; $3 = -_{2\,m2}$
$\sqrt{3}\cos 60^{\circ}\ 3_{r} = 2\times_{2\,m\,2} =_{2\,m}$

It can be observed from Fig. 5.3 to 5.6 that the resultant flux remains constant and is equal to $^{3}{}_{m}$. It rotates in space in the clockwise direction. Thus, the three-phase stator windings$_{2}$
supplied with three-phase balanced voltage produces a flux with the following properties:

• The resultant flux is constant and 1.5 times the maximum value of the flux due to any phase
$\sqrt{}\ 3_{\varphi m}\ ^{\sqrt{}\,3}\ \varphi_{m}\ 2^{2}$
Figure 5.6 *Fluxes when* wt = 180°.
• The resultant flux rotates around the stator in the air gap, at synchronous speed given by $NS = {}^{120f}{}_{.P}$
If the phase sequence is reversed, the direction of rotation of the flux is reversed.

5.4. Principle of operation

When the stator windings are fed by the three phase supply, a rotating magnetic flux is set up. The flux passes through the air-gap, and cuts the rotor conductors which are still stationary. An e.m.f is induced by induction in the rotor conductors, because of the relative speed between the rotating flux and the rotor conductors. The frequency of the induced e.m.f is same as the supply frequency and its magnitude is proportional to the relative velocity between the flux and the conductors. Since the rotor bars are short circuited in cage motors and are closed in slip-ring motors, rotor current is produced. The direction of the current is such as to oppose the cause, according to Lenz's law. The cause here is the relative speed between the flux and the conductors. The rotor starts rotating in the same direction as the rotating flux, thus reducing the relative speed between them.

The rotor never catches up with the rotating flux. If it did so, the relative speed which is the cause for the rotation would become zero. Hence, the induced e.m.f and current in the rotor would become zero momentarily. The flux continues to rotate, giving a relative speed again. Thus, the rotor always runs at a speed less than the flux which is rotating at the synchronous speed.

The difference between the synchronous speedN_s and the actual speedN is called the slip. It can be measured in r.p.m. More often it is expressed as % slip defined as

$$\% \text{ slip } s = \frac{Ns - N}{Ns} \times 100 \quad (5.2)$$

Note that the rotating flux (also called revolving field) is rotating at synchronous speed with respect to stator and at slip speed ($N_S - N$) with respect to rotor.

5.5. Frequency of rotor current

The frequency depends on the relative speed between the flux and the rotor. When the rotor is stationary the frequency of the rotor current is same as the supply frequency. When the rotor starts rotating, the frequency f of rotor currents is given by

$$f = sf$$

Example 5.1 Obtain the synchronous speed for different number of poles.

Solution: The synchronous speed is given by

$$Ns = \frac{120f}{P}.$$

f = 50 Hz in India. So we can calculate the synchronous speed for different number of poles.

No. of poles	N_S rpm
2	3000
4	1500
8	750
12	500
16	375
20	300

Remember that the rotor always runs close to the synchronous speed, but always less than that.

Example 5.2 A slip ring induction motor runs at 285 r.p.m when connected to a 50 Hz supply. How many poles does it have? What is the slip?

Solution: Since the rotor speed is close to synchronous speed, we pick the speed N_s, close to it (refer Example 5.1). N_s = 300 r.p.m. corresponding to P = 20.

$$\%s = \frac{300 - 285}{300} \times 100 = 5\%$$

Example 5.3

Example 5.3 induction motor has 2 slots/pole/phase. If supply is 50 Hz calculate (i) number of poles (ii) total stator slots (iii) speed of the rotating flux.

Solution:
(i) Number of poles = 2× slots/pole/phase = 2× 2 = 4
(ii) Total number of slots = slots/pole/phase× number of poles× number of phases = 2× 4× 3 = 24
(iii) Speed of rotating flux = $N_s = \frac{120f}{P} = \frac{120\times 50}{4} = 1500$ r.p.m.

Example 5.4 A 4-pole, A 4-pole, induction motor operates from a 50 Hz supply. Calculate

(i) Speed of the rotating flux
(ii) Speed of rotor when slip is 0.05
(iii) Rotor current frequency at a slip of 0.05
(iv) Frequency of rotor current at standstill (rest)

Solution:
(i) Field revolves at synchronous speed,$N_s = \frac{120f}{P} = \frac{120\times 50}{4} = 1500$ r.p.m
(ii) $Ns - N = 0.05Ns$
$\frac{1500 - N}{1500} = 0.05$; N = 1425 r.p.m.
(iii) f = sf = 0.05× 50 = 2.5 Hz
(iv) At standstill N = 0 and s = 1. f = 50 Hz.

Example 5.5Example 5.5 induction motor runs at 1740 r.p.m. What is the frequency of rotor currents if supply frequency is 60 Hz?
Solution: $N_s = \frac{120\times 60}{4} = 1800$ r.p.m.
$\% s = \frac{1800 - 1740}{1800} \times 100 = 3.33\%$
f = sf = 0.033× 60 =2Hz

Example 5.6 A 12 pole alternator driven at 500 r.p.m supplies power to an 8-pole A 12 pole alternator driven at 500 r.p.m supplies power to an 8-pole . IM whose slip is 2.5%. Calculate the speed of induction motor.
Solution:

Alternator
Induction motor

12 pole 8 pole, 500 r.p.m. 2.5% slip N = ?
The frequency of the voltage generated by the alternator is got from,

$500 = \frac{120 \times f}{12} \Rightarrow f = 50$ Hz.

This is the supply frequency to the induction motor. The synchronus speed of the induction motor

$N_s = \frac{120 \times 50}{8} = 750$ r.p.m.

$\frac{0.025}{} = \frac{}{750 - N}$

$\frac{}{750}$

N = 731.25 r.p.m.

5.6. Torque developed

The torque developed by the rotor of the induction motor is given by

T ∝ I2 cos 2 = KI2 cos 2. (5.3)

where I_2 = rotor current at standstill, $\cos_2$ = rotor power factor and K = a constant. If the rotor induced e.m.f. at standstill isE_2, then

∝ E2 and

T = K E2I2 cos 2 (5.4)

As$_2$, the angle between the rotor induced e.m.f and rotor current increases, $\cos_2$ decreases and hence torque is reduced.

5.6.1. Starting torque

The torque developed by the rotor at starting is called the *starting torque*. Let

rotor induced e.m.f/phase at standstill E_2 =

I_2 = rotor current/phase at standstill

X_2 = rotor reactance/phase at standstill

(Note that since rotor frequency f = sf, is a function of slip the reactance is not a constant)

Z_2 = rotor impedance/phase at standstill

$\frac{}{R}$

$\frac{}{2} + X^2$

$\frac{2}{E2} = \frac{2}{E2}$;I2 = $\frac{}{Z2}$ R2 + X2

$\frac{2}{R2} = R2\cos 2 = Z2$ $R^2 + X^2$

2 2

Standstill torque or starting torque

Tst = K1E2I2 cos 2
E2× R2= K1E^{2}R2(5.5)= K1E2 2
2
+
X
2
R
2
+
X
2
R
^{2} + X^2
R2 2 2 2$^{2\ 2}$
3
3 E
22
R
2 2
The constant $K_1 = 2\pi n_s$, n_s is synchronous speed in r.p.s..∴ $T_{st} = \frac{2\pi n_s}{R2 + X2}$.
2 2
If supply voltage is a constant, the flux and hence E_2 is a constant. Therefore,
$R2\ T_{st} = K2\frac{}{R2 + X2}$ (5.6)
2 2
Thus, the starting torque depends on the rotor resistance and standstill reactance.

The resistance of a cage motor is small compared to its reactance. Hence, the power factor is low. Therefore the starting torque of a cage motor is small. Hence, such motors cannot be used where the motor has to start against a heavy load.

Slip-ring motors on the other hand have a facility to insert an external resistance. Hence the power factor can be improved by adding external resistance as shown in Fig. 5.2. The external resistance is gradually cut out as the motor picks up speed.

5.6.2. Condition for maximum starting torque

We differentiate (5.6) to obtain the condition for maximum starting torque as
follows:

follows:

$$\frac{dT}{dR_2} = K_2 \frac{R_{22}^2 + X^2 - R_2(2R_2)}{(R_2 + X_2)^2} = 0$$

Or

$$R_2^2 + X^2 = 2R_2^2$$

Or $R_2 = X_2$ (5.7)

Thus the starting torque is maximum when the rotor resistance is equal to the standstill rotor reactance.

5.6.3. Torque under running conditions

When the rotor starts running, the relative speed decreases. Therefore, the induced e.m.f. in the rotor also decreases. Let

E_r = rotor e.m.f./phase under running conditions.

I_r = rotor current / phase under running conditions.

$E_r = sE2$ (5.8) $X_r = sX2$; (5.9)

Due to decrease in the rotor current frequency by a factor s, the reactance is decreased. Similarly, since the induced e.m.f. in the rotor is proportional to the relative speed, it is also decreased by a factor of s.

$T_r \propto E_r I_r \cos_{2r}$ (5.10)

Now $I_r = \frac{Er}{Z} = \frac{sE2}{\sqrt{R_2^2 + (sX2)^2}}$ (5.11)

$\cos_{2r} = \frac{R2}{X} = \frac{R2}{\sqrt{R_2^2 + (sX2)^2}}$ (5.12)

Substituting (5.11) and (5.12) in (5.10) we get

$$Tr = K_1 sE_2^2 R2 \ (5.13) \frac{}{R^2 + (sX2)^2}_2$$

The constant

$$K_1 = \frac{}{2\pi n}$$

^{3} where
$_{s\ s}$ is synchronous speed in r.p.s. n
5.6.4. Condition for maximum torque under running conditions From (5.13) we set
A
=
1
=
KR
^{2} + (sX2)^{2}
2
T SE22R2
R
^{2} sX^{2}
=
K
2 2
sE22 $^{+}$ E22R2
$^{2\ X2dA}$ = K$^{-R2\ +2}$
E22R2s^{2}E22$^{=\ 0}$ds
R
2
=
X
2
^{2} ⇒ R^{2} = s^{2}X^{2} or R2 = sX2.$_{s2E22\ E22R2\ 2\ 2}$
Hence, under running conditions the torque is maximum at the slip where
R2 = sX2 (5.14)

Since, sX_2 is the rotor reactance per phase under running condition, the maximum torque is obtained at that value of slip which makes the rotor reactance per phase equal to the rotor resistance per phase.

Substituting $R_2 = sX_2$ in (5.13) we get the maximum torque as
$^{K\ R2E22R2}$ KE^{22} $_{(5.15)Tmax\ =}$ ^{X2} $_{=\ 2R2\ +\ R2\ \times\ X}$
^{2} 2X
^{22} X2

From (5.15) we can see that
• The maximum torque is independent of the rotor resistance.

• The speed at which maximum torque occurs is determined by the rotor resistance [i.e. $R_2 = sX_2$]. Therefore by varying the rotor resistance it is possible to vary the speed at which maximum torque occurs. This is possible only in slip ring motors.

• The maximum torque varies as square of the applied voltage.
• The maximum torque is inversely proportional to the rotor standstill reactance. Hence, it must be kept as low as possible.
• To obtain maximum torque at starting, the rotor resistance must be equal to rotor standstill reactance.

5.6.5. Break down Torque

The maximum torque is also called the *breakdown torque*,T_b. The slip at which it occurs is called the breakdown or pull-out slip s_b. The torque at any slip s is given by

$$T = T_b \frac{2}{(s_b/s) + (s/s_b)} \quad (5.16)$$

5.6.6. Full load torque

The full load torque is given by

$$T_f \propto \frac{sfR2}{R^2 + (sfX2)^2} \quad (5.17)$$

where s_f is the slip at full load torque.

$$T_{max} \propto \frac{1}{2X2} \text{ (from 5.15)} \quad (5.18)$$

Dividing (5.17) by (5.18)

$$\frac{T_f}{T_{max}} = \frac{2sfa}{a2 + s2} \quad (5.19)$$

where $a = \frac{R2}{X2}$ To generalize, the operating torque at any slips is given by,

$$\text{Torque at slip } s = \frac{2as}{a^2 + s^2} \quad (5.20) \text{Maximum torque}$$

In particular at starting s = 1.

$\therefore$

$$T_{st} = \frac{2a}{1 + a^2} \quad (5.21) T_{max}$$

Example 5.7

Example 5.7 star connected rotor has an induced e.m.f. of 75 volts at standstill. The rotor resistance = 1 and reactance = 3.5 . Calculate the current/phase and power factor, when (i) rotor is short-circuited (ii) the slip rings are connected to an external resistance of 2.8 per phase.

Solution:

E

2

=

^{75} = 43.3 V Since its star connected phase voltage = $^{\text{line voltage}}$ $\sqrt{}_3$ $\sqrt{}_3$ (i) R2 $^{= 1}$;

X2 = 3.5

Z2 = $\sqrt{}1^2 + 3.5^2 = 3.64$

Rotor current/phase = $^{43.3}$ = 11.89 $A_{3.64}$

P .f = ^{R2} =^{1} = $0.274_{Z2\ 3.64}$

(ii) R_2 = 1+ 2.8 = 3.8

X

2

= 3.5

= $\sqrt{}3.8^2 + 3.5^2 = 5.166$ .Z_2

Rotor current/phase =$^{43.3}$ = 8.38 $A_{5.166}$

p.f =$^{3.8}$ = $0.736._{5.166}$

Example 5.8 A 415 V, 50 Hz delta-connected induction motor has a star-connected slip-ring rotor. The phase transformation ratio is 3.5. The rotor resistance is 0.012 and standstill leakage reactance is 0.25 per phase. Determine

(i) Rotor current at start when slip-rings are shorted.
(ii) Rotor p.f when slip rings are shorted ands = 1.
(iii) Rotor current and p.f at 4% slip with slip-rings shorted.
(iv) The external resistance per phase required to obtain a starting current of 50A in

stator supply lines.

Solution: The stator is delta connected.

∴ phase voltage is equal to line voltage.

Transformation ratio is 3.5.
$\therefore$ rotor voltage = $\frac{415}{3.5}$ = 118.57 V.
(i) = $\sqrt{0.012^2 + 0.25^2}$ = 0.2503 Z_2
Rotor current at start = $\frac{118.57}{0.2503}$ = 473.71 A
(ii) p.f = $\frac{R2}{Z2}$ = $\frac{0.012}{0.2503}$ = 0.048 lag.

(iii) s = 0.04; $X_r = sX_2 = 0.04 \times 0.25 = 0.01$
$Z_r = \sqrt{0.012^2 + 0.01^2} = 0.0156$
$E_r = sE_2 = 0.04 \times 118.57 = 4.743$ V

$\frac{4.743}{0.0156}$ = 304.038I_2 =
p.f = $\frac{0.012}{0.0156}$ = 0.77 lag.
(iv) when supply current is 50A, rotor current = 50×3.5 = 175 A (like in a transformer $I_2 = \frac{I1}{K}$). E_2 = 118.57 V.

$\therefore$ Z2 = $\frac{118.57}{175}$ = 0.678 .
X_2 = 0.25 (since it is starting current we take standstill reactance). $R_2 = \sqrt{0.678^2 - 0.25^2} = 0.63$.
Extra resistance = 0.63− 0.012 = 0.618 .

Example 5.9 A three phase slip ring induction motor has a star connected rotor with an induced e.m.f. of 100 Volts between slip rings at standstill when normal voltage is applied to the stator. The rotor winding resistance per phase is 0.3 and standstill leakage reactance pe phase is 1.5 . Calculate (i) rotor current/phase when running at a slip of 5% (ii) the slip and rotor current/phase when maximum torque is developed.

Solution: E_2/phase = $\frac{100}{53}$ = 57.735 V
(i) $E_r = sE_2 = 0.05 \times 57.735 = 2.89$ V $X_r = sX_2 = 0.05 \times 1.5 = 0.075$ $Z_r = \sqrt{0.3^2 + 0.075^2} = 0.309$ Rotor current/phase = $\frac{2.89}{0.309}$ = 9.35A

(ii) when maximum torque is developed
R2 = sX2.

$\frac{R2}{X2}$ = $\frac{0.3}{1.5}$ = 0.2s =

$X_r = 0.2 \times 1.5 = 0.3$

$0.3^2 + 0.3^2 = 0.42 Z_r =$

$E_r = sE_2 = 0.2 \times 57.735 = 11.547$ V

$I_2 = \frac{11.547}{0.42} = 27.49$ A.

Example 5.10 Calculate the torque exerted by a 4-pole, 50 Hz, 3-phase induction motor operating at 3% slip, which develops a maximum torque of 150 kg-m at a speed of 1440 r.p.m.

Solution:

S_b (break sown slip) $= \frac{1500 - 1440}{1500} = 0.04$

From (5.16)

$\frac{T}{T_b} = \frac{2}{\frac{s}{s_b} + \frac{s_b}{s}} = \frac{2}{\frac{0.04}{0.03} + \frac{0.03}{0.04}} = 144$ N − m.

Example 5.11 A 3-phase 400/200 V, Y-Y connected induction motor has a rotor resistance of 0.06 and reactance of 0.3 standstill per phase. What additional resistance must be added to make starting torque equal to the maximum torque.

Solution:

$\frac{T_{st}}{T_{max}} = \frac{2a}{1 + a^2}$; $T_{st} = T_{max}$

$\frac{2a}{1 + a^2} = 1 \Rightarrow a = 1.$

$a = \frac{R_2}{R_2}$;

$\frac{R_2}{X_2} = 1$

$X_2 = 0.3$;

External resistance per phase = 0.3−0.06 = 0.24 (subtract the rotor resistance/phase).

Example 5.12 A 3-phase, 50 Hz, 8 pole induction motor has a full load slip of 2%. The rotor resistance per phase is 0.01 and reactance per phase at standstill is 0.05 . Find ratio of maximum to full-load torque and speed at which maximum

torque occurs.

Solution:

$N_s = \frac{120 \times 50}{8} = 750$ r.p.m.

$s_b = \frac{R2}{X2} = \frac{0.01}{0.05} = 0.2$. (Slip for maximum torque)

Corresponding speed = $(1 - 0.2) \times 750$

= 600 r.p.m.

$R2 = 0.2$; sf = 2% = 0.02a = $X2$

$T_f = 2s_f a$

$a2 + s2 = \frac{2 \times 0.02 \times 0.2}{0.22 + 0.022} = 0.198 Tmax$ f

$Tmax = 5.05 . Tf$

Example 5.13 A 12-pole, 3-phase, 600-V, 50 Hz, star-connected, induction motor has rotor-resistance and standstill reactance of 0.025 and 0.6 per phase respectively. Calculate (i) speed at which maximum torque occurs (ii) ratio of full-load torque to maximum torque, if the full load speed is 490 r.p.m.

Solution:

$N_s = \frac{120 \times 50}{12} = 500$ r.p.m.

$R_2 = 0.025$; $X_2 = 0.6$;

$R2 = \frac{0.025}{0.6} = 0.0416a = X2$

(i) Slip at which maximum torque torque occurs is when $R_2 = s_m X_2$ or

$s_m = \frac{R2}{X2} = 0.0416$.

Corresponding speed = $500(1 - 0.0416) = 479.2$ r.p.m (ii) Full load speed = 490 r.p.m.

$sf = \frac{500 - 490}{500} = 0.02$.

Full-load torque $= \frac{2}{a^2 + s^2} = \frac{2 \times 0.0416 \times 0.02as_f}{0.0416^2 + 0.02^2} = 0.781$.

Maximum torque$_f$

Example 5.14 A 1000 HP, 3 phase, 50 Hz, 16-pole induction motor has a rotor resistance of (0.02+ j0.15) at standstill. Full-load torque is obtained at 360 r.p.m. Calculate (i) the speed at which maximum torque occurs (ii) ratio of maximum torque to full-load torque (iii) rotor resistance to be added to get maximum starting torque.

Solution:

$N_s = \frac{120 \times 50}{16} = 375$ r.p.m.

$N_{FL} = 360$ r.p.m.

$375 - 360 = 0.04 sf = 375$

$R2 = \frac{0.02}{X2\ 0.15} = 0.1333a =$

(i) The slip at maximum torque = a = 0.1333

$N_m = N_s(1 - s) = 375(1 - 0.1333) = 325$ r.p.m.

(ii)

$Tf = 2asf$

$a2 + s2 = \frac{2 \times (0.1333) \times 0.04}{(0.1333)2 + (0.04)2} = 0.55 Tmax\ f$

or $\frac{Tmax}{Tf\ 0.55} = \frac{1}{0.55} = 1.82.$

The maximum torque is 1.82 times the full load torque.

(iii) For maximum starting torque $R_2 = X_2$.

Present resistance is 0.02 . Reactance is 0.15 . Extra resistance needed is 0.15− 0.02 = 0.13 .

Example 5.15 A 4 pole, 50 Hz, 3-phase induction motor has a rotor resistance and reactance of 0.02 and 0.1 per phase respectively. What is the additional rotor resistance per phase required to give three-fourth of the maximum torque at starting.

Solution:

$$Tst = \frac{2}{1 + \frac{a}{2}} = \frac{3a}{Tmax\ 4}$$

$3a^2 - 8a + 3 = 0 \Rightarrow a = 0.45$ and 2.214.
We reject 2.214 since the ratio is too large to be practical.
$\therefore a = 0.45 = \frac{R2 + r}{X2}$ [r is additional resistance]
$R_2 + r = 0.45 \times 0.1 = 0.045.$
Addition resistance = 0.045 − 0.02 = 0.025 per phase.

Example 5.16 An 8-pole, 50 Hz, 3-phase slip ring induction motor has effective rotor resistance of 0.075 /phase. Stalling speed is 648 r.p.m. How much resistance must be inserted to obtain maximum torque at starting.

Solution:
$N_s = \frac{120 \times 50}{8} = 750$ r.p.m.
Starting speed corresponds to the maximum torque.
$sm = \frac{750 - 648}{750} = 0.136$
Now under maximum torque conditions $m = \frac{R2}{X2}$
$0.136 = \frac{0.075}{X_2} \Rightarrow X_2 = 0.551$
To obtain maximum torque at starting, the resistance/phase= reactance/phase.$\therefore$additional rotor resistance needed = 0.551 − 0.075 = 0.476 .

Example 5.17 A three phase induction rotor having a 6-pole star-connected stator winding, runs on 240 V, 50 Hz supply. The rotor resistance and standstill reactance are 0.1 and 0.8 /phase respectively. The ratio of stator turns to rotor turns is 2.0. Full load slip is 4%. Calculate the full load torque, maximum torque and speed at maximum torque.

Solution: The turns ratio gives the induced e.m.f in rotor [similar to transformer]
Turns ratio
=
rotor turns/phase $= \frac{1}{1.8}$
stator turns/phase
$\therefore E2 = E1 \times \frac{1}{1.8}$
$E2 = \frac{240}{\sqrt{3}} = 77$ V; $s = 0.04E1 = \frac{240}{\sqrt{3}}$
$N_s = \frac{120 \times 50}{6} = 1000$ r.p.m.
$1000 = \frac{50}{3}$ r.p.s.
$ns = 60$
$sE22R2Tf = \frac{3}{2\pi ns}$
$= 0.04 \times 772 \times 0.13 \times \frac{(0.1)^2 + (0.04 \times 0.8)^2}{2\pi(50/3)}$

= 61.63 N-m
For maximum torque the slip
sm(or sb) = ^{R2} = $^{0.1}$ = $0.125_{X2\ 0.8}$
$^{Tmax = 0.125\times\ 772\ \times\ 0.13\ \times}$ $(0.1)^2 + (0.125\times\ 0.8)^2 2\pi(50/3)$
= 106.158 N-m.
We can also find T_{max} as
3
E
22
3
77
2
T_{max} = $_{2\pi ns\ 2X2}$ = $_{2\pi(50/3)\ 2\times\ 0.8}$ = 106.158 N-m.
Speed corresponding to maximum torque is
N_m = 1000(1− 0.125) = 875 r.p.m.

5.7. Torque-slip characteristics (or Torque-Speed Characteristics)

We have seen $_{KsE2R2T = R2 + (sX2)22}$
when s =0(N = N_s), the torque is zero.
At speeds close to synchronous speed,s is small. Hence,sX_2 is small compared toR_2 and can be neglected.
∴ T ∝ ^{s} or T ∝ s(if R2 is a constant)$_{R2}$

So for low slips, the torque-slip curve is a straight line. As the slip increases (with decreasing speed) the torque also increases and reaches a maximum at a slip where R_2 = sX_2. This torque is the pull-out torque, stalling torque or breakdown torque. When slip further increases, then R_2 becomes negligible compared tosX_2. Hence for large slips,

T ∝^{s} ∝ ^{1}.$_{(sX2)2\ s}$
6R
Torque 3R N − M_R
1.0 0.5 0 Slip
0 50% 100% Speed as % of N_s **Figure 5.7** *Torque-slip characteristics.*

This gives a rectangular hyperbola. Thus, if the load torque increases beyond the

maximum torque of the induction motor, the motor stalls. Hence the name *stalling torque*. The torque-slip characteristics is shown in Fig. 5.7.

It can be seen that the value of the maximum torque does not depend onR_2. However, the slip at which the maximum torque occurs depends on the value ofR_2.

Since T ∝ sE22 andE_2 ∝ V (the supply voltage) the torque at any speed is proportional to the square of the applied voltage. The change in the supply voltage affects the starting torque and the running torque. When the voltage falls the torque also falls. To maintain the same torque, the speed decreases and slip increases.

If V changes to V , slip s to s and torque T to T then
$T_=$ sV ^{2} ^{2}T s V

5.8. Power stages in an induction motor The various power stages are shown in Fig. 5.8. We can also depict it as shown in Fig. 5.9.
Motor input in stator Stator Cu
Rotor loss and $_{\text{inputcore loss}}$

Rotor Mechanical copper power in rotor loss gross torpue T_g Friction and windage

loss
Rotor output
Figure 5.8 *Power stages in Induction motor.*
Motor input (P1)
Stator copper and iron losses Rotor input (P2)
Rotor Cu loss Mechanical power

developed (Pm) or
Gross torque (T_g)

Friction and windage losses Rotor output (P_{out})

Figure 5.9 *Alternative representation of power stages.*

The iron losses in the stator (or core losses) are practically constant. They constitute the eddy current and hysteresis losses, which depend on the supply

frequency and the flux density. The rotor iron loss is negligible because the rotor current frequency is very small (sf) under running conditions.

The gross torque is due to the mechanical power,P_m developed in rotor. It is given by

P_m T_g $=_{2\pi N/60}$ (in terms of P_m) (5.22) or

T

g

=

P2

$_{2\pi Ns/60}$ (in terms of rotor input) (5.23)

(N—actual speed; N_s—synchronous speed). The shaft torque is given by

Pout

T_{sh} $=_{2\pi N/60}$ (5.24)

The difference between T_g and T_{sh} is due to the friction and windage losses.

5.8.1. Relationship between various power stages

Stator input P_1 = Stator output+ stator losses

The rotor input P_2 = stator output.

Rotor gross output or the mechanical power developedP_m = P_2−rotor copper losses. This power gives rise to the gross torque.

P_m = $^{2\pi NTg}$ Watts.$_{60}$

If copper losses in rotor were absent then the mechanical power developed is equal to the rotor input and rotor would run at synchronous speed.

∴ P2 = $^{2\pi NsTg}$ Watts.$_{60}$

P_2 is nothing but power across the air gap.

Rotor copper loss

=

P

m

−

P

2

=

T

g

×
$2\pi(N_s - N)$
60

Rotor copper loss$_{= N_s - N_{= \text{slip}}}$Rotor input Ns
∴ Rotor copper loss = s × rotor input
Rotor gross output = Rotor input− rotor copper loss (5.25) = P2 − P2s = P2(1− s)
∴ P_m = P2(1− s) (5.26) Rotor gross output,$P_{m = 1- s = N_{(5.27)}}$Rotor input,P2 Ns
Pm is nothing but efficiency.$_{P2}$
∴ rotor efficiency = ^{N} (5.28)$_{Ns}$

Thus if a power P_2 is delivered to the rotor across the air gap,sP_2 is lost as copper loss and (1− s)P_2 appears as gross output. Therefore, it is necessary to maintain the slip as low as possible. The useful torque is the gross torque minus the friction and windage losses.

Example 5.18 The power input to the rotor of a 440 V, 50 Hz, 6 pole, 3-phase induction motor is 100 KW. The rotor e.m.f alternates at 100 cycles per minute. Calculate the slip, the rotor speed and the rotor copper loss per phase.

Solution:
Rotor frequencyf = 100cycles per minute
^{100} 1.667 Hz.=$_{60 =}$
f = sf
1.667 = s × 60 ⇒ s = 0.033 or3.3% Rotor speed N = (1− s)N_s = (1− 0.033) × 1000 = 966.67 r.p.m.
$^{s \times \text{rotor input}}$Rotor copper loss/phase =$_3$
0.033× 100$_{= 1.1 \text{ kW}.= 3}$

Example 5.19

Example 5.19 , 50 Hz, 4-pole, Y connected induction motor has a full load speed of 1430 r.p.m. The rotor impedence is 0.4 + j4.0 per phase. The rotor/stator turn ratio is 0.8. Calculate (i) Full load torque (ii) rotor current (iii) full load rotor copper loss (iv) power output if friction and windage loss is 450

W (v) maximum torque and speed at which it occurs (vi) starting current (vii) starting torque

Solution:

$N_s = \frac{120\times 50}{4} = 1500$ r.p.m

$s = \frac{1500 - 1430}{1500} = 0.0467$

$E_1 = \frac{415}{\sqrt{3}} = 239.6$ V; $E_2 = 0.8\times 239.6 = 191.68$ V

(i) $T_f = \frac{3 sE_2^2 R_2}{2\pi n_s [R_2^2 + (sX_2)^2]}$

$= \frac{3\times 0.0467\times 191.68^2 \times 0.4}{2\pi \times 25 [0.42 + (0.0467\times 4)^2]}$

= 67.25 N-m.

(ii) $I_r = \frac{sE_2}{\sqrt{[R_2^2 + (sX_2)^2]}} = \frac{0.0467\times 191.68}{\sqrt{0.42 + 0.1868^2}}$

= 20.27 A.

(iii) Total rotor copper loss = $3I^2R_{2r}$

= $3\times 20.27^2 \times 0.4 = 493$ W (iv)

$P_m = 2\pi NT = \frac{2\pi \times 1430\times 67.25}{60 60}$

= 10, 070.64 W.

(v)

At maximum torque $s_b = \frac{R_2}{X_2} = \frac{0.4}{4} = 0.1$

$T_{max} = \frac{3 s_b E_2^2 R_2}{2\pi n_s [R_2^2 + (s_b X_2)^2]}$

$= \frac{3\times 0.1\times 191.68^2 \times 0.4}{2\pi \times 25 [0.42 + (0.1\times 4)^2]}$

= 87.71 N-m.

Speed at maximum torque = $(1 - s_b)N_s = (1 - 0.1) \times 1500 = 1350$ r.p.m.

(vi)

Starting current $= \frac{E_2}{\sqrt{[R_2^2 + X_2^2]}} = \frac{191.68}{\sqrt{0.42 + 4^2}}$

= 47.68 A.

(vii)

At starting s = 1.

$\therefore$

T

st
=
3
2
π
×
25

×
$191.68^2 \times 0.4\ 0.4^2 + 4^2$

= 17.37 N-m.
It can be seen that the starting torque is 25.8% of full load torque and the maximum torque is 130.42% the full load torque.

5.8.2. Synchronous watt

Synchronous watt is the torque which would develop a power of 1 Watt, at synchronous speed.

$2\pi N_s$ Rotor input = T_{sw} 60

$\therefore T_{sw} = {}^{P2 \times 60}$ (synchronous Watts)$_{2\pi Ns}$

Example 5.20 A 25 HP, 4 pole induction motor has an efficiency of 90% and speed of 1440 r.p.m at full load. If mechanical losses are 25% of total losses calculate the synchronous Watt.

Solution:

Motor output = 25× 746 = 18.65 KW

Motor input $^{18.65}$ = 20.722 KW $=_{0.9}$

Total losses = 20.722− 18.65 = 2.072 KW 25% Mechanical loss = 0.25× 2.072 = 0.518 KW = 518 W. P_m, Rotor mechanical power developed = output+mechanical losses= 18.65+0.518 = 19.168 KW

Tsw = 19, 168× 60$_{= 122 \text{ N-m.} 2\ 1500}$

Example 5.21 The power input to a 415 V, 50 HZ, 4 pole, 3-phase induction motor running at 1460 r.p.m. is 25 KW. The stator copper losses are 750 W and the mechanical losses amount to 1000 W. Calculate (i) slip (ii) rotor copper loss (iii) shaft torque (iv) efficiency.

Solution:

(i)
N_s = 1500 r.p.m.
$\frac{1500-1460}{1500}$ = 0.0266 or 2.66%s =
(ii)
Motor input, P_1 = 25 KW.
Stator copper loss = 750 W
Rotor input = $P_2 = P_1 - P_{scu}$ = 25000− 750 = 24250 W = 24.25 KW Rotor copper loss = sP_2 = 0.0266× 24.25 = 0.645 KW
(iii)
$P_m = P_2$ − rotor cu loss = 24.25− 0.645 = 23.605 KW

$P_{out} = P_m$ − mechanical losses
= 23.605− 1 = 22.605 KW
Tsh × $2\pi N_{Pout = 60}$

$\frac{22605\times 60}{2\pi \times 1460}$ = 147.85 N-M.T_{sh} =
(iv)

% η = $\frac{22.605}{25} \times 100 = 90.42\%$
Example 5.22Example 5.22 , 50-Hz,15 kW induction motor drives a constant torque load at rated frequency, rated voltage and rated kW output and has a speed of 970 r.p.m. and an efficiency of 92%. Calculate (i) the new operating speed if there is a drop of 10% in voltage and 5% drop in frequency (ii) the new output power. Assume losses are constant.

Solution:
V_2 = 0.9× 230 = 207 V (reduction of 10 %) f_2 = 0.95× 50 = 47.5 Hz (reduction of 5%)
$\frac{120\times 50}{6}$ = 1000 r.p.m.N_{s1} =
$\frac{120\times 47.5}{6}$ = 950 r.p.mN_{s2} =
1000− 970$_{= 0.03;s1 = 1000}$
Since load torque remains constant,
sv^2
$_f$ remains constant. S1V12$_=$ S2V22 f1 f2
s
2

=
s
1
$V1^2$ f2
V2 f1
$^{230\ 2\ 47.5}$ = 0.0351= $0.03_{207\ 50}$
N2 = (1− s2)N_s2 = 950(1− 0.0351) = 916.65 r.p.m.
P = $^{2\pi NT}$; since torque is constant,$_{60}$
^{P1} = ^{N1}; P1 = 15KW; N1 = 970 r.p.m;$_{P2\ N2}$
N_2 = 916.65 r.p.m.
^{N2} = 15× $^{916.65}$P2 = P1 ×$_{N1\ 970}$ = 14.175 KW.

Example 5.23 A 400-V, 60 Hz, 4 pole, 3-phase induction motor runs at a speed of 1750 r.p.m when connected to 400-V. What is the speed if voltage increases to 475 V.

Solution:

N_s = $^{120\times\ 60}$ = 1800 r.p.m_4
1800− $1750_{= 0.0278.s1 = 1800}$
s
1
=
V
1
^{2} $400^{2\ =}$ $475s_2$ V_2
s
2
=
0.0278

×
400^2
$_{475}$ = 0.0392.

N2 = (1− s)N_s = (1− 0.0392)1800 = 1729.4 r.p.m.

Example 5.24 A 150 HP, 3300-V, 50 Hz, 3-phase star connected induction motor has a synchronous speed of 500 r.p.m. The full load slip is 2% and full

load p.f is 0.85. Stator copper loss = 2.5 KW; stator iron loss = 3.5 KW and rotational loss = 1.2 KW. calculate (i) rotor copper loss (ii) Line current (iii) Full-load efficiency. Draw the power flow diagram.

Solution:

Motor output = 150 H.P = $\frac{150\times 746}{1000}$ = 111.9 KW

Mechanical power developed in rotor, P_m = output + rotational losses = 111.9+ 1.2 = 113.1 KW.

(i) s 0.02 Rotor copper loss = $P_m \times_{1-s} =_{1-0.02} \times 113.1 = 2.31$ KW

(ii)

P_2, Rotor input = P_m + rotor copper loss = 113.1+ 2.31 = 115.41 KW Stator input = P_2 + stator copper loss+ Iron loss = 115.41+ 2.5+ 3.5 = 121.41 KW.

Input current$\frac{121410}{\sqrt{3\times 3300\times 0.85}}$ = 24.99 A

(iii)

$\%\eta = \frac{111.9}{121.41} \times 100 = 92.16\%$

The power flow diagram can be depicted as follows.

Stator Rotor input input 121.41 kW 115.41 Mechanical power developed, P_m

113.1 kW
Output 111.9 kW

Stator copper loss + iron loss 2.5 + 3.5 = 6 kW Rotor copper loss 2.31 kW Mechanical losses 1.2 kW

Example 5.25 An 18 KW, 4 pole 50 Hz, 3 phase induction motor has friction and windage loss of 2.5% of output. The full load slip is 4.2%. Compute the full load (a) rotor copper loss (b) rotor input (c) shaft torque (d) gross torque.

Solution:

Motor output P_{out} = 18 KW

Mechanical power $P_m = P_{out}$ + friction and windage losses = 18+ (0.025× 18) = 18.45 KW

(a) s 0.042 rotor copper loss = $P_{m1-s} =_{1-0.042} \times 18.45 = 0.809$ KW

(b)

rotor input = P_m + rotor loss = 18.45+ 0.809 = 19.259 KW

(c)

N_s = 1500 r.p.m; s = 0.042

N = (1− 0.042)1500 = 1437 r.p.m.

$^{Pout} = 9.55\times {}^{1800} = 119.62$ N-m$T_{sh} = 9.55_{N\ 1437}$

(d)

Gross torque = $9.55^{Pm\ 9.55\times\ 18450} = 122.61$ N-m$_N =_{1437}$

5.9. Starting of induction motors

Just like a d.c. motor, if full rated voltage is applied across the induction motor stator, a large inrush current flows. This starting current is around 5-7 times the rated current. However, the starting torque is only about 1.5 to 2.5 times the full-load torque. The large starting current causes a drop in the line voltage which is undesirable. Also the very high starting current may damage the station windings. Therefore it is not advisable to start large motors (above 25 KW) directly. Different starting methods are discussed below.

5.9.1. Direct-on-Line [DOL] starting

Let us see the relationship between starting torque and starting current with respect to the full load quantities.

We know rotor input = $^{2\pi NT} = KT_{60}$

Rotor copper loss = s × rotor input = sKT Rotor copper loss = $3I22R_2$ = sKT

$^{I22}\therefore T \propto_s$

Now the rotor current, $I_2\alpha$ stator current, I_1

T

∝

I

2

1

s

At starting s = 1; $\therefore T_{st} = KI_{st}^2$ (5.29) KI

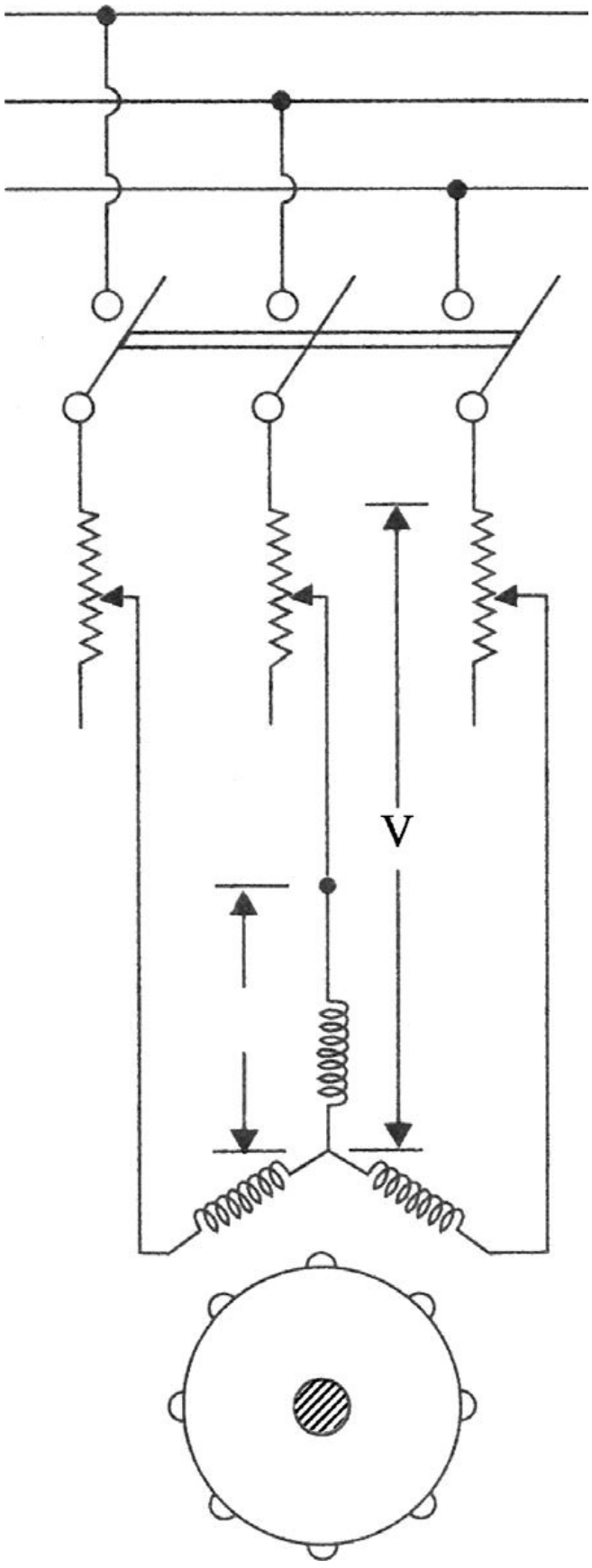
2

Similarly

T

f

=
f
$_{sf}$ (5.30)
Where I_f is the full load current and s_f is the full load slip. Dividing (5.29) by (5.30) we get
T
st
=
I
st
2
sf
Suppose T_f I_f
$_{Ist}$ = 7 and slip s_f = 0.04, then$_{If}$
$_{Tst}$ = 7^2 × 0.04 = 1.96.$_{Tf}$


V

V

xV
3-f
supply

Switch
Primary resistors
Stator
Rotor

Figure 5.10 *Primary resistor starting.*

Thus we can see that while the starting current is 7 times the full load current, the starting torque is only 1.96 times. Hence, DOL starting is not preferred except in case of small motors.

5.9.2. Stator rheostat starter or primary resistor starter

The connection for connecting resistors is shown in Fig. 5.10. In this method variable resistors are connected to the stator coils as shown in Fig. 5.10, such that the value of the resistance is maximum at starting and slowly reduced while the motor picks up its full speed.

Reduced voltage is applied to the stator because of the drops in the resistors. If I_{sc} is the starting current with full voltage and I_{st} is the starting current with reduced voltage and the voltage applied is xV, then $I_{st} = xI_{sc}$ $T_{st} = x^2T_{sc}$

$Tst_{= x}2\ Isc^2$
T
f
I
s
f^f
This method is suitable only for small machines. The advantages are

• High power factor during start because of the resistors.
• smooth acceleration
• Less expensive than auto transformer starter

Disadvantages are

Disadvantages are

• Heat dissipated in the resistors increases with the rating of the motor.
• Power wasted in resistors.
• Low torque efficiency.

5.9.3. Auto transformer starter

The auto transformer starter is shown in Fig. 5.11.

In this method a reduced voltage is obtained by taking tappings from a three phase auto transformer. They have 50%, 60% and 80% tappings so that the voltage can be adjusted to get the required starting torque. The switches shown may be air-break or air-immersed (for large motors).

Suppose the motor is started with an auto transformer having a transformation ratio K.If I_{sc} is the starting current with normal voltage and applied voltage is KV, then the starting current I_{st} is given by

Ist = KIsc (5.31)

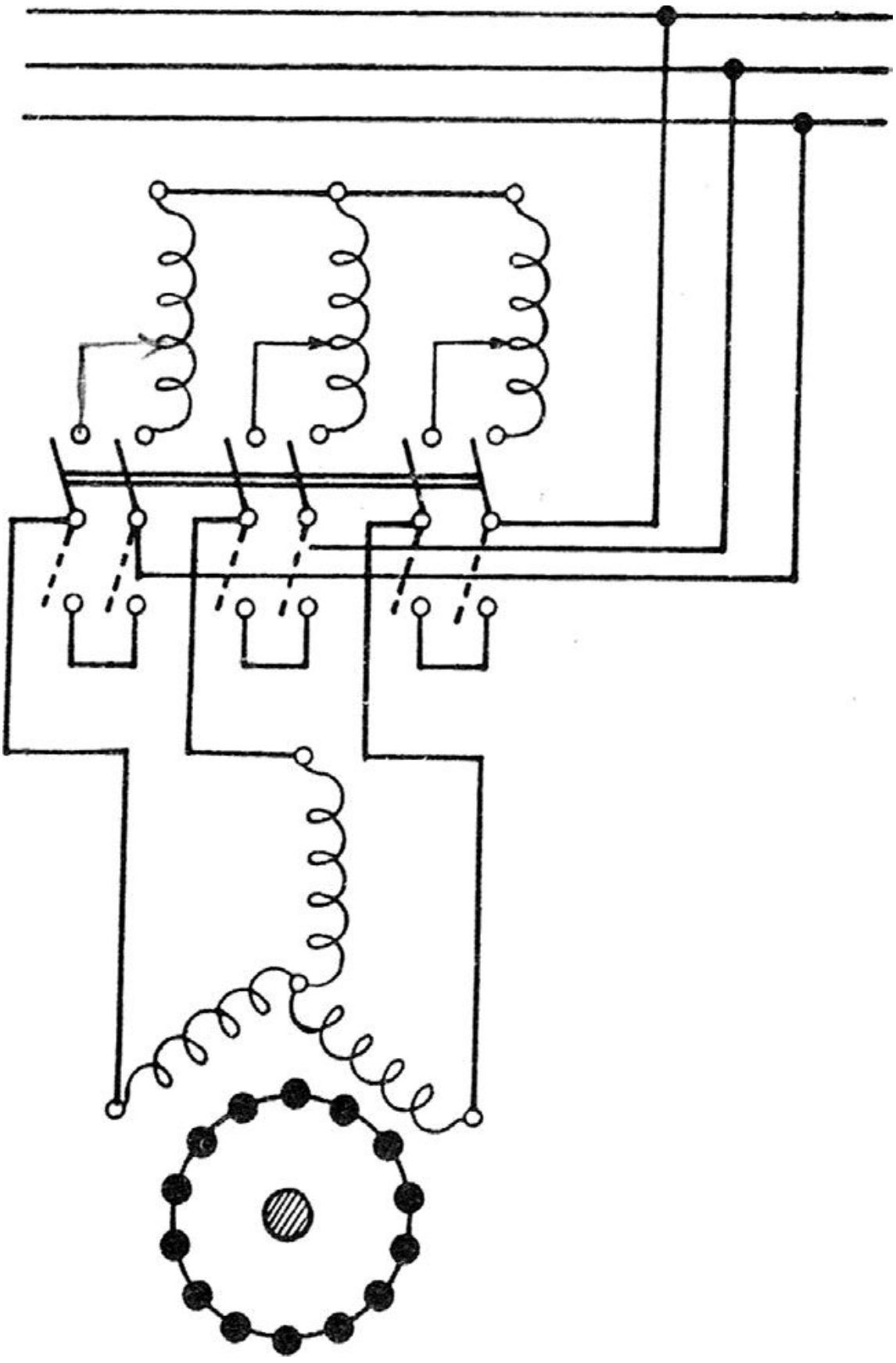
I_{sc} is also the short circuit current.

3 − φ

supply Autotransformer

Start

Run



Stator
Rotor
Figure 5.11 *Auto transformer starter.*

This is the motor input current. The autotransformer primary winding current which is nothing but the supply current is given by supply current = KI_{st}. This ratio of the staring torque with reduced voltage to the full load torque is given by

2 2
$T_{st} = I_{st} s_f = K^2 I_{sc} s_f$ (5.32)
Tf If If

Advantages of autotransformer starter

• Voltage is reduced by transformation and therefore current and power drawn from supply mains is also reduced in comparison to resistor starting.

• Adjustment of starting voltage by selection of proper tap on the auto transformer.
• Availability of the highest torque per ampere of supply current.
• Suitable for long starting duration.
• Can be used to start star or delta connected motors.

Disadvantages

• Low power factor
• Higher cost for low rating motors.

5.9.4. Star-delta starter

In this method the stator windings are connected in star at starting and in delta while running. The applied voltage is reduced by a factor of
1
√ when connected in star. The current is reduced by
1 and the torque by 1 3
√3 3. The connections are shown in Fig. 5.12.

$T_{st} = 1 I_{sc}^2$
T
f
3
I
s
f f

Here, I_{sc} and I_f represent the phase currents.

The star-delta starter is equivalent to an autotransformer starter with a ratio of $\frac{1}{\sqrt{.3}}$

The above three methods can be used for either squirrel cage or slip ring motors. However, since additional resistance can be included in the rotor of the slip ring motors we have rotor rheostat starting for these motors.

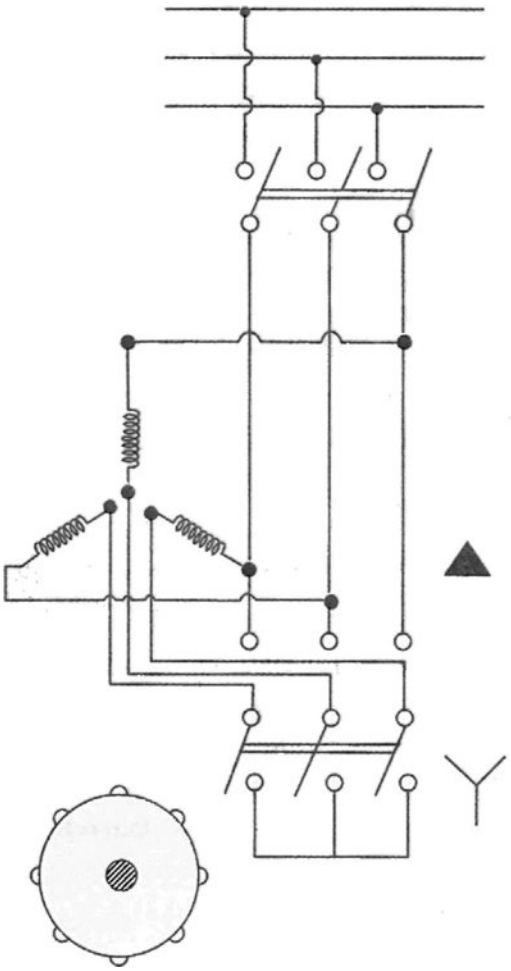
5.9.5. Rotor rheostat control

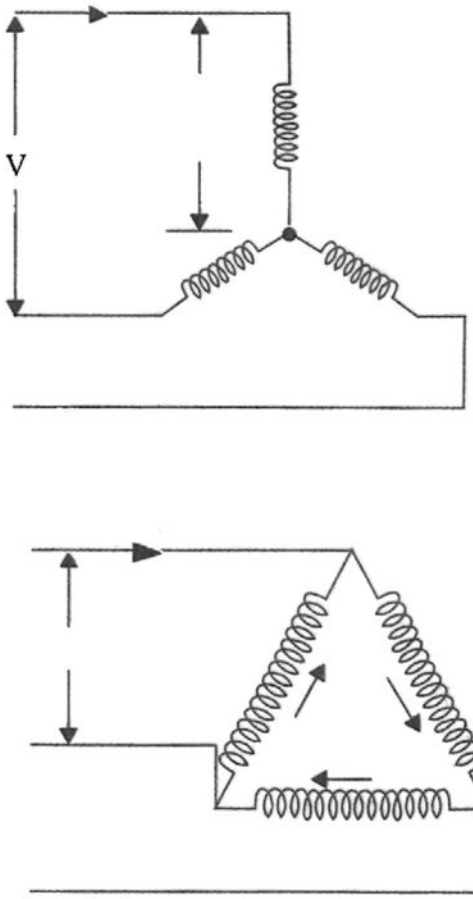
These motors are started with full applied voltage. The control resistor is connected in a star as shown in Fig. 5.13.

The resistance is introduced during starting and then gradually cut off. The additional resistance enables to develop a high starting torque with reasonable starting current. Hence, these motors can be started on load.

Example 5.26 The short circuit current of a small 3-phase induction motor is 3 times the full-load current. Determine the starting torque as a percentage of full-load torque if full-load slip is 3%.

3-f supply$_{\text{Ist}}$
V 3 V





Stator
I
st
/Phase = I
st
/Line = V Stator$_{\sqrt{}}$ 3 Z$_{\text{Run }\sqrt{}\text{ 3 Isc}}$
Rotor ^{V} Z Z Start$_{\text{Isc}}$
Z
^{Isc}/Phase = Z^{V}
I
sc
/Line =
√
3 V Z **Figure 5.12** *Star-delta starter.*

Solution: The short circuit current is nothing but the starting current, since its the current drawn when rotor is at rest.

$\therefore I_{sc} = sI_f$; $s_f = 0.03$.
2
Starting torque
T
$_{st} = T_f$
Isc_{sfIf}
2
=
T
f
$^{3If}\ 0.03 = 0.27T_{fIf}$
Starting torque is 27% of full-load torque.
Start Run
Figure 5.13 *Rotor rheostat starter.*

Example 5.27 A three phase, squirrel cage induction motor has a short-circuit current equal to 4 times the full load current. Find the starting torque as a percentage of full load torque if the motor is started by

(i) direct switching to supply
(ii) star-delta starter (iii) autotransformer
In (iii) the stator current is limited to twice the full load current and full load slip is 2.8%
Solution: $I_{sc} = 4I_f$; s = 0.028 (i) Direct switching.
T
st
=
Isc^2
$_{If\ f}T_f = (4)^2 \times 0.028 \times T_f = 0.448T_f = 44.8\%$ of T_f.s
(ii) Star-delta starter
T
st
=
1
I
2
$^{sc}\ s_fT_f = 0.149T_f$ or 14.9% of $T_{f\cdot 3\ If}$

(iii) Autotransformer
$K^2 Isc = 2If\ K$

$=$
$2\ 1$
I
$If = {}^{2If} = \sqrt{}_{2sc\ 4If}$

$2_1\ 2$
$Tst = K^{2\ Isc}\ sfTf = \sqrt{}_2\ (4)^2 \times 0.03\ Tf_{If}$
$= 0.224\ T_f$ or 22.4% of T_f

Example 5.28 Find the percentage tapping required on an autotransformer required to start a squirrel cage induction motor against ^{1} full-load torque. The short circuit$_4$
current is 4 times the full-load current and full load slip is 3%.

Solution:
T
st
1
$= 4;\ I_{sc} = 4I_f;\ s_f = 0.03_{Tf}$
T
st
$=$
k
2
I
2
$^{sc\ sf}T_f\ I_f$
$^1 = K^2 \times [4]^2 \times 0.03 \Rightarrow K = 0.722.{}_4$
The % tap is 72.2%.

Example 5.29

Example 5.29 , 50 HZ, 6-pole cage motor takes 6 times full load current at standstill and develops 1.75 times the full load running torque. If the full load

current is 28 A,

(i) What voltage must be applied to produce full-load torque at starting. (ii) What current does this voltage produce?
(iii) If autotransformer starter is used what is the line current?
(iv) What is the starting torque if starting line current is limited to full load current?

Solution:
(i) $T \propto V^2$

$1.75\ T_f \propto 415^2$ (first case)
$T_f \propto V^2$ (starting torque is equal to full load torque)

$$\frac{1.75\ T_f}{415^2} = \frac{T_f}{V^2}$$
$\Rightarrow V = 313.71$ V.
(ii) The current is proportional to voltage.
(iii)

(iv) $6I_f \propto 415$ (first case) $I \propto 313.71$

$\frac{6\times 313.71}{415} \times 28 = 130$ A.I $=$
$313.71 = 0.756$ K $= 415$

Line current $= K^2 I_{sc} = 0.756^2 \times 6 \times 28 = 96$ A $\frac{120\times 50}{6} = 1000$ r.p.m $N_s =$
$1000 - 950 = 0.05$sf $= 1000$

Given that line current is full load current.
$28 = K^2 I_{sc} = K^2 \times 6 \times 28$ 1
$K = \sqrt{6}$
Tst $= K^2$ Isc 2 1$(6)^2 \times 0.05$ Tf I s
f f $= 6$

= 0.3

The starting torque is 30% of full load torque. **Example 5.30** The full-load efficiency and power factor of 12 KW, 440 V, 3-phase induction motor are 85% and 0.8 log respectively. The blocked rotor line current is 45 A at 220 V. Calculate the ratio of starting to full-load current, if the motor is provided with a star-delta starter. Neglect the magnetising current.

Solution: Short circuit current with full voltage is

$I_{sc} = 45 \times \frac{440}{220} = 90$ A

Full load Input $= \frac{12 \text{ KW}}{0.85} = 14.12$ KW. $\sqrt{}$

$3 \times 440 \times I_f \times 0.85 = 14,120\ I_f = 23.1$ A.

In star delta starter,

$I_{st} = \frac{Isc}{\sqrt{3}} = \frac{90}{\sqrt{3}} = 52$ A

$\frac{5.2}{23.1} = 2.256 I_{st} I_f =$

Example 5.31

Example 5.31 50 Hz, 400 V, connected induction motor runs at 960 r.p.m. on full-load. It takes 86.4 A on direct starting. The full load efficiency is 88% and p.f. is 0.85. Find ratio of starting torque to full-load torque with a star-delta starter.

Solution:

Isc/phase $= \frac{86.4}{\sqrt{3}} = 49.88$ A

$\frac{1}{\sqrt{3}} \times$ Isc/phase $= 28.8$ A. Ist/phase $=$

Full load input $= \frac{15 \times 746}{\sqrt{}} = 12,716$ W.

0.88

$3 \times 400 \times I_L \times 0.85 = 12,716$

$I_L = 21.59$ A (This is line current on full load)

Full load $\frac{21.59}{\sqrt{3}} = 12.46$ A. I_L/Phase $=$

$\frac{1000 - 950}{0.05} = 1000$

T
st
=
I
st
^{2} 28.8^2
$s_f = {}_{12.46} \times 0.05 = 0.267_{Tf\ If}$
T_{st} = 26.7% of full load torque.

5.10. Equivalent circuit of induction motor

The energy transfer from stator to rotor in an induction motor takes place entirely inductively through the magnetic flux linking the two. Essentially therefore it is like a transformer. On no-load, the motor draws a current I_0. I_0 is made up of two components (i) the working or iron loss component I_w, which supplies the no-load motor losses and (ii) the magnetizing componentI_μ which sets up magnetic flux in the core and the air gap.

On load the primary currentI_1 consists of two componentsI_0 andI_2. We can write the following equations:
Rotor input/phase = $E_1 I_2 \cos\varphi$
Total rotor input = $3E_1 I_2 \cos\varphi$
Loss in rotor = $3I^2R_2 = 3I_2 E_r \cos\varphi_2$
I
2
=
KI
2
⇒
I
2

=
I_2
K

Er = sE2 and E2 = KE1 ∴ E_r = KsE1
∴ Loss in rotor = $3\times^{I2}$ $(KsE_1)\cos\varphi_K$

= (3E1I_2 cosφ) × s = s × rotor input
R
1
X
1
I
1
I
, R′ X′
2 2 2
+
I
0
I
′
2 = KI_2
R2I_w I_μ R′ = K2 $V_{1\ R}$
2
X0 E_2 = E_1
0
R′ X_{2L} X′ 2 = K2
R′ = R′ 1 − $1_{L\ 2\ s}$ −

Figure 5.14

Rotor output = rotor input− rotor loss
= (1− s) rotor input
rotor efficiency rotor output 1− s = N (5.33) = rotor input = Ns
When motor is loaded the rotor current
sE2 =E2 (5.34)I2 = 2 + (sX2)^2 R2^2 2$R_{2\ s}$ + X2
We can view the rotor circuit as being connected to a source $E_r = sE_2$, consisting of a fixed resistance R_2 and variable reactance sX_2 or as being connected to a source E_2, consisting of a variable resistance R2 and a reactanceX_2.Now R2 can be split as follows:s s
R2 1 1 R2 (5.35)$_s$ = R2 +$_s$ −
1 The component$R_{2\ s}$ − 1 is called the load resistance. It is the electrical equivalent of the mechanical load on the rotor. The equivalent circuit is shown in

Fig. 5.14. If K = 1, $E_2 = E_1$, the equivalent circuit becomes as shown in figure Fig. 5.15.

R_1 $X_{1\ I1\ I'}$ R_2 X_{22}

I0

I_w I_μ ^{X0} $R_L = R_2$ $^{1} - 1_{R0\ s}$

Figure 5.15

5.11. Tests on induction motor

The no-load and blocked-rotor tests can give the values of the elements of the equivalent circuit.

5.11.1. No-load test

In this test, the motor is run on no load at rated voltage. The currentI_0 drawn on no-load, the voltage, V, and the power drawn, W_0 are measured. Now

Y

0

=

I0

V0

Here V_0 is the voltage. If star windings are used in stator, the measured voltage should be divided by $\sqrt{}3$. The reading of the watermeter W_0, gives losses on no-load, since the output is zero. These losses are

• Stator copper loss on no load, $3I^2R_{,0}$

•

Stator core loss

=

3

G

0

V

2

0

• Loss due to friction and windage loss.

The core loss and mechanical losses together are called fixed losses. In no-load test we can separate out the stator copper losses and mechanical losses from W_0 to get the core loss. Then core loss

=
3
G
0
V
2 0
B
0
=
Y
2 – G2
0 0
R
0
and
X
0
are found from
Z
0
=
1
G
0
–
jB
0
. The real part ofZ_0 isR_0 and the imaginary part of Z_0 is X_0.

5.11.2. Blocked rotor test

This test is also called locked rotor test or short-circuit test. The rotor is locked (not allowed to move) and voltage is applied till rated currents flow through the stator windings. The power drawn under this condition is also measured. If the voltage, current and power are respectively V_{sc}, I_{sc} and W_{sc},weget

Z
01
=

V_{sc}
Isc
$^{Wsc}R01 =_{3I2}$
sc
X
01
=
Z
0
1
2
− R
2 01
Here R_{01}, X_{01} and $Z_0$1 are quantities referred to stator.
$R_2 = R01 - R1$ and $X_2 = X01 - X1$ where R_1 and X_1 are stator resistance and reactance respectively.
Example 5.32Example 5.32φ induction motor are,
V_0 = 110 V(Line-Line) I0 = 10 A
W0 = 450Watts.
The stator resistance/phase is 0.05 ; friction and windage losses are 135 W and the motor is star connected. Calculate R_0 and X_0.
Solution:
Stator copper loss = $3I^2R_1 = 3\times 10^2 \times 0.05 = 15$ W_0
$^{110} = 63.5$ VV0 = $\sqrt{}_3$
$3G0V^{\ 2} = 300_0$
$300\ G0 =_{(63.5)2 \times 3} = 0.0248$
$Y0 =^{10} = 0.157_{63.5}$
$B_0 = 0.157^2 - 0.0248^2$ 0.155=

1 $1^{Z0 =}$ 0.0248− j0.155 $^{=}$ 0.157− 81° $^{= 1+ j6.29}$ R0 = 1
X_0 = 6.29 .

Example 5.33

Example 5.33 , star connected induction motor takes 25 A at a line voltage of 30 V with blocked rotor. The power input is 420 W, and core loss is 30 W at this

voltage. The d.c. resistance between a pair of stator terminals is 0.1 . The ratio of a.c to d.c resistance is 1.6. Find the equivalent leakage reactance/phase of the motor and the stator and rotor resistance per phase.

Solution:

$V_s = {}^{30} = 17.3$ V.$\sqrt{}_3$

$I_s = 25$ A

$^{17.3}$ 0.692Z01 $=_{25}$ =

stator and copper losses = 420− 30 = 390 W 3× 25^2 × R01 = 390. .∴. R_{01} = 0.208 .

R_{01} is equivalent resistance of motor referred to the stator.

Leakage reactance/phaseX_{01} = $0.692^2 - 0.208^2$ = 0.66 .

Stator resistance is measured between two terminals.

A

r

B C

The resistance ‘r’ between two terminals will include the resistance of two phases in series. Therefore resistance of stator/phase = $^{0.1}$ 0.05 .$_{2 =}$

a.c resistance = 0.05× 1.6 = 0.08

Hence, effective resistance of rotor reflected to stator is 0.208− 0.08 = 0.128 .

Questions

1. Explain the constructional features of the slip ring and squirrel cage induction motors.
2. Clearly explain how the rotating magnetic field is produced.
3. Briefly explain the principle of operation of the induction motor.
4. Define slip.
5. Obtain the expression for the torque developed by the induction motor.

6. An 8-pole induction motor, fed from a 50 Hz supply runs at 735 r.p.m. What are the magnitudes of (i) synchronous speed (ii) slip speed (iii) per unit slip (iv) percentage slip.

Ans: [750 r.p.m; 15 r.p.m; 0.02; 27.]

7. A 6-pole 50 Hz slip ring induction motor runs with a speed of 970 r.p.m. Calculate the percentage slip and frequency of rotor currents. [3%; 1.5 Hz]

8. An 8-pole altenator runs at 750 r.p.m and supplies a 6 pole induction motor

running at a slip of 3%. What is the speed of the induction motor and the frequency of the rotor induced e.m.f? [970 r.p.m; 1.5 Hz]

rotor induced e.m.f? [970 r.p.m; 1.5 Hz]

induction motor are 0.024 and 0.12 respectively. Find the speed at maximum torque. Also find the additional rotor resistance per phase required to develop 80% of maximum torque at starting. [1200 r.p.m; 0.036] 10. Obtain the condition for (i) maximum torque during start (ii) maximum torque under running conditions.

11. Draw the torque-slip characteristics of a three-phase induction motor.
12. Clearly explain the effect of supply voltage and supply frequency on torque and speed.

13. A 4-pole, 50 Hz, 3-phase induction motor has a slip-ring rotor with a resistance of 0.04 and standstill reactance of 0.2 per phase. Find the amount of resistance to be inserted in each rotor phase to obtain full-load torque at starting? What is the power factor in the rotor at starting? Slip at full load is 3% [0.084 ; 0.516 p.f]

; 0.516 p.f]

inductor motor has a rotor resistance per phase of 0.1 and standstill reactance per phase of 0.4 . The rotor induced e.m.f per phase is 100 V when stationary. Calculate the rotor current and rotor power factor (a) when stationary (b) when running with a slip of 5%.

15. A 4-pole, 3-phase, 50 Hz induction motor has a voltage between slip-rings on opencircuit of 520 V. The star connected rotor has a standstill reactance and resistance of 20 and 0.4 per phase respectively. Determine:

(a) The full load torque if full-load speed is 1,425 r.p.m.
(b) ratio of starting torque to full-load torque

(c) Additional rotor resistance required to give maximum torque at standstill
[200 N-m; 0.82; 1.6]

16. A 6-pole, 3 phase induction motor has a rotor resistance and reactance of 0.02 and 0.1 respectively per phase. At what speed would it develop maximum

torque? Find out the value of resistance necessary to give half of maximum torque at starting. [800 r.p.m; 0.007]

17. Show that the rotor efficiency is given by $^{N}._{Ns}$

18. Define synchronous watt.

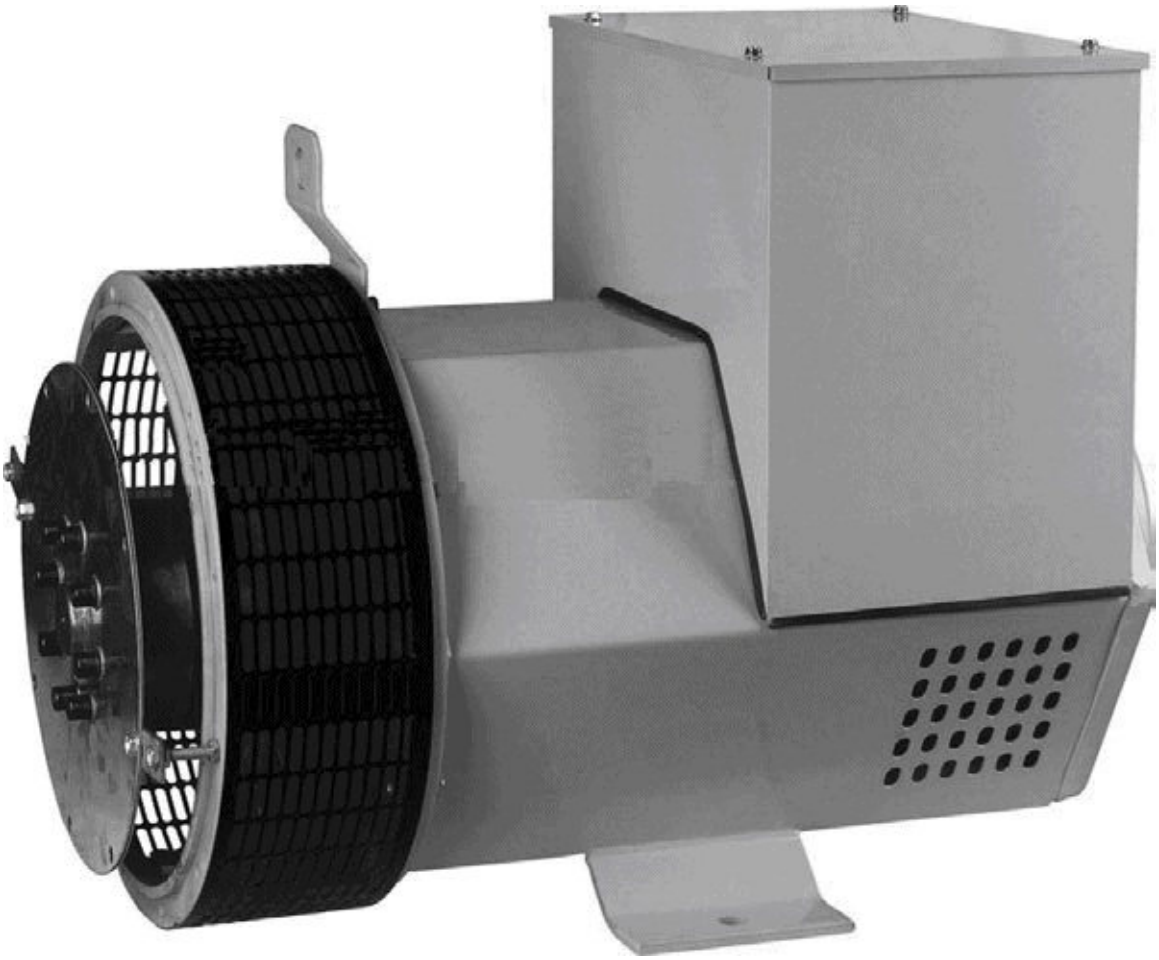
19. Draw a neat diagram to show the power stages in a three phase induction motor. 20. Obtain the equivalent circuit of an induction motor.

21. The power input to the rotor of a 440-V, 50-Hz, 3-phase, 6-pole induction motor is 60 kW. It is observed that the rotor e.m.f. makes 90 cycles per minute. Calculate (a) slip (b) rotor speed (c) rotor copper loss per phase (d) mechanical power developed (e) rotor resistance per phase if rotor current is 60 A. [0.03; 970r.p.m; 600W; 58.2 kW; 0.167]

22. A 500 V, 50 Hz, 3 phase induction motor develops 14.92 kW inclusive of mechanical losses when running at 995 r.p.m, with 0.87 p.f. Calculate slip; rotor copper loss; total input if stator losses are 1,500 W; line current; number of cycles per minute of the rotor e.m.f. [0.005; 75W; 16.5 kW; 22 A.

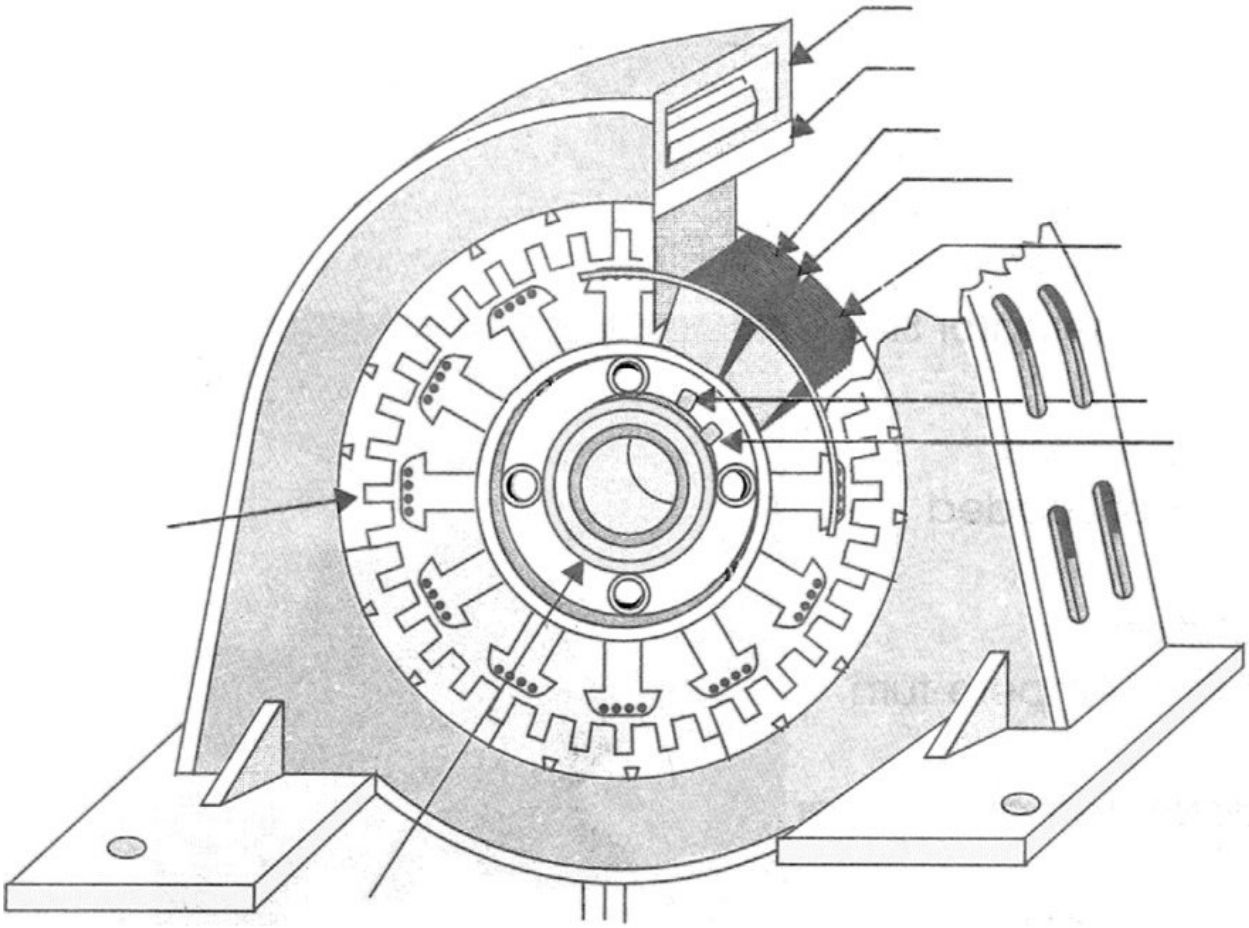
CHAPTER - 6

Alternators



Learning objectives

Principle of operation
Armature windings
Winding factor
Induced e.m.f.
Alternator on load
Synchronous reactance
Vector diagram of loaded alternator OC and SC tests
Voltage regulation by synchronous impedance method
Synchronization of alternator



Stator frame Stator core
Cage winding End rings
Spider Cage bars
Exciter
Brushes Slots $\text{Stator}_{\text{Rotor}}$
Rings Stator terminals **Figure 6.1** *Alternator construction.*

6.1. Introduction

Alternators are a.c. generators. They are also called synchronous generators. They run at a constant speed called the synchronous speed. They operate on the same fundamental principle of electromagnetic induction as the d.c. generators (remember the e.m.f. induced in the armature of a d.c. generator is a.c?). In a d.c. generator the armature rotates and the field is stationary. In an alternator the field rotates and the armature is stationary. The field is called the rotor and the armature is called the stator. The stator has three phase windings which produce balanced three phase voltages.

6.2. Construction

The constructional details are shown in Fig. 6.1.

6.2.1. Stator frame

The stator consists of a cast-iron frame, which supports the armature core. It is used for holding the armature stampings and windings in position. Ventilation is maintained with the help of holes cast in the frame itself.

Wide-open Semi-closed Closed **Figure 6.2** *Stator slots.*

6.2.2. Stator core

The core is supported by the stator frame and is built up of laminations of special magnetic iron or steel alloy. The core is laminated to minimise loss due to eddy currents. The laminations are insulated from each other and have spaces between them for following the cooling air to pass through. The slots for housing the armature conductors lie along the inner periphery of the core. The slots are of different shapes as shown in Fig. 6.2.

6.2.3. Salient pole rotor

Alternators have two types of rotors:

- Salient pole rotor
- Non-salient pole rotor

The salient pole rotor is also called the projecting type pole. It is used in low and medium speed alternators. It has a number of projecting (salient) poles, having their cores bolted or dovetailed onto a heavy magnetic wheel generally of cast iron as shown in Fig. 6.3.

Such rotors have short axial length and large diameter. The poles and the pole-shoes are laminated to minimize heating due to eddy currents.

6.2.4. Non-salient pole or smooth cylindrical rotor

Non salient pole rotors are used in high speed turbo alternators which are driven by steamturbines. The rotor consists of a smooth solid forged steel cylinder, having a number of slots milled out at intervals as shown in Fig. 6.4.

N Field winding Projected pole

S S
Bolt Mechanical$_{Shaftsupport}$
N

Figure 6.3 *Salient pole rotor.*
Non-salient Polar area Pole
^{N} Slots
D.C. supply
S
Field windings (a) (b)
Figure 6.4 *Non salient pole rotor.*

They are designed for 2 pole or 4 pole turbo-generators, running at 3600 r.p.m. or 1800 r.p.m. respectively. The polar regions are left unslotted. The field coils are placed in the slots and are so arranged that the flux density is maximum on the polar central line and gradually falls away on either side. The poles do not project out from surface of rotor. Hence, they are called non-salient poles. Such rotors are characterised by small diameters and long axial length. This construction gives better balance and lesser windage losses.

6.2.5. Damper windings

These are copper bars which are short-circuited at both ends by heavy copper rings located on the pole-shoes. They are useful in preventing hunting; which is speed fluctuations, in alternators and provide starting torque in synchronous motors. Turbo-generators usually do not have damper windings because the solid field poles act as dampers. Under normal running conditions, damper windings do not carry any current because the rotor runs at running conditions, damper windings do not carry any current because the rotor runs at φ voltage under unbalanced conditions.

6.2.6. Armature windings

D.C. machines have closed windings, but alternator windings are open, which means that no closed path exists for the armature currents in the winding it self. Generally, alternators are connected in star. One end of each phase winding is joined to the neutral point and the other end brought out. There are two popular types of windings used for three phase alternators:

• Single layer winding

• Single layer winding
• Double layer winding

With respect to windings we define the following terms.

Pole pitch: It is defined as the distance between two adjacent poles, expressed as the number of armature conductors/pole or slots per pole. For example, if we have 12 slots, then the pole pitch for a 4-pole alternator is 3 slots/pole.

Coil span: The distance between two sides of a coil is the coil-spam.

Full pitch: If the coil spam is equal to the pole-pitch, then the winding is said to have full-pitch. This means if, one side of the coil is under the center of a N-pole, then the other side of the same coil should be under the center of S-pole, which is 180◦ electrical apart. When the coil-span is full-pitched, the e.m.f's induced in the two sides of the coil are added together.

Pitch-factor or chording-factor

Consider a 4-pole alternator with 24 slots. The pole-pitch is 24 *4 = 6 slots*pole. Therefore, a coil whose first coil side is in slot number 1, will have the second coil side in slot number (1+ 6) 7. This is shown in Fig. 6.5.

N S
24 12 34 5 67 8 …
Pole pitch
Figure 6.5 *Pitch-factor.*

If the coil sides are placed in slots 1 and 6, then it is said to be fractional-pitched or short-pitched coil. The coil span is now $\frac{5}{6}$ of pole pitch. It falls short by $\frac{1}{6}$ pole pitch or

$\frac{180^\circ}{6} = 30^\circ$ electrical. Short pitch windings are used for the following reasons:

• Save copper of end connections.
• To reduce harmonics in generated e.m.f.
• Reduce eddy current and hysteresis losses.

The total voltage induced in the coil is reduced, because the voltages induced in the two coil sides are out of phase and their vectorial sum is less than their

arithmetic sum. The pitch-factor or chording factor (K_p or K_c) is defined as

$$K_p = \frac{\text{Vector sum of induced e.m.f. in coil sides per coil}}{\text{Arithmetic sum of induced e.m.f. in coil sides per coil}} \quad (6.1)$$

It is always less than 1. Suppose E_s is the induced e.m.f. in each side of the coil. If the coils are full-pitched, the total e.m.f. is $2E_s$ as shown in Fig. 6.6(a). Now consider a short-pitched coil, which is short-pitched by 30◦, as shown in Fig. 6.6(b).

The resultant voltage E is given by

E

=

$_{2Es}$ cos 30˚ ◦

$_2$ = $2E_s$ cos 15

e.m.f. induced in each coil side

E_s E_s E

$2E_{s\ Es}$

(a) Total e.m.f. (b) 30°Es

Figure 6.6 *Vector sum of e.m.fs.*

K

p

=

Vector sum $_=$ $2E_s$ cos 15˚ Airthmetic sum

2

E

= 0.966

s

pitch-factor = 0.966

In general, if the coil-span falls short of full pitch by an angleα; then

$K_p(K_c)$ = cos $^{\alpha}$ $(6.2)_2$

α is called the chording angle.

Distribution-factor or breadth-factor

It is also called winding-factor or spread-factor. The coils in each phase are not bunched is one slot, but are distributed in a number of slots. They are displaced from each other by a certain angle. The result is that the e.m.f's induced in coil sides constituting a group are not in phase with each other but differ by an angle

sides constituting a group are not in phase with each other but differ by an angle equal to the angular displacement of the slots.

Lets take a simple example. Consider a 4 pole alternator which has 36 slots. Therefore the pole-pitch is 9 slots/pole. Angular displacement between any two adjacent slots $= {}_{180^\circ} = 20^\circ$ (electrical). We have three phases and hence 3 slots/phase/pole. If the three9

coils (assuming there is one coil in one slot) were bunched together in one slot, then the arithmetic sum would be $3E_s$, whereE_s is the e.m.f. induced in one coil side as shown in Fig. 6.7(a).

E_s E_s E_s E Es 20° E_s 20° (a) $3E_s$ (b)^{E}s

Figure 6.7 *Distribution factor.*

Since the coils are distributed, the individual e.m.f's have a phase difference of 20° with each other as shown in Fig. 6.7(b). The distribution factor,K_d, is defined as

Kd = e.m.f. with distributed winding (6.3)e.m.f. with concentrated winding

The angular displacement between slots,β,isgivenby

$$\beta = \frac{180^\circ}{180^\circ}$$

No. of slots/pole = n(6.4)

The distribution factor,

$$K_d = \frac{\sin(m\beta/2)}{\sin\frac{\beta}{2}} \quad (6.5)$$

where m is the no. of slots/phase/pole.

We can calculate K_d for different values of m as shown in Table 6.1.

6.3. Advantages of stationary armature

As mentioned in section 6.1, alternators have a stationary armature and a rotating field. The advantages are

The output current can be led directly from fixed terminals on the stator (armature) to•

the load circuit, without having to pass it through brush contacts.

• The slip-rings are on the low-voltage, low-power d.c. field circuit, which can

be easily insulated.

Table 6.1 Distribution factor.

Slots/pole	mβ°	K_d
3	160°	1.0
6	230◦	0.966
9	320◦	0.960
12 4	15◦	0.958
15 5	12◦	0.957
18 6	10°	0.956
24 8	7.5	0.955

• It is easier to insulate stationary armature windings for high a.c. voltages, which may have as high a value as 30 kV.
• The windings can be braced to prevent deformation.

Example 6.1 Calculate the pitch factor for a winding with 36 stator slots, 4 poles and a coil-span of 1 to 8.

Solution:

No. of slots/pole = 36/4 = 9

Therefore coil span in full pitch winding is 1 to 10. With a coil span of 1 to 8, it falls short by ^{2} pitch. *i.e.* $^{2} \times 180^{\circ} = 40^{\circ}$.$_{9\ 9}$

pitch factor

K

c

=

cos

$^{\alpha}{}_{=}$ cos 40°

$_{2\ 2}$ = 0.94.

Example 6.2 Calculate the pitch factor for a winding with 72 slots, 6 poles and coil span 1 to 10.

Solution:

No. of slots/pole = ^{72} $12_{6\ =}$

The coil span falls short by (^{3}) × 180° = 45°$_{12}$

K

c

=

cos 45°
$_2$ = 0.924

Example 6.3 Calculate the distribution factor for a 36-slots, 4-pole, single layer three phase winding.

Solution: 36 n = No. of slots/pole =$_4$ = 9 180° 180° $_\circ\beta$ = n = 9 = 20

m = No. of slots/pole/phase = 9/3 = 3.

sin$^{m\beta}$ sin$^{3\times20\circ}$

K

d

=

2 2

sin

β

=$_{3 \sin 20\circ}$ = 0.96 m$_2$ 2

Example 6.4 In an alternator, a part of the winding consists of six coils in series, each having an e.m.f. of 12 volts induced in it. The coils are placed in successive slots and between each slot and the next, there is an electrical phase displacement of 30°. Find the e.m.f. of the coils in series.

Solution:

sin

6×30◦ 2

β = 30°; m = 6; K_d =$_{6\times \sin 30\circ}$ = 0.644

2

Arithmetic sum of voltages = 6× 12 = 72.

Vector sum = K_d × arithmetic sum = 0.644× 72 = 46.37 V.

Example 6.5 An alternator has 9 slots/pole. FindK_d if we have (i) one winding in all the slots (ii) one winding using the first ^{2} of the slots/pole (iii) three windings placed in 60◦ group.^{3}

Solution:

β

=

180° $_\circ$ 9 = 20

(i) One winding;

m
=
9;
K
d
=
sin
9×20°
2 = 0.64 9 sin 20°
2
(ii) One winding in 2 of the slots per pole. 3
m = 2 × 9 = 6 K_d =
sin
6×20°
2
6 sin 20° = 0.83 3 2
(iii) m = 9 = 3; 3
sin 3×20°
K
d
=
2
3 sin
20° = 0.96.
2

6.4. Induced e.m.f. in an alternator

Let Z = No of conductors (coil sides) in series per phase. One turn is two coil sides.

∴
T
=
Z

2 P = No. of poles
f = frequency of induced e.m.f. in Hz.
φ = flux/pole in webers
K

d
=
distribution factor
=
sin(mβ/2)

$^{m(\ sin\beta/2)}$ $^{\alpha}K_c$ (or K_p) = pitch factor or coil factor = $\cos_2$
K_f = form factor = 1.11 (for sinusoidal voltage as form factor is 1.11) N = rotor speed in r.p.m.

One revolution of the rotor takes ^{60} seconds. Each stator conductor cutsφp webers.$_N$
dφ = φP
^{60}dt =$_N$
∴
Average e.m.f. induced per conductor
=
dφ $_=$ NφP dt 60
Now N = ^{120f} .$_P$
Average e.m.f. per conductor
=
φP 120f 60 P = 2fφ volts.
If there are Z conductors in series/phase, then average e.m.f./phase = 2fφZ volts = 4fφT volts.

R.m.s. value of e.m.f./phase = 1.11 × 4fφT = 4.44fφT (Note that this is same as the e.m.f. induced in a transformer winding.) This would be the voltage if all the coils of a phase were full-pitched and concentrated or bunched in one slot. But this is not so and the actual voltage induced is reduced by a factor ofK_c and K_d.

Voltage induced/phase = 4.44$K_c$$K_d$fφT
= 4$K_C$$K_d$$K_f$fφT volts. (6.6)

If the alternator is star connected, the line voltage is$^{\sqrt{}}$3 times the phase voltage.
Example 6.6 A 3-phase, 16-pole alternator has a star-connected winding with 144 slots and 10 conductors/slot. The flux per pole is 28 mWb and the speed is 375 r.p.m. Find the frequency and phase and line e.m.f. assuming full pitched

winding.

Solution: $f = \frac{PN}{120} = \frac{16 \times 375}{120} = 50$ Hz

slots/pole $= \frac{144}{9}$; $\beta = \frac{180^\circ}{n} = \frac{^\circ}{16} = 20^\circ$

$m = \frac{\text{slots/pole/phase}}{3} = \frac{9}{3}$

$K_d = \frac{\sin 3 \times 20^\circ}{3 \sin 20^\circ} = \frac{2}{2} = 0.96$

No. of conductor $= 144 \times 10 = 480$/phase $Z = \text{Phase } 3$

$T = \frac{480}{2} = 240$/phase

$E_{ph} = 4.44 \times 1 \times 0.96 \times 50 \times 0.028 \times 240 = 1432$ V

$E_{line} = \sqrt{3} E_{ph} = 2480.6$ V.

Example 6.7 Find the line voltage of a star-connected 3-phase, 6-pole alternator which runs at 1200 r.p.m. having a flux per pole of 0.08 Wb, sinusoidally distributed. It has 54 slots having a double layer winding. Each coil has 8 turns and is chorded by 1 slot.

Solution:

n = slots/pole $= \frac{54}{6} = 9$; $\beta = \frac{180^\circ}{9} = 20^\circ$; m = slots/pole/phase = 9/3 = 3. Since winding is chorded by one slot, it is short pitched by $\frac{1}{9}$ or $\frac{180^\circ}{9} = 20^\circ$.

$K_c = \cos \alpha = \cos 20^\circ$

$K_c = 0.98$

$K_d = \frac{\sin 3 \times 20^\circ}{3 \sin 20^\circ} = 0.96$

2
^{8} = 144 (Here since its a double layer winding we willZ = 54×$_{3}$ have 4 turns per winding or 8 conductors)
^{144} = 72T =$_{2}$
6× ^{1200} = 60 Hzf =$_{120}$
E
ph
= 4.44× 0.98× 0.96× 60× 0.08× 72 = 1443.6 V = √3× 1443.6 = 2500.4 VE$_{L}$

Example 6.8 The stator of a 3-phase, 16-pole alternator has 144 slots, with 4 conductors/slot connected in two layer and the conductors of each phase are connected in series. The alternator runs at 450 r.p.m. Flux is 0.075 Wb/pole and is distributed sinusoidally. Calculate the e.m.f. induced per phase if the coil span is 150° electrical.

Solution: Coil span is 150°. It falls short by (180° − 150°) = 30°.
∴
K
c
=
cos 30°
$_{2}$ = 0.966
^{144} 9; m = ^{9} = 3; β = $^{180°}$ °n =$_{16}$ =$_{3\ 9}$ = 20 Kd = 0.96
16× 450$_{= 60\ Hzf\ =\ 120}$
No. of slots/phase = ^{144} = 48$_{3}$
No. of conductors/phase = 48× 4 = 192
No. of turns/phase ^{192} = 96 =$_{2}$
E$_{ph}$ = 4.44× 0.966× 0.96× 60× 0.075× 96 = 1778.75 volts

Example 6.9 A 4-pole, 50-Hz, star-connected alternator has 15 slots/pole and each contains 12 conductors. The conductors of each phase are connected in series. The winding factor is 0.95. The terminal voltage was 1800 volts. Find the flux/pole. If the same windings were lap-connected in a d.c. machine. What would be the induced e.m.f.?

Solution:

$K_d = 0.95$

$K_c = 1.0$ (assumed) Total number of slots = 15× 4 = 60

slots/phase $= \frac{60}{3} = 20$

Number of conductors/phase = 20× 12 = 240

Number of turns/phase $= \frac{240}{2} = 120$

Phase voltage $= \frac{1800}{\sqrt{3}} = 1039.23$ volts

1039.23 = 4× 1.11× 1× 0.95× φ × 50× 120 φ = 0.041 Wb = 41 mWb.

When connected in a d.c. generator

$E_g = \frac{ZN\varphi P}{60 A}$

Z = 60× 12 = 720 $N = \frac{120f}{P} = \frac{120\times 50}{4} = 1500$ r.p.m.

$E_g = 60\times 4 = 720\times 1500\times 0.041\times 4 = 738$ volts.

Example 6.10 How are the pitch factor and distribution factors affected by harmonics?

Solution: The third harmonic has three times the fundamental frequency, the fifth harmonic has five times the fundamental frequency and so on.

(i) If the chording angle is α degrees electrical for the fundamental flux wave then,

Pitch factor $K_c = \cos\frac{\alpha}{2}$ (for fundamental)

$\cos\frac{3\alpha}{2}$ (for 3rd harmonic)

$\cos\frac{5\alpha}{2}$ (for 5th harmonic)

and so on. (ii)

The distribution factor

$K_d = \frac{\sin\frac{nm\beta}{2}}{m\sin\frac{n\beta}{2}}$

where n is the order of the harmonic.

Example 6.11 An alternator has 18 slots/pole and the first coil lies in slots 1 and 16. Calculate the pitch factor for (i) fundamental (ii) 5th harmonic.

Solution:
coil span = 16− 1 = 15 slots It falls short by 3 slots for full pitch
α
=
180
$^{\circ} \times 3 = 30^{\circ}$ 18
(i) K_c (fundamental) = $\cos 30^{\circ} 0.966_2$ =
(ii) K_c (5th harmonic) = $\cos 5 \times 30^{\circ} = 0.259_2$

6.5. Alternator on load

As the load on an alternator is varied, its terminal voltage is also found to vary. This is due to the following:

- Voltage drop due to armature resistanceR_a
- Voltage drop due to armature leakage reactanceX_L
- Voltage drop due to armature reaction.

6.5.1. Armature resistance and leakage reactance

The armature resistance R_a, causes a voltage dropIR_a which is in phase with the armature current$I$. SinceR_a is very small, this drop is note significant. When current flows through the armature conductors, fluxes are set up which do not cross the air-gap. They are called leakage fluxes. These cause an induced e.m.f. called the reactance e.m.f., which leads the armature current I by 90°. Hence, the armature winding can be said to possess a reactance X_L, such that the voltage drop isIX_L and equals the induced e.m.f. due to leakage flux. X_L is called the leakage reactance or potier reactance.

The phasor diagram is shown in Fig. 6.8.
$E = V + I(R_a + jX_L)$

6.5.2. Armature reaction

The armature reaction is the effect of armature flux on the main field flux. The power factor of the load has an effect on the armature reaction. For unity power factor load, the armature reaction is cross-magnetizing. This means that it has a distortional effect. For zero p.f.

E

$_V$ IX_L IR_aI

Figure 6.8 *Phasor diagram of alternator on load.*

$_{E0}$ E_0 − No-load e.m.f. IX

a

generated V− Termincal voltage

^{E} IX_s $_V$ IX_L IR_aI

Figure 6.9 *Phasor diagram with armature reaction.*

lagging loads, the armature flux has a demagnetising effect. This decreases or weakens the main flux. Hence, the generated e.m.f. is reduced. To keep the value of the generated e.m.f. same as that at no load, the field excitation has to be increased to compensate for the weakening. For zero p.f. leading loads, the armature flux has a magnetizing effect, which results in a greater induced e.m.f. For values of power factor in between zero and unity, the armature flux has both a cross-magnetising as well as a demagnetising effect (if load is having lagging p.f.).

6.5.3. Synchronous reactance

The drop in the terminal voltage due to armature reaction can be taken into account by assuming the presence of a fictitions reactance X_a in the armature winding. IX_a is the drop due to the armature reaction. The phasor diagram is now drawn as in Fig. 6.9, to include the drop due to armature reaction.

The sum of the leakage reactance X_L and the armature reactance X_a, is called the synchronous reactanceX_s. On load, the total drop is given by$I(R_a+jX_s) = IZ_s$, where Z_s is the synchronous impedance. The phasor difference balance the terminal voltage V and the no-load induced e.m.f. E_0 isIZ_s.

6.5.4. Phasor diagrams of a loaded alternator

The phasor diagrams are drawn for different power factors of the load. In the diagrams,

E_0 = No-load induced e.m.f.

E = Induced e.m.f. on load after allowing for armature reaction drop V = Terminal voltage

I = Armature current

The phasor diagrams are shown in Fig. 6.10.

$E = V + I(R_a + jX_L)$ (6.7) $E0 = E + IX_a = V + I(R_a + jX_s) = V + IZ_s$ (6.8)

6.5.5. Voltage regulation

When the alternator is loaded, there is a drop in the terminal voltage, which depends on the magnitude of the load and also its power factor. The voltage regulation of an alternator is defined as the rise in voltage when the full load is removed, with the field excitation and speed remaining the same as a ratio of the rated terminal voltage

$$\% \text{ Regulation} = \frac{E0 - V}{V} \times 100 \quad (6.9)$$

A couple of points to be noted are

- $E_0 - V$ is the arithmetic difference
- The rise in voltage when load is removed/thrown off, is not same as the voltage drop when alternator is loaded
- When the load has a leading p.f., the terminal voltage will fall when the load is removed. Hence, the regulation will be negative.

E_0 E0 IXa IZs IX_a IZ_s

E

IX_L E

V φ V IX_L IR_aI IRa I

(a) Unity p.f. (b) Lagging p.f.

E_0 IX_a IX_L E IZ_sI

φ V IR_a IR_a

(c) Leading p.f. **Figure 6.10** *Phasor diagrams of alternator on load.*

6.5.6. Regulation by direct loading

In small machines the regulation can be determined by direct loading. Here the desired load (magnitude and p.f.) is applied and the excitation is adjusted such that the terminal voltage is the rated value. The load is now suddenly thrown off, keeping the field excitation and speed constant. The voltageE_0 is taken. The regulation is then calculated using (6.8).

For larger machines indirect methods are used to predetermine the regulation. The methods commonly used are the

(i) Synchronous impedance or E.M.F. method (ii) The Ampere-turn or M.M.F. method (iii) Zero power factor or portier method.

$_{E2}$ O.C.C
E_1
I_a E_0 S.C.C
Ia2
^{I}a1
B I_{f2}
$^{I}f1^{I}f$
Figure 6.11 *O.C.C. and S.C.C. of alternator.*

6.6. Synchronous impedance method or E.M.F. method To apply the synchronous impedance method we need to conduct two tests.
(i) The open circuit test (ii) The short circuit test.

6.6.1. Open circuit test

This test is used to obtain the open circuit characteristic (O.C.C.) similar to that of a d.c. generator. The alternator is run on no-load and the field excitation is varied and the induced e.m.f. noted for different values of field current. A plot of the induced e.m.f. versus the field current gives the open circuit characteristic.

6.6.2. Short circuit test

Here, the stator windings are short circuited and the field current is adjusted to let rated current pass through the stator windings. The short circuit characteristic (S.C.C.) is a plot of the armature current versus field current. It is a straight line passing through the origin. The O.C.C. and S.C.C. curves are shown in Fig. 6.11.

6.6.3. Computation of synchronous impedance The synchronous impedance Z_s is defined as

$$Z_s = \frac{\text{Open circuit voltage}}{\text{Short circuit current If constant}} \quad (6.10)$$

Referring to Fig. 6.11, when the field current is I_{f1}, the synchronous impedance is given by ^{E1}; at a field current of I_{f2} it is E_2/I a_2. This means that Z_s is not a constant$_{Ia1}$
and depends on the point at which it is computed. In predicting the regulation, the value of Z_s is usually taken to be that obtained from full-load current in the short circuit test.

The synchronous reactance is obtained from the synchronous impedance as,

The synchronous reactance is obtained from the synchronous impedance as,

$X_s = Z^2 - R^2$ (6.11)$_{s\ a}$

where R_a is the armature winding resistance/phase.

6.6.4. Computation of regulation

Once the value of R_a and X_s are obtained the regulation can be calculated.

Consider the phasor diagram shown in Fig. 6.12, for a lagging load.

For the figure,

$$E\ 0 = OC = OB^2 + BD^2 = (OA + AB)^2 + (BC + CD)^2 = (V\cos\varphi + IR_a)^2 + (V\sin\varphi + IX_s)^2 \quad (6.12)$$

$$\%\ \text{Regulation} \frac{E0 - V}{V} \times 100 =$$

When we have leading p.f.,

$$E = (V\cos\varphi + IR_a)^2 + (V\sin\varphi - IX_s)^2 \quad (6.13)$$

In the synchronous impedance method of computing the regulation, the armature reactance X_a (accounting for armature reaction) is clubbed with the leakage reactance X_L,togive

D

$_{E0}$ IX_s

$_V$ $IZ_{s\ C}$ IR_a

V sin φ φ A

O I_B V cos φ_{IRa}

Figure 6.12 *Phasor diagram for regulation.*

the synchronous reactance. This method gives a simple way to calculate the regulation for any load power factor.

Example 6.12

Example 6.12 φ alternator has an armature resistance of 0.3 per phase. The full load current flows on short circuit at a field current of 0.75 A. The open circuit voltage induced at this value of field current is 375 V. Find the synchronous impedance and synchronous reactance.

Solution: Three phase alternators are normally star connected. WhenZ_s andX_s have to be calculated, it is done on a single phase basis.

Full load current = $\frac{5\times 1000}{\sqrt{3}\times 415}$ = 6.96 A

R a = 0.3

E_0 = 3.75 V; E_o/phase = ^{275} = 158.77 V$_{\sqrt{3}}$

$Z_s = {}^{158.77}_{= 22.816.96}$

$X_s = 22.81^2 - 0.3^2 = 22.8$

Example 6.13Example 6.13φ alternator has an effective resistance of 0.02 and an armature leakage reactance of 0.07 . Compute the voltage induced in the armature and the percentage regulation when the alternator is delivering rated current at a load power factor (i) u.p.f. (ii) 0.8 p.f. lagging (iii) 0.8 p.f. leading.

Solution:

Solution:

Full load current

=

50×10^3

$_{220}$ = 227.27 A

$^{\sqrt{}}$3)φ alternator we don't divide by

$IR_a = 227.27\times 0.02 = 4.54$ V $IX_s = 227.27\times 0.07 = 15.91$ V

(i) *At u.p.f.* φ = 0

$E = (V\cos\varphi + IR_a)^2 + (V\sin\varphi + IX_s)^2 = (220+ 4.54)^2 + (0+ 15.91)^2$ 225.1 V=

% R = $^{225.1- 220}$ × 100 = 2.318%$_{220}$

(ii) *At 0.8 p.f. lag*

$E = (220\times 0.8+ 4.54)^2 + (220\times 0.6+ 15.91)^2$ 233.4 V=

% R = $^{233.4- 220}$ × 100 = 6.09%$_{220}$

(iii) *At 0.8 p.f. lead*

$E = (220\times 0.8+ 4.54)^2 + (220\times 0.6- 15.91)^2$ 214.64 V=

(Note negative sign in second term).

% R = $^{214.64- 220}$ × 100=−2.44%$_{220}$

Example 6.14 A 50-kVA, 415 V, 3-phase, 50 Hz alternator has an effective armature resistance of 0.25 *phase. The synchronous reactance is 3.2* phase and leakage reactance is 0.5 /phase. At u.p.f. and rated load, determine the (i) Internal e.m.f.E_a (ii) no-load e.m.f.E_0 (iii) percentage regulation on full-load (iv) value of synchronous reactance which represents armature reaction.

Solution: The internal e.m.f. E_a is the sum of the terminal voltage,IR_a drop

andIX_L drop. It does not account for the armature reaction. The no load voltage E_0 is the sum of E_a and drop due to armature reaction.

$V = {}^{415} = 239.6\ V_{\sqrt{3}}$
50
×
10
3
$I_{FL} = {}_{\sqrt{3}\times\ 415} = 69.56$ A

$IR_a = 69.56\times\ 0.25 = 17.39$ V $IX_a = 69.56\times\ 0.5 = 34.78$ V $IX_s = 69.56\times\ 3.2 = 222.592$ V

(i)
$E_a = (V + IR_a)^2 + IX^2$ (since it is u.p.f. V $\sin\varphi = 0)_L$
$239.6+\ 17.39)^2 + (34.78)^2 = ($
= 259.33 V
(ii)
$E0 = (V + IR_a)^2 + (I\ X_s)^2 = (239.6+\ 17.39)^2 + (222.592)^2 = 340$ V (iii)
$\%R = {}^{340-\ 239.6} \times 100 = 41.9\%_{239.6}$
(iv)
$X_a = X_s - X_L = 3.2-\ 0.5 = 2.7$
Example 6.15Example 6.15φ alternator. S.C. test: $I_f = 2.5$ A; $I_a = 100$ A
O.C. test: $I_f = 2.5$ A; E = 500 V
$R_a = 0.8$. Determine the voltage regulation and draw the phasor diagram for (i) u.p.f. (ii) 0.8 p.f. lead (iii) 0.8 p.f. lag.
Solution:
$Z_s = {}^{500} = 5_{100}$
$X_s = 5^2 - 0.8^2 = 4.936$
(i) *u.p.f.*
$E_0 = (2000+\ 100\times\ 0.8)^2 + (100\times\ 4.936)^2$ 2137.76 V=
$\%R = {}_{2137.76-\ 2000} = 6.89\%_{2000}$
(ii) *0.8 p.f. lead*
$E_0 = (2000\times\ 0.8+\ 80)^2 + (2000\times\ 0.6-\ 439.6)^2$ 1822.47 V=
$\%R = {}_{1822.47-\ 2000\times\ 100} = -8.87\%_{2000}$

$36.86°^{I}$
^{E0} IX_s ^{E0} 493.6 V V IR_a 493.6 V
$_{I}$ 2000 V 80 V
100 A (a) (b) 80 V
E_0 493.6 V
IZ_s 2000 V
80 V
36.86°
(c) I **Figure 6.13** *Example 6.15.*
(iii) *0.8 p.f. lag*
$E_0 = (2000\times 0.8+ 80)^2 + (2000\times 0.6+ 493.6)^2$ 2385.5 V=
$\%R = {}^{2385.5-\ 2000} \times 100 = 19.27\%_{2000}$

Example 6.16

Example 6.16 φ alternator with an effective armature resistance of 0.25 , a field current of 38 A produces a short circuit current of 200 A and an open circuit voltage of 1040 V. Calculate the full-load voltage regulation at 0.8 p.f. lag and 0.8 p.f. lead.

Solution:
1040
$^{\sqrt{3}} = 3Z_s =_{200}$
$X_s = 3^2 - 0.25^2 = 2.989$
Full load current = $^{100\times 103/\sqrt{3}} = 19.2$ A.$_{3000}$

voltage/phase $IR_a = 19.2\times 0.25 = 4.8$ V $IX_s = 19.2\times 2.989 = 57.39$ V

^{3000} 1730 V $=_{\sqrt{3}\ =}$
(i) *0.8 p.f. lag*
$E = (1730\times 0.8+ 4.8)^2 + (1730\times 0.6+ 57.39)^2$ 1768.8 V=
$\%R = {}^{1768.8-\ 1730} \times 100 = 2.24\%_{1730}$
(ii) *0.8 p.f. lead*
$E = (1730\times 0.8+ 4.8)^2 + (1730\times 0.6- 57.39)^2$ 1700.1 V=
$\%R =$ $1700.1- 1730_{\times\ 100=-1.73\%1730}$

Example 6.17 A three phase, 10 kVA, 415 V, 50 Hz alternator supplies rated load at 0.8 p.f. lag. $R_a = 0.5$ and $X_s = 8$. Find the voltage regulation at full load. Draw the phasor diagram and find the power angle.

C
E_0 IX_s
V IR_a δ
36.86°
0 IAB **Figure 6.14** *Example 6.17.*

Solution:

$I_{FL} = \frac{10\times 10^3}{\sqrt{3}\times 415} = 239.6$ V; $V = \frac{415}{\sqrt{3}} = 13.91$ A

$IR_a = 13.91\times 0.5 = 6.955$ V

$IX_s = 13.91\times 8 = 111.28$ V

$E_0 = (239.6\times 0.8 + 6.955)^2 + (239.6\times 0.6 + 111.28)^2$

= 323.27 V

$\% R = \frac{323.27 - 239.6}{239.6} \times 100 = 34.92\%$

The power angle is the angle between E_0 and V. Referring to Fig. 6.17, we get

tan
(
36.86
$\circ + \delta) = \frac{BC}{OB} = \frac{V\sin(36.86^\circ) + 111.28}{V\cos(36.86^\circ) + 6.955} = 1.2836$

$\tan^{-1}(1.2836) = 52^\circ$

$\delta = 52 - 36.86 = 15.14^\circ$ **Example 6.18Example 6.18**φ, 50 Hz star connected 1000-kVA, 2300 V alternator gives a short circuit current of 251 A at a certain field current. The open circuit voltage for the same excitation is 478.3 V. The armature resistance between a pair of terminals is 0.2 . Find the full load regulation at (i) u.p.f. (ii) 0.7 p.f. lag (iii) 0.85 p.f. lead.

Solution:

V
=
2300 1327.9 V; $I_{FL} = \frac{1000\times 10^3}{\sqrt{3}\times 2300} = 251$ A $\sqrt{3}R_s = 1.1Z_s = 478.3/$
251

Between two terminals, two phase windings are connected in series. $\therefore R_a = \frac{0.2}{2} =$ 0.1 /phase.

$X_s = 1.1^2 - 0.1^2 = 1.095$ $IR_a = 251 \times 0.1 = 25.1$ V $IX_s = 251 \times 1.095 = 274.845$ V

(i) *u.p.f.*

$E = (1327.9 + 25.1)^2 + (274.845)^2$ 1380.63 V=

$\%R = \frac{1380.63 - 1327.9}{1327.9} \times 100 = 3.97\%$

(ii) *0.7 p.f. lag*

$E = (1327.9 \times 0.7 + 25.1)^2 + (1327.9 \times 0.714 + 274.845)^2$ 1551.44 V=

$\%R = \frac{1551.44 - 1327.9}{1327.9} \times 100$

= 16.83%

(iii) *0.85 p.f. lead*

$E = (1327.9 \times 0.85 + 25.1)^2 + (1327.9 \times 0.526 - 274.845)^2$ 1229.13 V=

$\%R = \frac{1229.13 - 1327.9}{1327.9} \times 100 = -7.43\%$

6.7. Synchronization

The process of connecting an alternator in parallel with other alternator or with the busbars is called synchronization. For proper synchronization the following conditions must be satisfied:

• The terminal voltages of the two alternators must be same.

• The frequency of the two alternators must be the same. Since, $N = \frac{120f}{p}$, this implies that the two alternators must be running at same speed.

• The phase sequence of the two alternators must be the same.

The same conditions are to be met for synchronization with the bus bar too.

Questions

1. Describe the various parts of the alternator.
2. Explain the terms (i) pitch factor (ii) distribution factor.
3. What is a short-pitch winding?
4. Derive the expression for the induced e.m.f. of an alternator.
5. What is the effect of harmonics on K_c and K_d?

6. A 4 pole alternator has a flux per pole of 0.1 Wb. It has 4 slots/pole/phase and 4 conductors/slot, with a coil spam of 150°. What is the line voltage? [**Ans:** 1366 V]

1366 V] φ, 16-pole alternator has 90 slots, each containing 12 conductors. If the speed is 600 r.p.m. and flux per pole is 0.1 Wb, calculate the line e.m.f. when the

speed is 600 r.p.m. and flux per pole is 0.1 Wb, calculate the line e.m.f. when the phases are (i) star-connected (ii) delta connected. Assume the winding factor is 0.96 and the flux is sinusoidally distributed.

Ans: (i) 6.93 kV (ii) 4 kV

(i) 6.93 kV (ii) 4 kV φ, 16-pole alternator has a star connected winding with 144 slots and 10 conductors/slot. The flux/pole is 0.04 Wb and speed is 375 r.p.m. Find the frequency and line voltage.

Ans [50 Hz; 3525 V]
9. Why is there a drop in the terminal voltage when an alternator is loaded? 10. Draw the phasor diagram of an alternator on-load for u.p.f.; lagging p.f. and leading p.f.

11. Define voltage regulation.
12. Explain the O.C.C. and S.C.C. of an alternator?
13. What is synchronous impedance of an alternator? Is it a constant?
14. How is the synchronous impedance of an alternator determined?

15. Explain the synchronous impedance method of determining the regulation of an alternator.

16. If a certain field current in an alternator gives a current of 150 A on short-circuit and a terminal voltage of 900 V on open-circuit, find the internal voltage drop with a load current of 60 A.

[**Ans:** 360 V]
17. What is synchronization?

17. What is synchronization?

φ, 50 Hz, 2000 kVA, 2300 V, star-connected alternator gives a short-circuit current of 600 A for a certain field excitation. For the same excitation the open circuit voltage is 900 V. The resistance between a pair terminals is 0.12 . Find the full-load regulation at (a) u.p.f. (b) 0.8 p.f. lag (c) 0.8 p.f. lead. **Ans:** (a) 7.3% (b) 23.8% (c) −13.2%

19. The full load current of a 2000-V single-phase alternator is 100 A. On short circuit a field current of 2.5 A produces 100 A. An e.m.f. of 500 V is produced on open circuit by a field excitation of 2.5 A. The armature resistance is 0.8 .

Determine the voltage regulation at 0.8 p.f. lead.

Ans: −8.9%

20. In a single-phase alternator a field current at 3 A produces an armature current of 250 A on short circuit and an induced e.m.f. of 1500 V on open circuit. What is the terminal voltage when a load of 250 A at 6.6 kV, 0.8 p.f. lag, is switched off. Also calculate the regulation.

Ans: 7,898 V, 197%.

CHAPTER - 7

Single Phase Induction Motors



Learning objectives

Double field revolving theory Split phase induction motor Capacitor-start motors
Capacitor-run motors
Shaded-pole motor
AC servomotors
Synchros, working and
application
Stepper motor
Characteristics and working of stepper motors

7.1. Introduction

In Chapter 5 we have studied the three-phase induction motors. Single phase motors are very commonly used to drive various electric appliances such as washing machines, mixers, drills, fans, vacuum cleaners, air conditioners *etc.* They are generally fractional H.P. motors. The basic principle of single phase motors are same as that of three phase machines. The induction principle is extensively employed because of the simplicity of the rotor and because it avoids commutation.

The performance requirements of various applications vary so widely, that there are different types of single phase motors designed to meet specific applications. The single phase motors broadly fall into the following types:

(i) Induction motors
(ii) Repulsion motors
(iii) AC series motors
(iv) Synchronous motors.

7.2. Single Phase Induction Motor

This motor is similar in construction to the three phase induction motor. The salient features of the construction are
(i) The stator is provided with a single phase winding called the *main winding*.
(ii) Since, the motor is not self starting, a starting mechanism has to be provided.
(iii) If an auxiliary winding is used, a centrifugal switch is needed to cut it out after starting.
(iv) The rotor is cage type.
The single phase induction motor is shown in Fig. 7.1

When fed from a single-phase supply, the starter winding produces an alternating flux, unlike the revolving (rotating) flux produced by a three phase winding. The alternating flux does not produce any torque as explained by the double field revolving theory. Hence, single-phase induction motors are not self-starting and need a mechanism to start the motor.

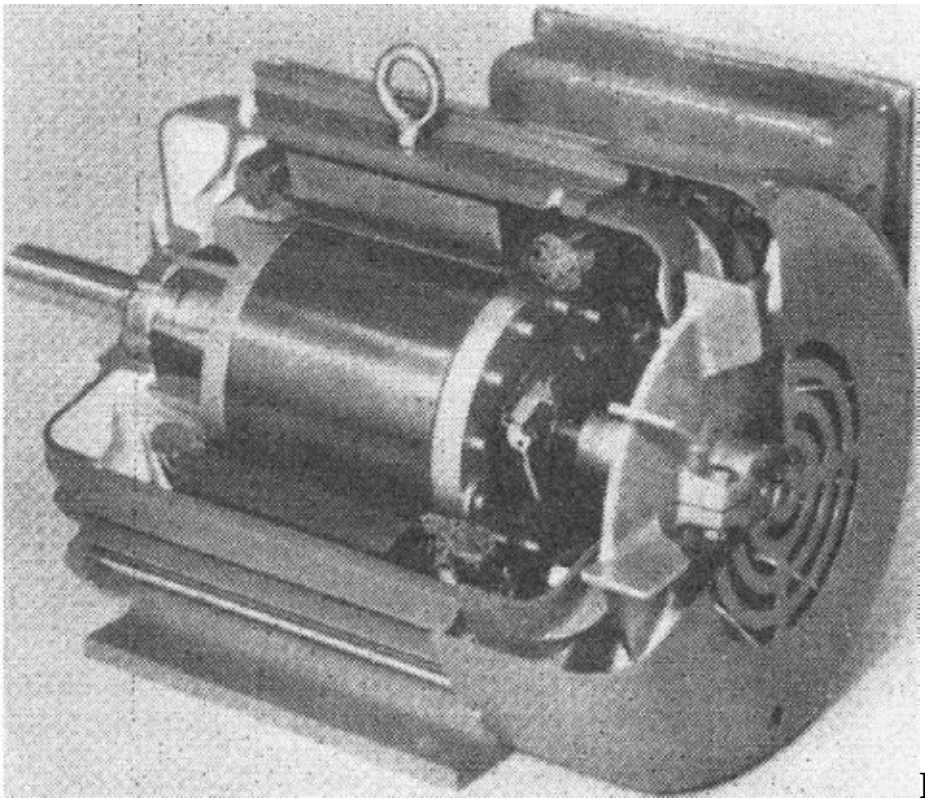


Figure 7.1 *Cut out view of single phase induction motor.*

7.2.1. Double field revolving theory

Any sinusoidally varying alternating quantity can be represented by two vectors, each having half the magnitude of the original and rotating in opposite directions, with a synchronous speed

$N_s = \frac{120f}{P}$. (7.1)

In (7.1) f is the frequency and P is the number of poles. This can be explained by considering a step by step revolution of the vectors.

Step 1. Consider an alternating flux with a maximum value φ_m. It can be split into two components φ_A and φ_B with amplitude $\varphi_m/2$ revolving in anticlockwise and clockwise directions. Start at the position shown in Fig. 7.2(a). Here, angle$\theta = 0^\circ$ and flux = φ_m.

Step 2. Consider after the rotation ofφ_A andφ_B through an angleθ, in opposite directions, as shown in Fig. 7.2b. The resultant flux is

$2 \frac{\varphi m}{2} \cos \theta . \times 2 = \varphi_m$

Step 3. After a rotation of 90° (quarter cycle)φ_A andφ_B will occupy the positions shown in Fig. 7.2c. Here, $\varphi = 0$:

φA =$^{\varphi m}$ φ_{A2}

$\varphi = \varphi_m$ $^{\theta}$ $\varphi_m \cos \theta$

$_{\varphi B =}$ φm φ_B

(a)^2 (b)

φA φB

φ_A φ

=

0

φ

^{m} φ_B φ = 0

(c)$^{\varphi B}$ (d) (e)$^{\varphi A}$ **Figure 7.2** *Double-field revolving theory.*

Step 4. After a rotation of 180° (half cycle) φ_A and φ_B occupy the positions shown in Fig. 7.2d. Here resultant flux $\varphi = \varphi_A + \varphi_B = -\varphi_m$.

Step 5. After three quarter cycle rotation,φ_A andφ_B reach the positions shown in Fig. 7.2e. Here, $\varphi = 0$.

If we plot the values of resultant flux againstθ betweenθ = 0° andθ = 360°, then we obtain the curve shown in Fig. 7.3.

It can be seen that we get a cosine curve. Hence, the alternating flux can be looked upon as being made up of two revolving fluxes, each with half the amplitude, revolving synchronously in opposite directions.

Each of the two components, cuts the rotor, inducing an emf and producing its own torque. These two torques, obviously, are oppositely directed and equal in magnitude. The net torque is the difference between the two torques.

φ

0° 90° 180° 270° 360°

θ **Figure 7.3** *Resultant flux.*

The slip of the flux component rotating in the same direction as rotor, called forward flux, has a slip s. Then the flux component rotating in the opposite direction, called backward flux, will have a slip 2–s. The torque components are given by

Forward torque,
$T_f = KI_2^{2R_2}(7.2)_s$
Backward torque
$T_b = -KI_2^{2R_2}(7.3)_{(2-s)}$
where I_2 is the current in the rotor and R_2 is the rotor resistance. The total torque $T = T_f + T_b$. All the three are shown in Fig. 7.4.

At standstill s = 1 and$(2-s) = 1$. Hence the forward and backward torques are equal and net torque is zero. Therefore, there is no starting torque. It is for this reason that single phase induction motors are not self-starting. If the rotor is somehow given an initial start in one direction, the net torque will accelerate the rotor to full speed, in the same direction.

Example 7.1 Prove that the slip of the backward flux is $2-s$, where s is the slip of the forward flux.
Solution: Let the speed of the rotor be N rpm. Then the slip, s, of the forward flux is given by
$s = \frac{N_s - N}{N_s} = 1 - \frac{N}{N_s}$ or $\frac{N}{N_s} = 1 - s$
where N_s is the synchronous speed.
$Torque_{T_f}$
$T_{s=2_{s=1}\ s=0}$
T_b
Anticlockwise Clockwise **Figure 7.4** *Torque components.*
The backward flux rotates with same speed in opposite direction. Hence, we can take the speed to be −N r.p.m. The slip is,
$s = \frac{N_s - (-N)}{N_s} = 1 + \frac{N}{N_s} = 1 + (1 - s) = 2 - s.$

7.3. Split-phase induction motor

Since the single phase induction motor is not self starting, it must be provided with a starting mechanism, so as to create a revolving flux, which will produce the necessary starting torque. The idea used in most methods are focused on temporarily converting it to a two-phase motor during the starting period.

To achieve this, the starter is provided with an extra winding, called the *starting or auxiliary winding*, in addition to the main winding. The phase difference of the currents in the two stator windings is very large, close to 90◦. These currents produce a revolving flux, and hence make the motor self starting.

In split phase machines, the main winding has a low resistance and high reactance. The starting winding has a high resistance and low reactance. The resistance can be increased by simply connecting a resistor in series as shown in Fig. 7.5(a).

(a) Split-phase windings

(b) Phasor diagram

Figure 7.5 *Split phase induction motor.*

The phasor diagram is shown in Fig. 7.5b. It can be explained as follows:

• V is the reference vector.

• The current in the starting winding, I_s, is almost in phase with V, since it is highly resistive.

• The current in the main winding,I_m, lagsV by almost 90◦, since it is highly inductive.

• The net current is given by the phasor sum ofI_s and I_m.

The phase angle between V and I_s is made as large as possible, since the starting

torque is proportional to sinα.

A centrifugal switch ‘S’ is connected in series with the starting winding and is located inside the motor. When the motor reaches 70%–80% of its full-load speed, the switch opens, disconnecting the starting winding. The starting torque is 1.5 times to 2 times the full load torque and the starting current is 6–8 times the full-load current. They are extensively used in fans, blowers, centrifugal pumps, washing machines, domestic appliances *etc.* Such motors are also called resistance–start split-phase motors. They have ratings in the range $\frac{1}{20}$ to $\frac{1}{3}$ H.P.

Example 7.2 A resistance split-phase motor is rated $\frac{1}{4}$ H.P., 1725 rpm, 115 V, 60 Hz.
When rotor is locked, the test results were as follows:
Main winding: V = 23 V; I_m =4A; P_m = 60 W
(starting winding) Auxiliary winding: V = 23 V; I_s = 1.5 A; P_s = 30 W. Draw the phasor diagram and find the current drawn from the line at 115 V under locked rotor.
49.6
°
29.6° V
I_s
I_m
Figure 7.6 *Example 7.2.*
Solution: The locked rotor test is similar to the locked rotor test in three phase induction motors. The rotor is locked and prevented from rotating. We calculate the phase angle$\varphi_m$between Im and V.

$\cos\varphi m = \frac{Pm}{VIm} = \frac{60}{23\times 4} = 0.65$

$\varphi_m = 49.6^\circ$

I_m lags V by 49.6°.
The angle φ_s, between I_s and V is given by

cos
φ
s
=

$P_s = 30$

$23\times 1.5 = 0.87 VI_s$

$\varphi_s = 29.6^\circ$

I_s lags V by 29.6°. The phasor diagram is shown in Fig. 7.6.

Total current = $I_m \angle \varphi_m + I_s \angle \varphi_s = 4 \angle 49.6^\circ + 1.5 \angle 29.6^\circ = 5.43 \angle 44.23^\circ$

The locked rotor current at 23V = 5.43A

The locked rotor current at $115^{115} = 27.15A.V = 5.43 \times_{23}$

7.4. Capacitor-start motors

The capacitor motor is identical to a split-phase motor, except that a capacitor is connected in series with the starting winding as shown in Fig. 7.7.

I Im

I_s Main

Starting winding

winding I_s

1φ $Vsupply_C$

I

S Rotor Im(a)(b)

Figure 7.7 *Capacitor-start induction motor.*

The capacitor is so chosen such that the angle α, between I_s and I_m is around 80°. Since torque is proportional to sinα, torque is increased. As a result, the current in the auxiliary winding is only about half that in a split-phase motors. Therefore, it gets heated less quickly.

The capacitor is electrolytic type and is designed for extremely short-duty service. When the motor reaches 75%–80% of full load speed, the centrifugal switch S opens. Owing to high starting torque and relatively low value of I_s, the capacitor-start motor is suited to applications involving frequent or prolonged starting periods. Their wide spread use is due to availability of small, reliable, low-cost electrolytic capacitors. However, they can be used only for short periods. They are guaranteed for not more than 20 periods of operation, each period not to exceed 3 seconds. They are built in sizes ranging from ^{1} H.P. to 10 H.P. Typical loads are compressors, large fans, pumps *etc.*[6]

7.5. Capacitor run motors

These motors are similar to capacitor start motors, except that the winding with the series capacitor is connected all the time as shown in Fig. 7.8.

Leaving the capacitor improves efficiency, power factor, overload capacity and leads to quieter running. Hence they are used in places like offices, hospitals *etc.* where silence is important. The starting torque is low and hence they are rated below 500 W.

The capacitor used is oil or paper type capacitor, since it needs to be of continuous-duty rating. Since same capacitor is used for starting and running condition, neither can be

I_s Auxillary$_{Main1\varphi}$
supply
C Rotor

Figure 7.8 *Capacitor run motor.*

optimum. Values used are in the range 2 to 20 μF. The starting torque is low because of low value of capacitor used. In a modification of the design, two capacitors are used, one for starting and a separate one for running. While starting, the two capacitors are connected in parallel, so that their combined capacitance is sum of the two. A centrifugal switch cuts off the starting capacitor after the motor reaches around 75% of its rated speed. This way both optimum starting and running torques can be achieved.

The direct ion of rotation can be reversed by interchanging the leads of either the auxiliary winding or the main winding. Capacitor-start motors cannot be reversed, under running conditions. Capacitor-run motors can be made to change direction of rotation while the motor is running.

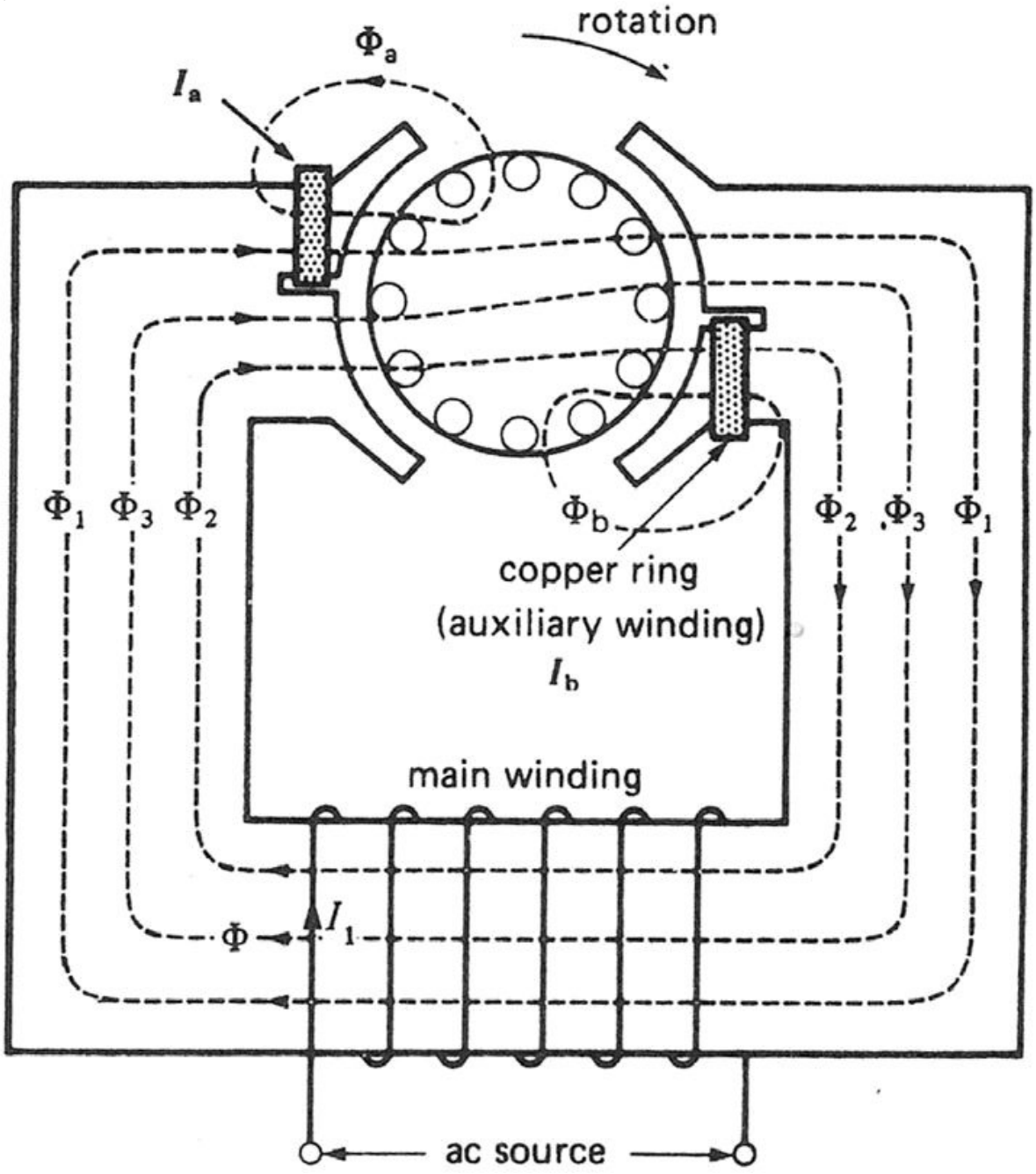
7.6. Shaded pole motor

Shaded-pole motors are very popular for low ratings of the order of 40–50W. These motors have salient poles on the stator and a squirrel cage rotor. A two pole and a four pole shaded-pole motor are shown in Fig. 7.9.

A single pole is shown in Fig. 7.9c. It has a slot cut at one third distance from one edge. Around the small part of the pole a short circuited copper ring, called the *shading coil*, is placed. This part of the pole is called the shaded part and the other part is called the unshaded part. The production of torque can be explained as follows.

• The stator winding is connected to an *a.c.* voltage, which makes an alternating

exciting current flow through it.
• When the current is increasing, an e.m.f. is produced in the shading coil, by transformer action in such a direction as to oppose the increase of exciting current. Hence, the flux
(a)



(b) (c)

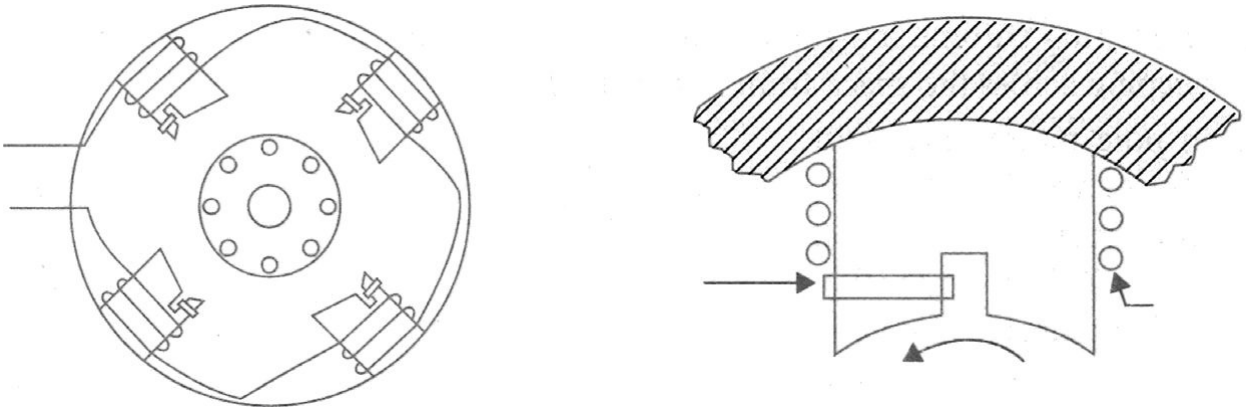


Figure 7.9 *Shaded-pole motor.*

mostly shifts to the unshaded part as shown in Fig. 7.10a. The magnetic axis is along the center of unshaded part.

• When exciting current is close to maximum value, rate of change is less and hence very low voltage is induced in the shading coil. The flux is more or less uniformly distributed over the pole face as shown in Fig. 7.10b. The magnetic axis is at the center of the pole.

• When the exciting current decreases, the voltage in the shading coil is produced so as to oppose the decrease in current. Consequently, the magnetic axis shifts to the middle of the shaded part as shown in Fig. 7.10c.

We can see that the magnetic axis moves from the unshaded to shaded part of the pole. This is equivalent to a revolving magnetic field. The rotor starts rotating in the direction from unshaded side to shaded part of the pole.

Magnetic axis
Magnetic axis
Crowding of flux in $_{(a)}$ unshaded part$_{(b)}$

Magnetic axis Uniform flux over

pole

(c) Flux crowded over shaded part

Figure 7.10 *Magnetic axis in shaded pole rotor.*

Though simple, these motors have low starting torque, very little overload

capacity and low efficiency. They are used in small fans, toys, hair dryers, clocks *etc.*

7.7. AC Servomotors

Servomotors , sometimes called *control motors*, are electric motors that are specially designed and built, primarily for use in feedback control systems as *output actuators*. Their power rating can vary from a fraction of a watt to a few hundred watts. Since, they are used in control systems, they need to have a high speed of response, which requires a low rotor inertia. These motors are therefore smaller in diameter and longer in length. They operate at low or zero speed and thus have a larger size for their torque or power rating

V
a
±90
°
High resistance cage rotor
Reference $_{\text{Vm } 0^\circ}$ phase

Figure 7.11 *AC servo motor.*

T NW-m

Low rotor High rotor resistance resistance

ωsyn ωM

Figure 7.12 *Torque-speed characteristic of ac servo motor.*

than conventional motors of similar rating. They are used in a variety of applications such as robots, radars, computers, tracking systems, machine tools and process controllers.

Most high power servo motors are *d.c.* servomotors. *A.C.* servo motors are used for low power applications. They are robust and have lower inertia than *d.c.* servomotors. However, they are non-linear and have lower-torque compared to *d.c.* servomotors of the same size.

Most *a.c.* servomotors used in control systems are of the two-phase squirrel-cage induction type. The frequency range is 60–400 Hz. A schematic diagram of a two phase *a.c.* servomotor is shown in Fig. 7.11 and the torque-speed characteristic in Fig. 7.12.

The stator has two distributed windings displaced 90° electrical, apart. One winding, called the *reference or fixed phase* is connected to a constant-voltage source, $V_m \angle 0^\circ$. The other winding, called the *control phase*, is supplied with a variable voltage of the same

Drag cup rotor
Stator

Shaft Stationary rotor core

Stator
Figure 7.13 *Drag-cup rotor construction.*

frequency as the reference phase, but phase-displaced by 90° electrical. The direction of rotation of the motor depends on the phase relation (leading or lagging) of the control phase voltage with respect to the reference phase voltage.

For balanced two phase voltages, $|V_a|=|V_m|$, the torque-speed characteristic of the motor is similar to that of three phase induction motor. The non-linear characteristic, when rotor resistance is low, is not acceptable in control applications. But, if the rotor resistance is high the torque-speed characteristic is linear over a wide speed range. Generally, the reference phase voltage is fixed and the control phase voltage is varied.

In low power control applications, a drag-cup rotor is used, to reduce the inertia. A thin cup of non-magnetic conducting material is used as rotor, as shown in Fig. 7.13. Because of the thin conductor, the rotor resistance is high, resulting in high starting torque. The stationary iron core in the center of the cup completes the magnetic circuit. A typical application of an *a.c.* servomotor is in the radar position control as shown in Fig. 7.14.

Two potentiometers are used. The reference potentiometer generates a voltage E_{ref} depending on the desired position θ_{ref}. The second potentiometer is connected to the shaft of the servomotor and produces a voltage E proportional to shaft position θ. The difference $(E_{ref} - E)$ is the error voltage and is proportional to the position error, $\theta_{ref} - \theta$. This error voltage is fed to a servo amplifier, which generates the necessary control voltage V_a, to produce a torque which would rotate the rotor to reduce the position error to zero.

θref θ
Radar
E_{ref} E
–
+ ∑ Servo Va ±90° amplifier
θ
V_m 0° **Figure 7.14** *Radar position control system.*

7.8. AC tachometer

A tachometer is a transducer which converts a mechanical position to an electric voltage. A drag-cup servomotor serves as a tachometer. The tachometer connections are shown in Fig. 7.15(a).

The drag-cup rotor is driven by a prime mover. The reference winding produces the reference fluxφ_{ref}. The quadrature output winding is connected to an*a.c.* voltmeter, which is calibrated directly in r.p.m.

The principle of operation can be explained referring to Fig. 7.15(b). Since the reference winding is highly inductive, its current lags the applied voltage by 90°, producing the flux φref. When the drag-cup is rotated clockwise (CW) by the prime mover, eddy currents are induced in the rotor producing a CW rotor flux. The phasor sum of φ_{ref} and rotor flux produces a resultant flux $\varphi_{out(CW)}$ which generates an induced *a.c.* voltage in the quadrature output winding. Similarly a rotation in counter clock wise (CCW) direction produces a rotor flux in CCW direction.

The magnitude of the rotor flux is a function of the rotor speed. The speed-output voltage characteristic is shown in Fig. 7.15c. At very high speeds saturation effect is produced. Thus the voltmeter, which reads the output voltage induced, can be directly calibrated in r.p.m., as the output voltage is linearly related to the speed. To ensure accuracy the upper limit of the calibrated voltmeter is kept below the speed at which saturation takes place.

Reference winding Drag cup φ_{rotor} (CW) φ_{rotor} (CCW) rotor
Input ac
φ_{out} (CW) $^{\varphi ref}$ φ_{out} (CCW)

Output (b) Iref winding

Prime
mover V_0(ac)

V_0
(a)
Speed (CCW) Speed (CW)
(c) V_0(ac) **Figure 7.15** *Tachometer.*

7.9. Synchros

Synchros are *a.c.* electromagnetic devices that convert a mechanical displacement into an electric signal. Synchros are widely used in control systems for transmitting shaft position information or for maintaining synchronism between two or more shafts. They are used primarily to synchronize the angular positions of two shafts at different locations, where it is not possible to make a mechanical inter connection of the shafts.

The synchro control transmitter (CX) has a balanced three-phase stator winding similar to the stator winding of a three-phase synchronous machine. The rotor is of the salient pole type using dumb bell construction with a single winding as shown in Fig. 7.16a.

S_2 α
e2n
Balanced three$_{\text{erphase stator}}$
winding
Rotor
winding$_{e3n}$ e_{1n}
S_3
(a) Construction (b) Schematic^{S1} **Figure 7.16** *Synchro control transmitter.*

If a single-phase *a.c.* voltage is applied to the rotor through a pair of slip rings, an alternating flux field is produced along the axis of the rotor. This induces voltages in the stator winding by transformer action. If the rotor is aligned with the axis of stator winding 2, flux linkage of this stator winding is maximum and this position is defined as the *electrical zero* position of the rotor. In Fig. 7.16(b), the rotor is shown displaced from the electrical zero by an angle α.

A single phase *a.c.* voltage is applied to the rotor winding and the rotor is

displaced by an angle α from its electrical zero position, and held fast. The rotor voltage is given by
er = √2Er sinωt

Single phase voltages having same frequency as the rotor voltages are induced in each stator phase by transformer action. The value of the induced voltage depends on the coupling between the stator phase and the rotor winding.

If
K

=
effective stator turns effective rotor turns then the induced stator voltages are given by

e 1
e2
e3

The r.m.s. voltages are
= $\sqrt{2}KE_r \sin\omega t \cos(\alpha + 120^\circ)_n$ = √2KEr sinωt cosαn = √2KEr sinωt cos(α − 120◦)n

$E\,1_n = KE_r \cos(\alpha + 120^\circ)$ $E2_n = KE_r \cos\alpha$
$E3_n = KE_r \cos(\alpha - 120^\circ)$

Each rotor position corresponds to a unique set of stator voltages. This is exploited in identifying the rotor shaft position.

The synchros control receiver (CR) has the same basic structure as the control transmitter. However, it is provided with a mechanical viscous damper to permit the receiver rotor to respond without causing rotor to overshoot its mask.

The synchro control transformer, is shown in Fig. 7.17(a). It has a uniform air gap because of the cylindrical shape of the rotor. This is important, to maintain a constant impedance irrespective of the rotor position, when the rotor terminals are connected to an amplifier. The stator has a balanced three phase winding.

The electrical zero is defined as the position of the rotor that makes coupling with stator winding equal to zero. This is as shown in Fig. 7.17(b). The impedance per phase of stator winding is greater in the transformer than in the

impedance per phase of stator winding is greater in the transformer than in the transmitter, allowing several control transformers to be fed from a single control transmitter.

7.9.1. Applications of synchros

Synchros are used to transmit torques over a long distance without a rigid mechanical connection. One such arrangement is shown in Fig. 7.18.

The rotor windings of the synchro transmitter and synchro receiver are connected to the same single-phase *a.c.* supply and the stator windings are connected as shown. Let the rotor of transmitter be displaced by an angle α, and its winding connected to the single phase *a.c.* by closing switchSW_1. Voltages are induced in stator windings of CX, which

S_2
3φ winding
Startor Rotor
1
φ
winding
S
3
S_1
(a) Construction (b) Schematic **Figure 7.17** *Synchro control transformer.*
makes currents flow in CR because of the electrical connections. These currents produce a flux field.

If the rotor of CR is now connected to *a.c.* voltage by closingSW_2, a field flux is created along the CR rotor axis, which interacts with the flux field produced by the stator windings to produce a torque. This torque, rotates the CR rotor to the same displacementα, as CX. In this position the stator voltages of CR have magnitude and phase as CX and hence no current flows and no torque is produced.

If now the transmitter rotor is displaced to a new position, the receiver rotor will take a similar corresponding position. The displacement of rotor of CR would also cause a similar displacement of rotor of CX.

7.10. Stepper motors

A stepper motor is also called a stepping motor. It rotates by a specific number of degrees, 2°, 2.5°,5°,15°,45°, etc., in response to an electrical pulse. It can be called an incremental actuator which converts digital pulse inputs to analog output shaft motion. It is therefore

αα
S_2 S_2
CX CR
S_1 S_1 S_3 S_3
Transmitter Receiver SW_1 SW_2
a.c. supply
Figure 7.18 *Synchro transmitter and synchro receiver arrangement for shaft alignment.*

used in digital control systems. Typical applications are printers, tape drives, disk drives, machine tools, X–Y recorders and robotics. Stepper motors are available with a few steps per revolution to high values of around 400 steps per revolution.

Two types of stepper motors are widely used:
• Variable-reluctance type
• Permanent magnet type.

7.10.1. Variable reluctance stepper motor

A four-phase two pole, single stack, variable reluctance stepper motor is shown in Fig. 7.19. When the stator phases are excited with *d.c.* current in proper sequence, the resultant air gap field steps around and the rotor follows the axis of the air gap field due to the

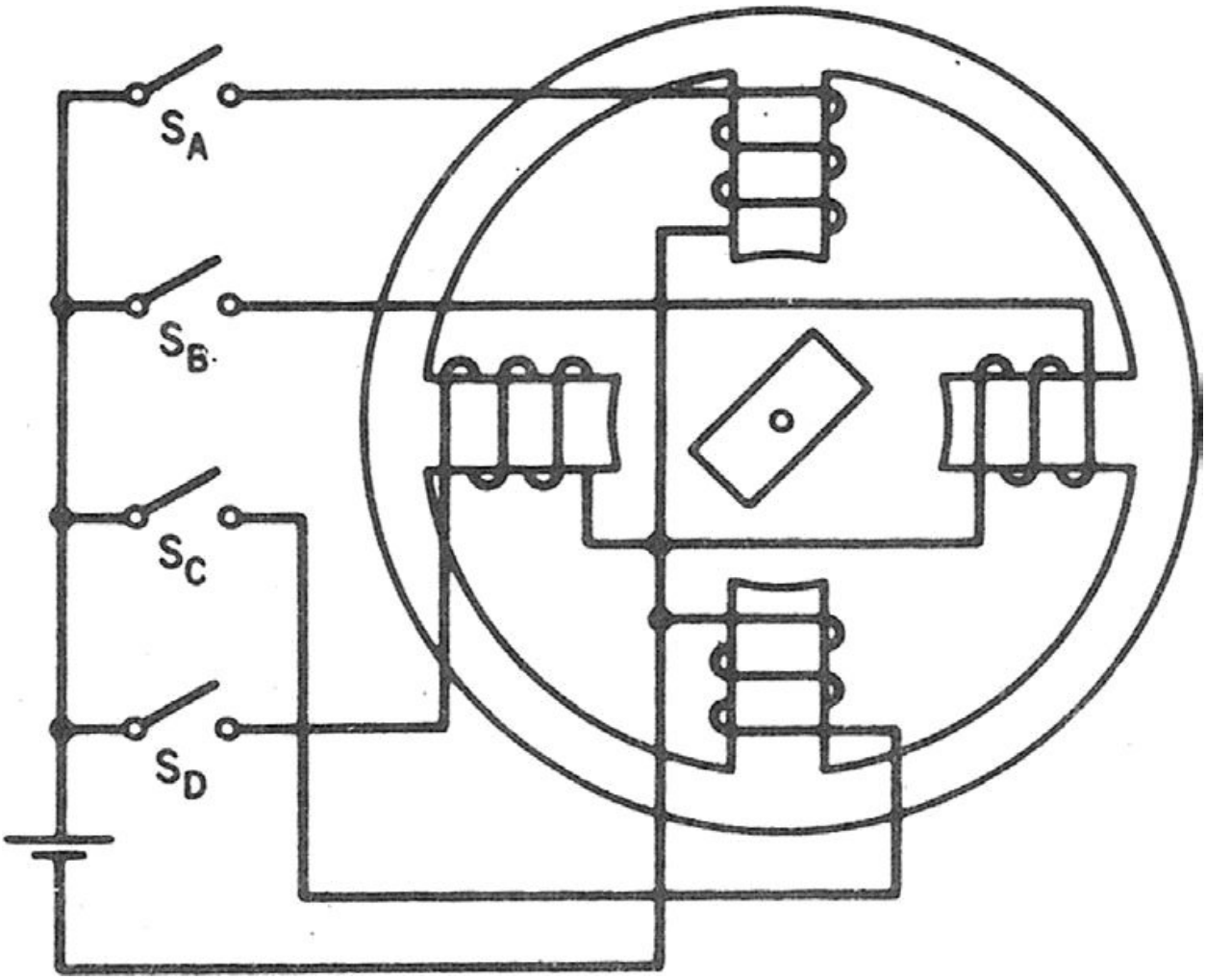


Figure 7.19 *Stepper motor*.

Table 7.1
ABCD

1 000 1 −→ Energized
1 100 0−→ Not energized
0 100
0 110
0 010
0 011
0 001
1 001
1 000 …

reluctance torque produced. The reluctance torque is produced because of the tendency of the ferromagnetic rotor to align itself along the direction of the

tendency of the ferromagnetic rotor to align itself along the direction of the resultant field.

Consider Fig. 7.20(a). Here only winding A is excited and the rotor aligns along axis of phase A. Next, both windings A and B are excited, which makes the resultant mmf axis move 45° in the clockwise direction. The rotor aligns along this axis as shown in Fig. 7.20(b). Next only coil B is energized, when the rotor aligns along its axis as shown in Fig. 7.20(c). At each transition of the excitation, the rotor moves through 45°. The sequence of switching can be shown in the form of a table as shown in Table 7.1.

i_a A Φ_A
D B
C
i_a Φ_AA
i_b Φ_R 45° Φ_B D B
C
A
i_b
Φ_B D B
C

Figure 7.20 *Excitation of stepper motor.*

The direction of rotation can be reversed by reversing the sequence of switching the windings, *i.e.* A A+D, D, D+C, *etc.* Each switching, causes the rotor to move 45°. To obtain smaller steps, multipole rotor construction is used. A four-phase six pole stepper motor is shown in Fig. 7.21.

A 15°$_{45°}$
$_{P1}$ 30°$_{P6\ P2}$

D B
P_5 P_3
P_4

C

Figure 7.21 *Multipolar rotor construction.*

The working is explained as follows:

• When phase A winding is excited, P_1 aligns with axis of phase A.
• Next phase A and B are excited. The resultant mmf moves 45◦.P_2, nearest to the new mmf axis, pulls to align with it.
• Motor steps in anticlockwise direction by 15°.
• A is de-excited. The mmf axis, along axis of B, pulls P_3 in line. So the winding sequence is A, A+B, B, B+C, C.... The rotor rotates anticlockwise in steps of 15°.

7.10.2. Permanent magnet stepper motor (PMSM)

It has a construction similar to the variable-reluctance type, but the rotor is made of a permanent magnetic material. It is shown in Fig. 7.22.

The motor is a two-pole permanent magnet stepper motor. The rotor poles align with two stator teeth (or poles) depending on the excitation. Figure 7.22, shows alignment when phase A is excited, with a positive current. If excitation is switched to phase B, the rotor steps by 90◦. The current polarity decides the direction in which the motor moves. A switch over to positive current will take the rotor forward clockwise by 90◦, a negative current will take the rotor 90° in anticlockwise direction.

PMSM have higher inertia and are hence slower than variable-reluctance stepper motors. The maximum pulse rate is around 300 pulses per second, whereas it can be around 1200 for a reluctance motor. However, PMSM produce more torque per ampere of stator current.

B
NS
*i*A
AA'
B' **Figure 7.22** *Permanent magnet stepper motor*.

7.10.3. Parameters affecting stepper motors

Torque versus current

The torque developed by a stepper motor depends upon the current. Figure 7.23 shows the characteristic.

The torque the motor can exert while moving from one position to the next, is called the *pull-over torque*. When the motor is at rest, a holding current must continue to flow in the last winding that was excited, so that the rotor remains locked in place.

Start–stop stepping rate

There is an upper limit to permissible stepping rate if the rotor stops at each step and starts again at next application of pulse. If the pulse rate is too fast, the rotor will be unable to accurately follow the pulses and steps will be lost. To maintain synchronism, the rotor must settle down before advancing to next position. So in such operations the speed is limited to ^{1000} = 167 steps per second, since the rotor needs at least 6 ms to be stable in$_6$
a step. Higher the load torque, lesser the number of steps.

N − M 4 Torque
3

2
Rated pulse current

1
2 4 6 8 10 12
Current (A) **Figure 7.23** *Stepper motor characteristic.*
Slow speed

A stepper motor can be made to run at a uniform speed, without stopping in between steps. This is called *slewing*. The inertia effect is absent and the motor can carry a greater load torque when it is slewing.

A stepper motor carrying a load, cannot suddenly go from zero to a stepping rate of 2000 sps. Similarly, a motor that is slewing cannot abruptly be halted. The motor must be gradually accelerated or deccelerated. This process is called ramping. The ramping phase is completed in a fraction of a second.

Questions
1 What is double field revolving theory? Explain briefly and use the theory to explain why the single phase induction motor is not self starting.

2 Explain the working of a split-phase induction motor.
3 Briefly describe the working of a capacitor start motor. What are its characteristics?

4 What are capacitor-run single phase induction motors? What advantages do they possess over capacitor-start motors?

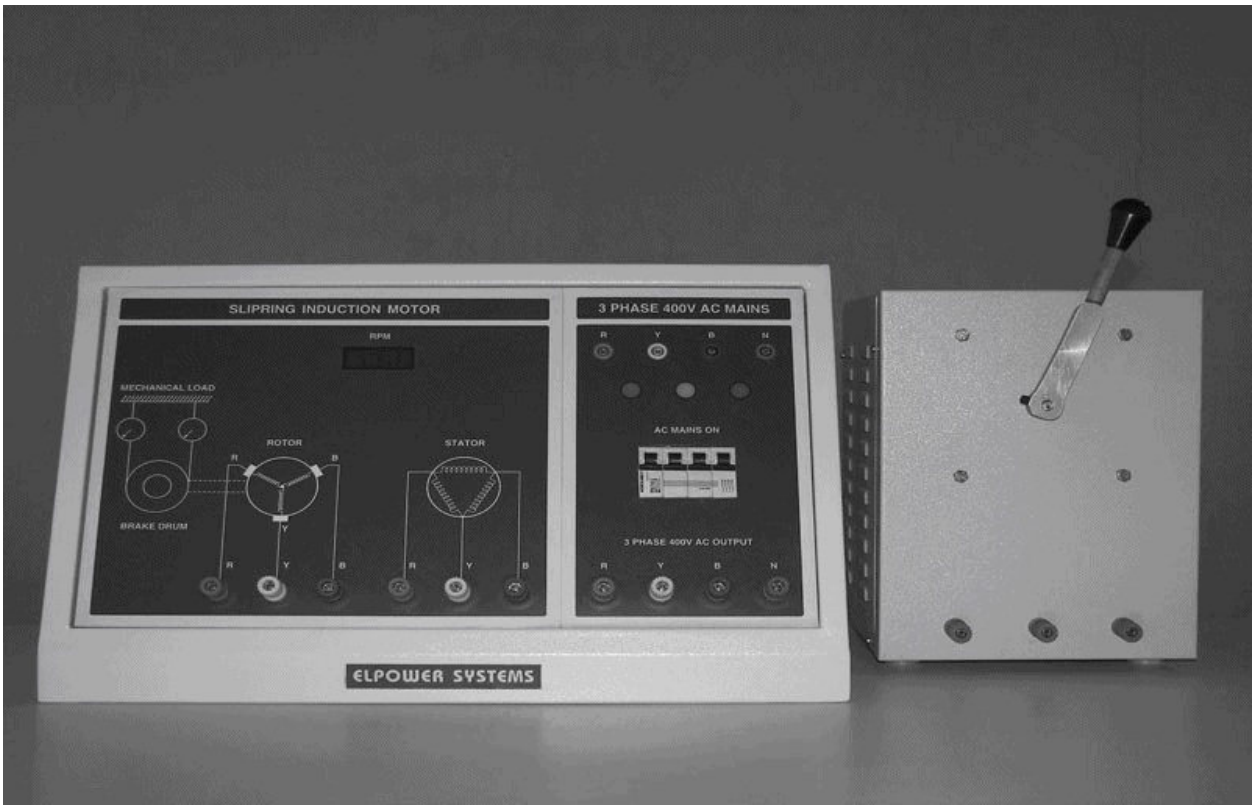
5 What is a shaded-pole motor? How does the shading coil help in producing a starting torque?
6 What are servomotors?
7 Explain with a diagram, the working and the torque speed characteristics of an *a.c.* servomotor.

8 Show how a drag-cup motor can be used as a tachogenerator.
9 What are synchros? Explain with a neat diagram any application of synchros.
10 What are stepper motors? What are the various types of stepper motors?

11 Give in a tabular form the sequence of pulses to be applied for a 4-pole stepper motor to get a step site of (i) 45° (ii) 90°.
12 What is slewing and ramping?

CHAPTER - 8

Lab Manual Electrical Machines Lab



633

In this chapter we discuss the conduction of some simple experiments on electrical machines. Before we discuss the experiments, there are certain common points to be noted for any machine. In any experiment, it is most important to choose the correct ratings of the meters and the rheostats. Otherwise the equipment may be damaged.

8.1. D.C. Shunt Motor

The rating of a typical d.c. shunt motor of Kirloskar make is as follows:

Rated capacity : 3.5 kW Rated I/P and O/P voltage : 220 V d.c. Armature current : 18.5 A Armature resistance : 0.35 Rated shunt field voltage : 220 V d.c. Rated shunt field current : 0.95 A Shunt field resistance : 174.1 Brake drum radius : 107 mm RPM : 1500

• SinceE_b = 0 at starting, a starting resistor has to be added in series with the armature. This can be done by connecting a three point starter or an external rheostat in series with the armature. The rating of the rheostat is important, since it has to limit the starting current to a maximum value equal to the rated current. In this case I_a = 18.5 A. The

starting current is equal to
I
s
=
V
Rs+Ra. SinceR_a is small we can neglect it. Hence, the minimum starting resistance
R
s
needed is given by
V
Ia(rated). Here,
R_s = 220 = 11.89 .18.5

We therefore need a minimum resistance of 11.98 , with a current carrying capacity of 18.5 A. Rheostats of high current rating are expensive. Instead we can use a rheostat of higher resistance value. We can use a standard 29 rheostat. The starting current would be 220 = 7.58 A. Hence we use a 29 , 10 A rheostat in series with the armature.29

The rheostat in series with the armature is always kept in “cut-in” (full resistance) position at the time of starting.

• The field current has to be decreased to decrease the flux and increase the motor speed upto rated value. We normally connect around 100 –230 rheostat in series with the field winding.

The rheostat in series with the field winding is completely “cut-out” (zero resistance) at the time of starting the motor.
• When the motor is loaded, the line current ammeter should be chosen to read the full load (rated) current. For the rating given we chose 0–20 A.
• The range of the voltmeter connected across the supply should be to read the rated voltage. We choose 0–250 V.

• The no-load current is around 10% of the rated current. Therefore, the no-load test we choose ammeters of a lower range, 0–2 A.
• The armature terminals brought out on the panel are normally marked A–AA or A^+–A^-. The field winding terminals are marked Z–ZZ or F^+–F^-.

8.2. D.C. Shunt Generator

The rating of the generator is similar to that of the d.c. shunt motor. Note the following in choosing the meters for the generator.
• The ammeter reading the load current should have a range capable of reading the rated current. We can use 0–20 A for ratings as in (8.1).
• The voltmeter reading the terminal voltage should have a range capable of reading the rated voltage. We use a 0–250 V meter.
• The field current ammeters have smaller range around 0–2 A. Using an ammeter of higher range will give inaccurate readings.
• A rheostat is connected in series with the generator field winding. We normally use rheostats around 230 –300 , for machines rated around 3–5 kW.
The field rheostat of the generator is initially “cut-in”. It is then gradually cut-out to build the rated voltage.

8.3. Three Phase Induction Motor

The typical rating of a three phase squirrel cage induction motor commonly used in the lab is as follows (Kirloskar make):

Rated voltage : 415 AC
Rated current : 7.9 A
Frequency : 50 Hz
RPM : 1430
Phase : 3
Output : 3.7 kW
Connected :
Insulation class : F
Make : Kirloskar
Brake drum radius : 107 mm

107 mm φ induction motor:

• The voltmeters measuring line-to-line voltage should have a higher range. We choose 0–500 or 0–600 V. If we read line-to-neutral voltage (in star connected systems) we can use a voltmeter of lesser range

V

LN
=
VLL
√3

• The autotransformer whose output is fed to the induction motor, must have a current rating about twice the rated current of the induction motor, to withstand high starting currents.

• The no-load current is about 40% of the rated full load current. Suitable range of ammeter has to be selected.
• In blocked rotor test, we apply a low voltage suitable range of voltmeter should be used.

• When rheostats are connected in series withe the slip-ring rotor, the current rating of the rheostat should be properly chosen.
• When wattmeters are used to measure no-load power, L.P.F. meters are used since the power factor is low. In blocked-rotor test and load test we use u.p.f. wattemeters.

8.4. Alternator
8.4. Alternator
φ alternator is as follows:

Rated O/P voltage : 415 V Current : 6.9 A RPM : 1500 Frequency : 50 Hz Phase : 3
Excitation : 1.9 A Rated KVA : 5.0 KVA Connection : Y
p.f. : 0.8 Make : Kirloskar

• The alternator field is d.c. Hence d.c. ammeters are used to measure the field current.
• The voltmeter should be of sufficient range to measure the line-to-line voltage of 415 V. Use 0–500 V (a.c.).

8.5. Transformer
The typical rating of a transformer is as follows:
Power : 1 kVA
Votage : 230 V/110 V
Either of the coils can be used as primary. The following points are to be noted.
• The polarities of the coils must be identified correctly (refer experiment 1)

•
Full load primary current
=
kVA
V1
•
Full load secondary current
=
kVA
V2
• No-load current is around 10%–12% of rated fullload current.
• L.P.F. wattmeters are used for no-load test and u.p.f. wattmeters for short-circuit test.

8.6. Loading Rheostats

These are used to load transformer, generator or alternator. The loading rheostats have resistors connected in parallel. Every time a load switch is closed, one resistor is added and the load current is increased. Another common loading arrangement is the lamp-load. When loads are used it is important to see that the rating is sufficient to load the machine to its rated value.

8.7. General Precautions

• Ensure that supply is off, before you make a connection.
• All connections should be made tightly.
• Choose proper ranges of meters and rheostats.
• Ensure rheostats are in correct positions before starting the experiment.

• Arrange the layout of meters and rheostats properly. Ensure that two rheostats are not so close that they touch each other.

• Ensure that the motors are running in the correct direction.
• Do not wear metallic objects like metal watches, rings, chains, bracelets *etc.*
• Wear shoes for safety.
• Do not wear loose garments near the machines.

EXPERIMENT 1
POLARITY TEST OF TRANSFORMER

Aim: To determine the polarity of a single phase transformer by conducting polarity test. **Apparatus required:**

Sl.No Apparatus Type Range Quantity
01 Voltmeter AC 0–300 V 01
02 Voltmeter AC 0–600 V 01

Rating of transformer: 110V/230 V, 1 KVA **Circuit Diagram:**
+ −

230 V, 50 Hz A.C.
Supply
V_1 0−300 V

P1 S1 −+ V_2

0−600 V P1 S2

Subtractive polarity Additive polarity

P2 S2 P2 S1
Procedure:

1. Connect the circuit as shown in circuit diagram.
2. Close supply switch (with auto transformer in zero position)
3. Apply some low voltage (say 100V)
4. If $V_2 > V_1$ then the transformer has additive polarity
5. If $V_1 > V_2$ then the transformer has subtractive polarity.

Result : The polarities are marked and the transformer has to be used with the correct polarities in any experiment.

EXPERIMENT 2
OPEN CIRCUIT AND SHORT CIRCUIT TEST ON A SINGLE PHASE TRANSFORMER

Aim: To predetermine the efficiency & regulation of a single phase transformer by conducting open circuit and short circuit tests.
Apparatus required:

Sl.no Apparatus Type Range Quantity
1 Voltmeter AC 0–250 V 1
2 Voltmeter AC 0–30 V 1
3 Ammeter AC 0–2 A 1

4 Ammeter AC 0–10 A 1
5 Wattmeter 150 V, 2.5 A, LPF 1
6 Wattmeter 75 V, 10 A, UPF 1
7. Auto transformer 0-230/270 V, 10 A 1

Ratings of transformer :230V/110 V; 1 KVA
Ratings of meters:

OC TEST : This test is conducted with the HV side open and meters are connected on LV side. The rating of the voltmeter is equal to the voltage rating of the LV side. The no-load current is about 10% of the rated full load current. Full load current on the LV side is kVA/Voltage, *i.e.* 1000/110 = 9.09 A. The wattmeter reads the iron loss. An LPF wattmeter is used, since power factor is low. Hence we can choose the meter ratings as follows:

Voltmeter : 0–150 V
Ammeter : 0–2.5 A
Wattmeter : 150 V, 2.5 A, LPF

SC TEST : This test is conducted with the LV side shorted and meters are connected on HV side. The applied voltage is low, about 10–15% of rated voltage. The full load current is passed through the HV winding. Full load current on the HV side is KVA/Voltage, *i.e.* 1000/230 = 4.348 A. The wattmeter reads the full load copper loss. An UPF wattmeter is used, since power factor is close to unity. Hence we can choose the meter ratings as follows:

Voltmeter : 0–30 V
Ammeter : 0–10 A
Wattmeter : 75 V, 10 A, UPF

Circuit Diagram:

230 V, 50 Hz A.C.
Supply
300 V, 2.5 A, LPF

A M L P1 S1
0−2 A
C V

V 0−300 V
Open circuited secondary winding

Open circuit test

230 V, 50 Hz A.C.
Supply
P2 S2

150 V, 10 A, UPF
A M L P1 S1
0−10 A
C V

V 0−30 V
Short circuited secondary winding

Short circuit test P2 S2
Procedure:
For open circuit test

1. Connect the circuit as shown in figure.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Adjust the output of the auto transformer such that rated voltage is applied to the transformer under test.
4. Note down the various meter readings.
5. Bring back the auto transformer to zero position and open the supply switch.

For short circuit test

1. Connect the circuit as shown in figure.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Slowly vary the output of the auto transformer such that rated current flows through the transformer under test.
4. Note down the various meter readings.
5. Bring back the autotransformer to zero position and open the supply switch.

Readings:

Open circuit test

Sl.no V_0 volts I_0 amps W_0 watts

1

Short circuit test

Sl.no V_{SC} volts I_{SC} amps W_{SC} watts

1

Calculations

1. Watt meter constant K = V
selected × Iselected × cos
full scale reading

2. $\cos_0 = {}^{W0}$, $I_W = I_0 \cos_0$, $I_\mu = I_0 \sin_{0,V0I0}$

3. R
0
=
^{V0},Xo $= {}^{Vo}$
I_w I_μ

4. Z ^{VSC},$R_{eq} = {}^{WSC}$,$X_{eq} = (Z_{eq})^2 - (R_{eq})^2$ $\cdot_{eq} = {}_{ISC\ I2}$
SC

5. % Regulation = $^{Isc\ Req\ cos\varphi \pm Xeq\ sin\varphi} \times 100_{V0}$
+ve for lagging P.F ; -ve for leading P.F.
x(
fullloadkVA
)
·
Cosφ
× 10
3

6. %Efficiency = $x(fullloadkVA)\cdot Cos\varphi \times 10^3 + W0 + x^2 W_{SC}$ where x is the fraction of full load output.

7. Power Factor at which regulation becomes zero:
The regulation can become zero only at leading power factor. × 100
IscReq cos − IscXeq sin = 0, IscReq cos = IscXeq sin tan = Req/Xeq, = $\tan^{-1}(Req/Xeq)$,

Tables :

Load P.F
% Regulation at full load Lagging P.F Leading P.F

01 0.2
02 0.4
03 0.6
04 0.8
05 1

Fraction of Efficiency
Sl. No. Full load x P.F=1 P.F = 0.8 01 0.25
02 0.5
03 0.75
04 1

Expected graph:
UPF
0.8 LAG
Lead Lag Load
Precautions:

1. Before closing switch ensure that the auto transformer is in minimum position.
2. Make all circuit connections tightly.
3. Ensure that the meter readings do not exceed their maximum values.

EXPERIMENT 2B
Aim : To determine the equivalent circuit of the transformer.

Procedure: The OC and SC tests are performed as mentioned above. The values of Ro, Xo, Req and Xeq are calculated as described above. The values of Ro and Xo are referred to the LV side, whereas the values of Req and Xeq are referred to the HV side. It is necessary to transfer them to the LV side. This is done as follows:

K = V(HV)/V(LV)
Req1 (referred to LV side) = Req/K^2
Xeq1 (referred to LV side) = Xeq/K^2

The equivalent circuit is shown below.
Req1 Xeq1
Load referred V_1 R0 X0 to primary
EXPERIMENT 3

LOAD TEST ON A SINGLE PHASE TRANSFORMER

Aim: To determine the efficiency and regulation of a single phase transformer by direct load test.

Apparatus required:

Sl.no	Apparatus	Type	Range	Quantity
01	Ammeter	AC	0–10 A	02
02	Voltmeter	AC	0–300 V	02
03	Wattmeter		300 V, 10 A, UPF	02
04	Loading Rheostat			01

Rating of the transformer : 1 KVA, 110 V/230 V

Rating of meters : Rated voltage is applied. The current is applied in steps, from no-load to full load. Hence the ammeters should read the rated current. The ratings are chosen accordingly.

Circuit Diagram:

150 V, 10 A, UPF 300 V, 10 A, UPF 230 V, 50 Hz_A M $L^{P1\ S1}{}_A$ M L $^{A.C.}$ 0−10 A 0−10 $A_{Supply\ C\ V\ C\ V}$
V 0−30 V V 0−30 V
P2 S2

Procedure:

1. Connect the circuit as shown in Fig.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Adjust the output of the auto transformer such that rated voltage is applied to the transformer under test. Note down the meter readings.
4. Vary the load in steps (till rated load) and in each step note down the meter readings. 5. Remove the load, bring back the autotransformer to zero position and switch off the supply.

Readings :

Sl.no V_1 I_1 W_1 V_2 I_2 W_2 %η

Calculations:

% Efficiency
=
Output power

Input power
$\% \eta = \frac{W2}{W1} \times 100\%$
$\% \text{ Regulation} = \frac{V2(\text{no load}) - V2(\text{load})}{V2(\text{no load})} \times 100\%$
Expected Gaph: p.f. = 1
Load
RESULT: Compare the results obtained with direct load test with those obtained by predetermination using OC and SC tests.

EXPERIMENT 4

SUMPNER'S TEST

Aim: To predetermine the efficiency and regulation of two identical transformers by conducting back to back test.

Apparatus required:

Sl.no	Apparatus	Type	Range	Quantity
01	Ammeter	AC	0–10 A	01
02	Ammeter	AC	0–2 A	01
03	Voltmeter	AC	0–600 V	01
04	Voltmeter	AC	0–300 V	01
05	Voltmeter	AC	0–30 V	01
06	Wattmeter	LPF	300 V, 5 A,	01
07	Wattmeter	UPF	150 V, 10 A,	01

Circuit Diagram:
300 V, 5 A, LPF 300 V, 10 A, UPF $_{A}$ M L P1 S1 L M_{A}
Switch 1
0−2 A 0−10 A
$^{C\ V\ V\ C}$V 0−300 V
Switch 2 230 V, 50 Hz
A.C.$^{P2\ S2}$ V0−30 V
Supply P1 S2

230 V, 50 Hz A.C.
Supply

SPST P2 S1 V
0−600 V

Procedure:
1. Connect the circuit as shown in Fig.

2. Ensure that the auto transformer is in zero position and SPST switch in open position and close the supply switch S_1.

3. If the voltage across the SPST switch is zero close the SPST switch. If not, open the supply switch S_1 and interchange the secondary terminals of any one transformer and close the supply switch S_1.

4. Close the supply switch S_2. Note down the readings on the primary side. 5. Slowly vary the output of the auto transformer such that rated current flows through the transformer.
6. Note down the different meter readings.V_0,I_0,W_{0T} are primary side reading,V_{sc},I_{sc}, W_{scT} are secondary side readings.
7. Bring back the auto transformer to minimum position and open the switch S_1,S_2 and SPST.

Tabular column:

Sl.no V_0 volts I_0 amps W_{0T} watts V_{sc} I_{sc} W_{scT}

01

Calculations:

1. Watt meter constant

K

=

Vselected × Iselected × cos

full scale reading

W 0T sc = WscT, Vsc = VscT2. W0 = 2,W 2 2

3. cos 0 = ^{W0} , IW = I0 cos 0, I_μ = I0 sin 0, $_{V0I0}$

^{V0} ,X_0 = ^{V0}4. R_0 = $_{Iw\ I\mu}$

5. Z ^{Vsc} ,Req = ^{Wsc} ,Xeq = $(Zeq)^2 - (Req)^2$ .eq = $_{Isc\ I2}$

sc

6. % Regulation = $^{Isc[Req\ cos\varphi \pm Xeq\ sin\varphi]} \times 100_{V0}$

+ve for lagging P.F

−ve for leading P.F

x(

full load kVA

)

.

cos

φ
× 10
3
7. % Efficiency = $\frac{X(\text{full load kVA}) \cdot \cos\varphi \times 10^3}{X(\text{full load kVA}) \cdot \cos\varphi \times 10^3 + W0 + x^2 WSC} \times 100$ x = fraction of full load.

TABULAR COLUMN FOR REGULATION:

% Regulation at full load Sl. No Load P.F Lagging P.F Leading P.F 01 0.2
02 0.4
03 0.6
04 0.8
05 1

TABULAR COLUMN FOR EFFICIENCY:

Fraction of Efficiency
Sl. No. Full load x P.F=1 P.F = 0.8 01 0.25
02 0.5
03 0.75
04 1

Expected graph:
%R UPF
η 0.8 LAG
01 $0_{\text{p.f.Lead Lag}}$
Output

EXPERIMENT 5
PARRALLEL OPERATION OF TWO DISSIMILAR SINGLE PHASE TRANSFORMERS

Aim: To determine the load shared by two dissimilar single phase transformers connected in parallel.

Apparatus required:

Sl.no Apparatus Type Range Quantity
01 Ammeter AC 0–5 A 02
02 Ammeter AC 0–10 A 01
03 Voltmeter AC 0–600 V 01
04 Voltmeter AC 0–30 V 01
05 Wattmeter 150 V, 10 A, UPF 01

Rating of the transformers: 230 V/230 V, 1 KVA;
230 V/230 V, 2 KVA.
Circuit Diagram:

P1 $S1_{A\ A}$
0−10 A 0−20 $A_{Switch\ 1}$
230 V, 50 $Hz_{P2\ S2A.C.\ 0-10\ ASupply\ P1\ S1\ A}$
Resistive load SPST
P2 $S2_{V}$
0−600 V
Procedure:
1. Connect the circuit as shown in Figure.

2. Keeping the SPST switch in open position close the supply switch. If the voltmeter across the switch reads double the secondary voltage of one transformer, open the switch and interchange the secondary terminals of any one transformer and then close the supply. If the voltmeter across the SPST reads the difference of the two secondary voltages, close the SPST switch.

3. Vary the load in steps and at each step note down the various meter readings.
4. Reduce the load and open the switch.
5. Conduct short circuit test to find the impedance of each transformer.

Readings:
Practical values Theoretical values Sl. no I_1 I_2 I I_1 I_2 I
I
1
=
I
Z1 +
$Z2_{,}$ Z1 $Z2^{I2\ =\ I}$Z1 + Z2 Z1 and Z2 are found from the short circuit test conducted on the individual transformers.
150 V, 10 A, UPF
230 V, 50 Hz_{A} M L P1 S1
$_{A.C.\ 0-10\ VSupply}$ C V Short circuited V 0−30 V secondary winding
P2 S2
S.C. Test (Transformer 1)

1. Connect the circuit as shown in Fig

1. Connect the circuit as shown in Figure.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Slowly vary the output of the auto transformer such that rated current flows through the transformer under test.
4. Note down the various meter readings.
5. Bring back the autotransformer to zero position and open the supply switch.
Sl.no Vsc1 volts Isc1 amps Wsc1 watts Z1 1
01
Z
1
=
Vsc1, $\varphi 1 = \cos^{-1}$ Wsc1 Isc1 Vsc1Isc1
S.C. Test (Transformer 2)
Sl. no V_{sc2} volts I_{sc2} amps $W_{sc2\ watts}$ $Z_{2\ 2}$ 01
Z
2
=
Vsc2, $\varphi 2 = \cos^{-1}$ Wsc2 Isc2 Vsc2Isc2
RESULT: Verify if the practical values match with the theoretical values.
EXPERIMENT 6
LOAD TEST ON THREE PHASE INDUCTION MOTOR
Aim: To plot the performance characteristics of three phase induction motor
Apparatus required:

Sl. no Apparatus Type Range Quantity
01 Ammeter AC 0–10 A 01
02 Voltmeter AC 0–600 V 01
03 Wattmeter 600 V, 10 A, UPF 02

Rating of the induction motor:
415 V, 5 HP, 1440 rpm.

Meter ratings: The meter ratings are chosen based on the rating of the induction motor. The voltmeter reads the line voltage. The rated full load current is calculated from the power rating and the voltage rating of the motor as follows

Rated current = Full load power/$\sqrt{}$3× Line voltage = (5× 746)/$\sqrt{}$3× 415 The

wattmeters are chosen to read the full load power. Upf meters are used. **Circuit Diagram:**

600 V, 10 A, UPF ML_A
415 V, 50 $Hz_{0-10\ A}$ R $^{A.C.}$ CV_{Supply}

R Y B
^{Y} 0−600 V
V CV_B

$^{3\text{-Phase ML}}$autotransformer
600 V, 10 A, UPF
Procedure:

1. Connect the circuit as shown in Figure.
2. Keeping the autotransformer in zero position, close the supply switch.
3. Vary autotransformer slowly and apply rated voltage to motor
4. Note down the no load meter readings and no load speed.

5. Vary the load in steps up to its rated load and at each step note down the meter readings, speed and S_1 and S_2
6. Release the load on the motor, bring back the autotransformer to zero position and open the supply switch.

Readings:
V W1 W2 S_1 S_2 N I/P o/p Torque volts I_{Amps} watts watts Kg Kg rpm watts watts %η pf % slip N-m
Radius of brake drum =Rm
Calculations: For each set of readings the calculations are done as below.

1. Input power = W1+ W2
2. T = 9.81× (S1 ∼ S2) ×RN-m
3. Output power = (2πNT)/60 Watts

4. % Efficiency $^{\text{Output power}}$ 100 $=_{\text{Input power}\times}$ 5. Synchronous speed Ns = 1500 rpm. % Slip = (Ns-N)/Ns× 100.

Expected graph:
% Slip p.f. T % $EFFI_{\text{Torque Torque}}$
% Efficiency

p.f.
% slip
Slip
Output

EXPERIMENT 7

SPEED CONTROL OF 3-PHASE INDUCTION MOTOR

Aim: To control the speed of three phase slip ring induction motor by
1. Stator voltage control and 2. Rotor resistance control

Apparatus required:

Sl. No
01
02

Apparatus Type Range Quantity Voltmeter AC 0–600 V 01
Ammeter AC 0–10 A 02

Circuit Diagram For: Stator voltage control and Rotor resistance control:
A A 415 V, 50 Hz 0−10 A 0−10 A $^{A.C.}$ V 0−600 V_{Supply}
R Y Y Y
B 3-Phase Rotor autotransformer Resistor Stator

Procedure For Stator Voltage Control:

1. Connect the circuit as shown in circuit diagram.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Adjust the output of the auto transformer such that rated voltage is applied for a particular value of rotor resistance R1.

4. Note down the voltmeter reading and record the speed.
5. Slowly decrease the voltage in steps and at each step note down the voltage and speed keeping the rotor resistance constant at R1 .
6. Repeat the steps 3, 4, 5 for a different value of rotor resistance R2.
7. Decrease the voltage to zero and open the supply switch.
8. Plot the graph of speed Vs voltage.

Tabular column:

Sl. No Voltage V (volts) Speed N (rpm) Procedure For Rotor Resistance

Control:

1. Connect the circuit as shown in circuit diagram.
2. Ensure that the auto transformer is in zero position and close the supply switch.
3. Adjust the output of the auto transformer such that rated voltage is applied V1.
4. Note down the speed and rotor current.

5. Vary the loading rheostats connected to rotor in steps and at each step note down ammeter readings and speed keeping the voltage constant at V1.

6. Repeat the steps 3, 4, 5 for a different value of voltage V2.
7. Decrease the voltage to zero and open the supply switch.
8. Plot the graph of speed Vs current

Tabular column:
Sl. No Current I (Amps) Speed N (rpm)
Expected graph:
Speed^{R1} R2 Speed_{V2} ^{V1} V1 > V2
R1 > R2
Voltage Resistance
EXPERIMENT 8
LOAD TEST ON SINGLE PHASE INDUCTION MOTOR
Aim: To determine the performance characteristics of single phase induction motor by conducting load test. **Apparatus required:**

Sl. no
01
02
03

Apparatus Type Range Quantity Ammeter AC 0–20 A 01
Voltmeter AC 0–75 V 01
wattmeter UPF 20 A, 75 V 01

Circuit Diagram:
300 V, 20 A, UPF 230 V, 50 Hz A
M A.C.
Supply^{0-}10 A C

V 0−300 V L $_{V\ CF}$ S1 S2

Auxiliary$_{\text{Mainwinding windingSC}}$
Single phase CF : Centrifugal Switch auto transformer SC : Starting Capacitor

Procedure: For load test

1. Connect the circuit as shown in Figure.
2. Ensure that the auto transformer is in zero position and close the supply switch.

3. Slowly vary the output of the auto transformer till rated voltage is applied to the motor (as shown in the name plate; *i.e.* 230V)
4. Note down the various meter readings (no load readings)
5. Load the motor in steps till the rated current is reached and in each step note down all the readings
6. Remove the load and bring back the autotransformer to zero position and open the supply switch.

Tabular column:
W N S1 S2 T
Sl. no V I (Input) rpm Kg Kg (N-m) O/P %η cos Slip

Calculations:
1. Watt meter constant
K
=
Vselected × Iselected × cos

full scale reading
2. T = 9.81 (S1 ∼ S2) r where r is the radius of the drum in mts
3. O/P = 2πNT/60
4. %efficiency = (Output/Input) × 100
5. cos = W/VI
6. Slip = Ns− N/Ns

Expected graph:
% Slip p.f. T % EFFI Torque
% Efficiency^{Torque} p.f.
% slip
Slip Output

EXPERIMENT 9
SWINBURNE'S TEST
Aim: To conduct Swinburne's test and predetermine the efficiency of the d.c. shunt machine as a generator and motor.
Apparatus Required:

Sl. no
1
2
Apparatus Type Range Quality Voltmeter DC 0–250 V 1 Ammeter DC 0–2 A 2

Machine ratings:
5 HP, 1500 rpm, 220 V

Meter ratings: Since it is a no-load test, the currents correspond to no-load values. The voltmeter reads the supply voltage. Therefore the ratings are
Voltmeter — 0–250 V Ammeters — 0–2 A
Circuit Diagram:
IL If
A
1
−
+−
2
230 V, 50 Hz
A
+D.C. 0−2 A 0−2 A
Supply 29 Ω 230 Ω
+ +
V 0−250 V A Z −−
AA ZZ
Circuit diagram with series armature rheostat
IL A If
A
1
−
LZ
+−
2

230 V, 50 Hz
A
+D.C.
Supply230 Ω
+
A Z −
AA ZZ
Circuit diagram with 3-point starter

Measurement of Armature Resistance

Ia
230 V, 50 Hz +A –
D.C. Loading0−20 ASupply rheostat A+ ++
A Va V 0−30 V
− −A−

Procedure:

1. Connect the circuit as shown in the circuit diagram.
2. Keep the armature rheostat in cut-in position and the field rheostat in cut-out position.
3. Close the supply switch.

4. Gradually cut-out the armature rheostat completely and cut-in the field rheostat till the motor runs at rated speed.
5. Note down the readings.
6. Bring back the rheostats to original position and switch off the supply.

Measurement of Armature Resistance

1. Connect the circuit as shown.
2. Close the supply switch.
3. Apply the load in steps and take down the readings.
4. Reduce load and open the supply.

Tabular Column:

V_L I_f I_L speed volts Amps Amps r.p.m.

To find armature resistance

Sl. No. V_a (V) I_a (A) R_a ( )
1

2
3

Average value of R_a = sl;dkfj .

The resistance obtained above is multiplied with a factor to account for increased resistance due to a.c. current flowing through the armature windings. The factor is normally taken to be 1.4.

Calculations:

Computation of constant loss

1. No load current = I_{L0}
2. No load voltage = V_{L0}
3. No load input = $V_{L0}I_{L0}$ = no-load losses (since o/p is zero)
4. Armature current$Ia_0 = I_{L0} - I_f$
5. Armature copper loss = Ia^2R_{a0}
6. Constant loss W_c = Input− Armature copper loss = $V_L0I_L0 - Ia^2Ra_0$

Computation of efficiency as a motor

For motor we calculate %η as $\frac{\text{Input}-\text{Losses}}{\text{Input}} \times 100$. We take values of I_L from no-load value to rated value. For a given value ofI_L, the efficiency is calculated as follows:

1. Input current = I_L A
2. Input voltage = V_L V
3. Input power = V_LI_L W
4. $I_a = I_L - I_f$ A

(Note that I_f is same as no-load value)

5. Armature copper loss = I^2R_a W_a
6. Total losses, $W_T = W_c + I^2R_a$ W_a
7. Output = Input − W_T W
8. $\%\eta = \frac{\text{Output}}{\text{Input}} \times 100$

Computation of Efficiency as a Generator
For a generator efficiency is calculated as
$\%\ \text{Output} \times 100\eta =$ Output + Losses
We again assume currents from no-load value to full load value.

1. Output current (load current) = I_L A
2. Terminal voltage V_L V
3. Output power $V_L I_L$ W
4. Ia = IL + If A
5. Armature copper loss = I^2R_a W_a
6. Total losses, $W_T = W_C + I^2R_a$ W_a
7. Input = Output + W_T W 8. $\%\eta = \frac{\text{Output}}{\text{Input}} \times 100.$

Expected graph:
Generator
%η
Motor
Output (W)
In subsequent experiments we show a rheostat in series with the armature. It can be replaced by a three point starter similar to what has been shown for this experiment.

EXPERIMENT 10
SPEED CONTROL OF DC SHUNT MOTOR
Aim: To study the speed control of a d.c. shunt motor by (a) armature control (b) field control.
Apparatus required:

Sl. No. Apparatus Type Range Quantity
1. Ammeter MC 0–2 A 1
2. Voltmeter MC 0–250 V 1

Circuit Diagram:
0−2 A
A
+−230 V, 50 Hz
D.C.
Supply100 Ω A 230 Ω

+
V 0−250 V Z −
100 Ω AA ZZ

Procedure:

(a) *Armature control method*

1. Make connections as shown in the circuit diagram.

2. Completely cutout the field rheostat and keep the potential divider at the position indicated, so that voltage across armature is zero.

3. Close the supply switch.
4. Adjust the field current to a constant value.

5. Increase the armature voltage using the potential divider. At each step, note down the voltage across the armature and the corresponding speed.
6. Repeat the experiment for different values of field current.

(b) *Field control method*

1. Make connections as shown in the figure.
2. Keep a fixed voltage across the armature.

3. Vary the field current using the field rheostat. At each step note down the field current and the corresponding speed.
4. Repeat the experiment for different values of armature voltage.

Tabular Column:

Armature voltage method

Armature Sl. No. Voltage V

I_{f1} = _____A I_{f2} = _____A Armature
N (r.p.m.) Voltage V N (r.p.m.)

Field current method Armature voltage = _____V Armature voltage = _____V
Field Field
Sl. No. Voltage A N (r.p.m.) Voltage A N (r.p.m.)

Expected graphs:

If_1
$^{If2\ V1}If_1 < If_2$
$^{N\ N\ V1 < V2}$(r.p.m.) (r.p.m.)

V2
V_a (V) I_f (A)

EXPERIMENT 11

O.C.C. OF D.C. SHUNT GENERATOR

Aim: To plot the open circuit characteristic of d.c. shunt generator.

Apparatus Required:

Sl. No. Apparatus Type Range Quantity
1. Ammeter MC 0–2 A 1
2. Voltmeter MC 0–300 V 1

Circuit Diagram:

230 V, 50 Hz
D.C.
Supply$_{230\Omega\ 230\Omega 29\Omega}$
$^{+}$ 0−250 V$_{+A+\ A+\ F+\ V\ --\ F+\ M\ G\ F-}$
$^{F-}$ A− A^{-} A^{+} −

Procedure:

1. Initially keep armature resistance of motor cut-in, field rheostat of motor cut-out and field rheostat of generator cut-in and close supply switch.
2. Cut-out armature rheostat of motor and cut-in field rheostat to bring motor to rated speed.

3. Cut-out the field rheostat of generator in steps. At each step note down the field current and the voltage developed.

4. Continue till the voltage developed is 10% above the rated voltage.
5. Bring back rheostats to original positions and open the supply switch.
6. Measure armature resistance.

Tabular column:

Sl. No. I_f (A) Voltage (V)

Calculation:

Armature current $I_a = I_f$ (since $I_L = 0$). Armature voltage drop = I_aR_a
Generated e.m.f. $E_g = V + I_aR_a$

Graph:

Critical resistance line

E_g (V)

Residual voltage I_f (A)

1. The voltage at $I_f = 0$, is the residual voltage.

2. Draw a line tangential to the linear part of the O.C.C. The slope of this line gives the critical resistance.

EXPERIMENT 12

LOAD CHARACTERISTICS OF D.C. SHUNT GENERATOR

Aim: To conduct load test on the given d.c. shunt generator and to plot internal characteristic (E_g vs. I_L) and external characteristics (V_L vs. I_L).

Apparatus Required:

Sl. No. Apparatus Type Range Quantity

1.

2.

3.

4.

Ammeter MC 0–2 A 1 Ammeter MC 0–20 A 1 Voltmeter MC 0–250 V 1 Loading rheostat 0–20 A 1

Circuit Diagram:

230 V, 50 Hz D.C.
Supply
0−20 A

+−
A
29Ω 29Ω 230Ω +
− A
+
A
+
F
+
0−250 V V F+ F−
F− A− A− A+ 0−2 A −

Procedure:

1. Keep rheostats in position described in experiment 11 and close supply switch.
2. Bring motor to rated speed.
3. Keep load switch open.
4. Adjust the field rheostat of the generator till rated voltage is built-up.
5. Note down the no-load readings.

6. Close the load switch. Apply the load in steps. At each step, note down the load current, field current and the terminal voltage.
7. Continue till rated load current flows.
8. Reduce the load gradually, bring back rheostats to original position and open the supply switch.
9. Measure the resistance of the armature. The speed has to be maintained constant at all times.

Tabular column:

Sl. No. I_f (A) V_L (Volts) N r.p.m I_L (A) $I_a = I_L + I_f$ $E_g = V + I_aR_a$ (V)

Expected Graph:

E_g

E_g (V) Volts V

I_L (A)

EXPERIMENT 13

LOAD TEST (BRAKE TEST) ON D.C. SHUNT GENERATOR

Aim: To conduct load test on d.c. shunt motor and draw its performance characteristics. **Apparatus Required:**

Sl. No.
1.
2.
Apparatus Type Range Quantity Ammeter MC 0–20 A 1
Voltmeter MC 0–250 V 1

Circuit Diagram:

230 V, 50 Hz + D.C.
Supply_ 0−20 A W1W2+−
A

A^+ $230\Omega_M$

A^- F^+
$29\Omega_{F-}$

Procedure:

1. Make connections as shown in the circuit diagram.
2. Keep armature rheostat cut-in and field rheostat cut-out.
3. Close the supply switch.
4. Cut out armature rheostat and gradually cut-in field rheostat till motor reaches rated speed.
5. Note down no-load readings.
6. Load the motor by tightening the brake. At each step note down the readings of the two spring balances, W_1 and W_2.
7. Continue loading till rated current is drawn by motor.
8. Release load. Bring back rheostats to original position and open the switch.

Tabular column:
IL VL N W1 W2 W1 ~ W2 T Sl. No. (A) (V) (r.p.m) (kg) (kg) (kg) N-M %η
Radius of brake drum, r = _____. **Calculations:**

1. $T = 9.81(W1 \sim W2) \times r$ N-M
2. output = $\frac{2\pi NT}{60}$ W
3. Input = $V_L I_L$ W
4. % $\eta = \frac{\text{Output}}{\text{Input}} \times 100$

Expected Graph:
η
T
N
Output
EXPERIMENT 14
REGULATION OF ALTERNATOR BY E.M.F. METHOD
Aim: Aim:φ alternator and predetermine the percentage regulation at different power factors by synchronous impedance (E.M.F.) method. **Apparatus Required:**

Sl. No

Sl. No.
1.
2.
3.
4. Apparatus Range Quantity A.C. ammeter 0–10 A 1 D.C. ammeter 0–2 A 1 A.C. voltmeter 0–500 V 1 D.C. voltmeter 0–300 V 1

Ratings of Machine:

DC Motor:
Rated capacity : 3.5 kW Rated I/P and O/P voltage : 220 V d.c. Armature current : 18.5 A Armature resistance : 0.35 Rated shunt field voltage : 220 V d.c. Rated shunt field current : 0.95 A Shunt field resistance : 174.1 Brake drum radius : 107 mm RPM : 1500

Alternator: Rated voltage : 415 AC Rated current : 7.9 A Frequency : 50 Hz RPM : 1430 Phase : 3
Output : 3.7 kW Connected :
Insulation class : F
Make : Kirloskar Brake drum radius : 107 mm

Circuit Diagram:
$^{0-2A}$ TPST Switch$+_{A}$ $-$
230 V, 50 Hz^{R} A.C.
Supply$_{29\Omega}$ 800$\Omega_{230\Omega}$
+
0−10 A 0−600 V A A$_{F+}$
$^{-\ F+}$ M^{Y}
F^{-} F− Y
B
OC and SC Tests
230 V, 50 Hz $+_{A}$ $-_{\text{Load RA.C.}}$
Supply 0−5 A
+
+
0−30 V A
$^{--}$ Y
Y$_{B}$ Measurement of armature resistance

Procedure:

1. Initially the field rheostat of d.c. motor is cut-out, armature rheostat cut-in and field rheostat of alternator cut-in. TPST switch is kept open.

2. The d.c. motor is brought to rated speed.
3. The field current of the alternator is increased in steps by varying the field rheostat. At every step the reading of the field rheostat. At every step the reading of the field current and the corresponding voltage generated is taken.

4. The steps are repeated till the alternator builds 10% above its rated voltage.

S.C. Test:

1. Bring back the field rheostat of alternator to original position.
2. Close the TPST switch.
3. Vary the field rheostat till the armature short circuit current equals the rated value.
4. Note down the short circuit current and the corresponding field current.
5. Bring back rheostats to original positions and open the supply switch.

Armature Resistance Measurement:

1. Make connections as shown in the circuit.
2. Close supply switch.

3. Apply load in different steps. At each step note down readings of the ammeter and voltmeter.
4. Remove load gradually and open the supply switch.

Tabular column:

OC Test

Sl. No. I_f (A) Terminal voltage (V) Terminal voltage/phase

=

Terminal voltage

$\sqrt{}_3$

Plot terminal voltage/phase vs I_f. *SC Test*

I_{fsc} (A) I_{sc} (A) Plot a straight line passing through origin and (I_{fsc}, I_{sc}). *Armature Resistance*

Sl. No.

I (A)

V (volts)

R
dc
=
V I
Average R_{dc} = _____ . R
dc
/
phase
=
Rdc
$^{2\ Rdc}$ .R_a = R_{ac}/phase = 1.6×$_2$ **Expected Graph:**
η
Voc
T
^{I}sc
I_f

Calculations:

1. Draw OC and SC characteristics.
2. From OC curve, find the open circuit voltageV_{oc} at a field currentI_{fsc} which produces the rated short circuit current.

3. $Z_s = {}^{Voc}$ (synchronous impedance)$_{Isc}$
$X_s = Z^2 - R^2$ (synchronous reactance)$_{s\ a}$

4. Generated e.m.f. $E0 = (V \cos\varphi + IRa)^2 + (V \sin\varphi \pm IXs)^2$
+ for lagging p.f. − for leading p.f.
V = Rated terminal voltage/phase. I = Rated current.

5. $\%R = {}^{E0\ -\ V} \times 100\%._V$
6. Calculate %R for various values of cosφ. 7. The regulation graph is as shown belows
%R Lag
Lead 01 0 cos $_{\varphi}$

CHAPTER - 9

Index

A
a.c. generators, 574
Absolute instruments, 313
AC servomotors, 618

Drag-cup rotor construction, 620
Torque-speed characteristic, 619
AC tachometer, 621
Active element, 88
Admittance, 183
Air friction damping, 318
All-day efficiency, 513
Alternating current, 5
Alternator, 574, 637
Armature reaction, 589
Armature windings, 577
Coil span, 577
Computation of regulation, 594
Computation of synchronous impedance, 594
Construction, 574
Full pitch, 577
On load, 589
Open circuit test, 593
Phasor diagram with armature reaction, 590
Phasor diagrams, 591, 592
Pole pitch, 577
Regulation by direct loading, 592
Short circuit test, 593
Stator core, 575
Stator frame, 574
Synchronization, 603
Synchronous reactance, 590
Voltage regulation, 591

Ammeters, 320
Amplitude, 153
Angular frequency, 153
Apparent power, 190
Application of DC generators, 398
Arcing time, 288
Argument, 153
Armature copper loss, 379
Armature core, 358
Armature diverters, 453
Armature reaction, 386, 589
Armature resistance, 589
Armature windings, 359, 577
Coil span, 577
Full pitch, 577
Leakage reactance, 589
Pole pitch, 577
Resistance, 589
Attraction type instruments, 328
Deflecting torque, 329
Auto transformer, 515
Auto transformer starter, 557

Average value, 170

B

B–H curve, 56
Batten wiring, 271
Bearings, 362
Bilateral element, 88
Biot-Savart’s law, 47
Blocked rotor test, 568
Brake test, 671
Breadth-factor, 579
Break down torque, 533
Breaking capacity, 289
British Standard wire guage, 273

Brush contact loss, 379

Brushes, 362
Generator characteristics, 386
Internal characteristic, 386
Magnetisation characteristics, 386
No-load saturation characteristic, 386

C
Cable Specification, 273
Capacitor run motors, 615
Capacitor-start motors, 614
Cartridge fuse, 290
Cfl Bulbs, 283
Characteristics of motors, 413

Characteristics, 413
d.c. compound motors, 419
dc series motor, 416
Performance curves, 419
shunt motor, 414
Charge, 4
chording-factor, 577
Cleat wiring, 270
Coefficient of coupling, 69
Coil span, 577
commercial efficiency, 513
Commutator, 362
Composite parallel circuits, 58
Composite series circuits, 57
Compound generator, 369
Long shunt compound generator, 369

Short shunt compound generator, 370

Compound motors, 413
Conduit wiring, 272
conservation of charge, 4
control motors, 618
Copper losses, 350, 379
Armature copper loss, 379
External characteristic, 386

Field copper loss, 379

Crest factor or peak factor or amplitude factor, 171

Critical resistance, 391
current, 4
current source, 90
current-controlled current source, 91
current-controlled voltage source, 90
Cut-off current, 288

D
d.c. compound generator, 391
Load characteristics, 391

d.c. generator, 356
Armature copper loss, 379
Armature core, 358
Armature reaction, 386
Armature winding, 359
Bearings, 363
Brush contact loss, 379
Brushes, 362
Characteristics, 386
Commutator, 362
Compound generator, 369
Construction, 356
Copper losses, 379
E.m.f. equation of d.c. generator, 363

Eddy current loss, 379
Efficiency, 380
External characteristic, 386
Field coils, 362
Field copper loss, 379
Internal characteristic, 386
Iron loss, 378
Losses, 378
Magnetisation characteristics, 386

Mechanical losses, 379
No-load saturation characteristic, 386

Pole core, 361
Pole shoe, 361
Power stages, 380
Self-excited generator, 366
Separately excited generator, 366

Series generator, 368
Shunt generator, 367
Voltage regulation, 389
Yoke, 358

D.C. Machines, 348, 354
Armature core, 358
Bearings, 363
Brushes, 362
Commutator, 362
Field coils, 362
Generator, 356
Pole core, 361
Pole shoe, 361
Principle of working, 355
Yoke, 358

D.C. Motor starters, 440
d.c. Motors, 431
Brake test, 432
Swinburne’s test, 435
D.C. Shunt Generator, 635
D.C. Shunt Motor, 444, 634
Armature control, 445
Flux control, 445
Speed control, 444
Damper windings, 576
DC motor, 402
Applications, 421
Condition for maximum power, 404
Efficiency, 421